Integrated Communication Circuits

Lecture 5 : Amplitude Modulation and Demodulation

Prof. LIN MIN

第五章　振幅调制与解调

5.1　调幅信号的基本特性

5.2　乘法器

5.3　平衡调制器

5.4　高电平调幅

5.5　包络检波

5.6　同步检波

5.1　调幅信号的基本特性

振幅调制的方式

1. AM(标准振幅调制)
2. DSB(抑制载波的双边带振幅调制)
3. SSB(抑制载波的单边带振幅调制)
4. VSBAM(残留边带振幅调制)

共同特性：

对频谱进行不失真的搬移。

Vestigial Sideband Suppressed Carrier

抑制载波 Suppressed Carrier

一、标准振幅调制
(Amplitude Modulation　AM)

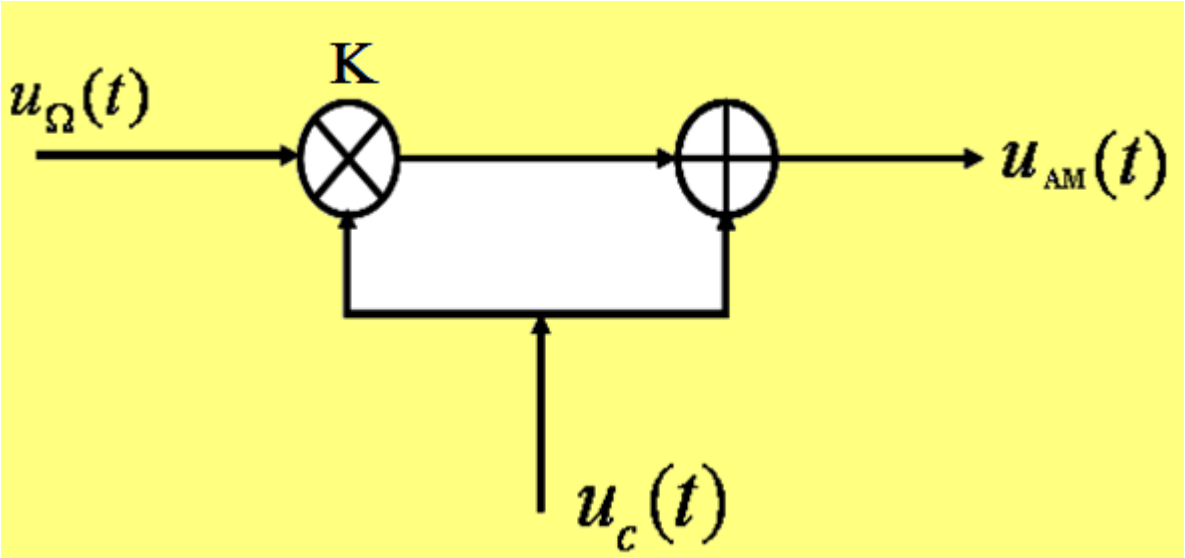
1. 组成模型

设载波信号为 $u_c(t) = U_{cm}\cos\omega_c t$

调制信号(基带信号)为 $u_\Omega(t)$

已调信号

$$u_{\mathrm{AM}}(t) = U_m(t)\cos\omega_c t$$
$$= \left[U_{cm} + k u_\Omega(t)\right]\cos\omega_c t$$



$$u_{AM}(t) = Ku_\Omega(t)u_c(t) + u_c(t)$$
$$= U_{cm}\left[1 + Ku_\Omega(t)\right]\cos\omega_c t$$

2. 单音调制

假设

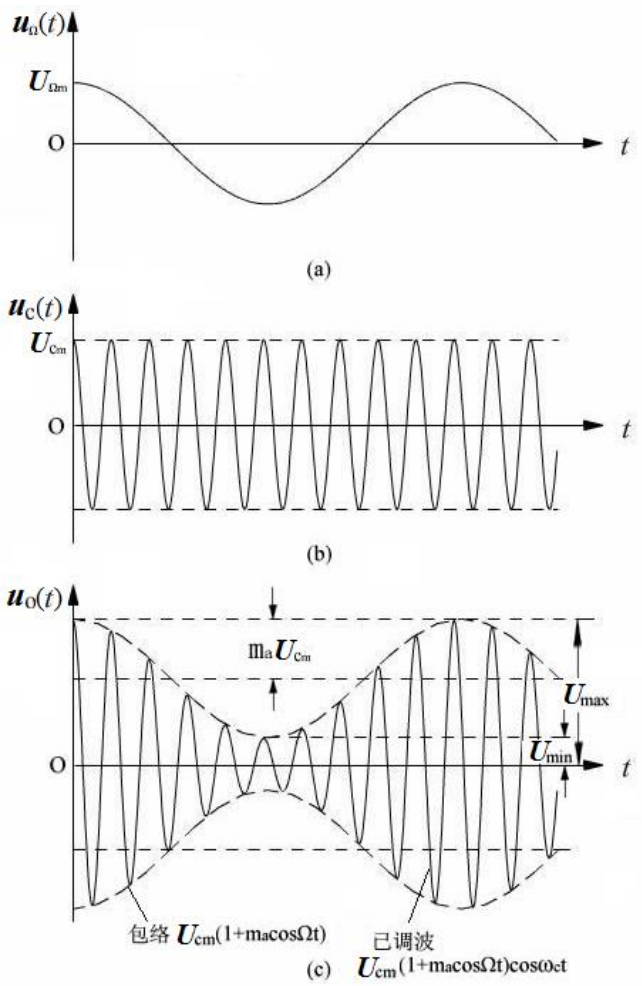
$$u_{\Omega}(t) = U_{\Omega \mathrm{m}} \cos \Omega t$$

$$u_c(t) = U_{cm} \cos \omega_c t$$

$$\begin{aligned} u(t) &= (U_{cm} + KU_{\Omega m} U_{cm} \cos \Omega t) \cos \omega_c t \\ &= U_{cm}(1 + KU_{\Omega m} \cos \Omega t) \cos \omega_c t \\ &= U_{cm}(1 + m_a \cos \Omega t) \cos \omega_c t \end{aligned}$$

$$\boxed{u_{\mathrm{AM}}(t) = U_{cm}(1 + m_a \cos \Omega t) \cos \omega_c t}$$

m_a **为调幅系数**

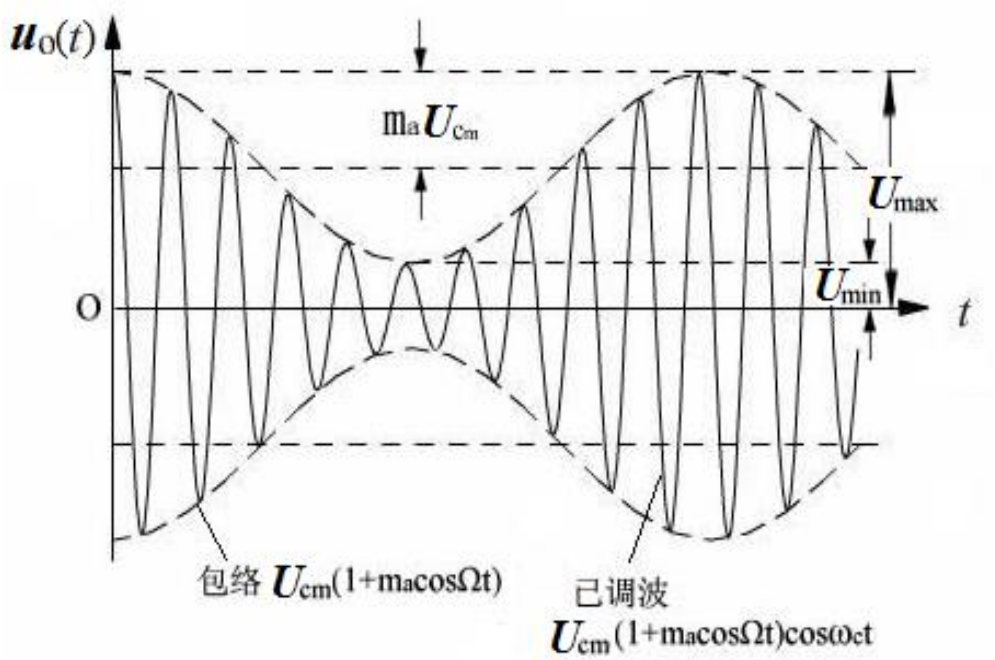


若 $0 < m_a < 1$ 的情况下

5-1.EXE

若 $m_a > 1$ (过调)

5-2.EXE



$$U_{\max} = (1+m_a)U_{cm}$$

$$U_{\min} = (1-m_a)U_{cm}$$

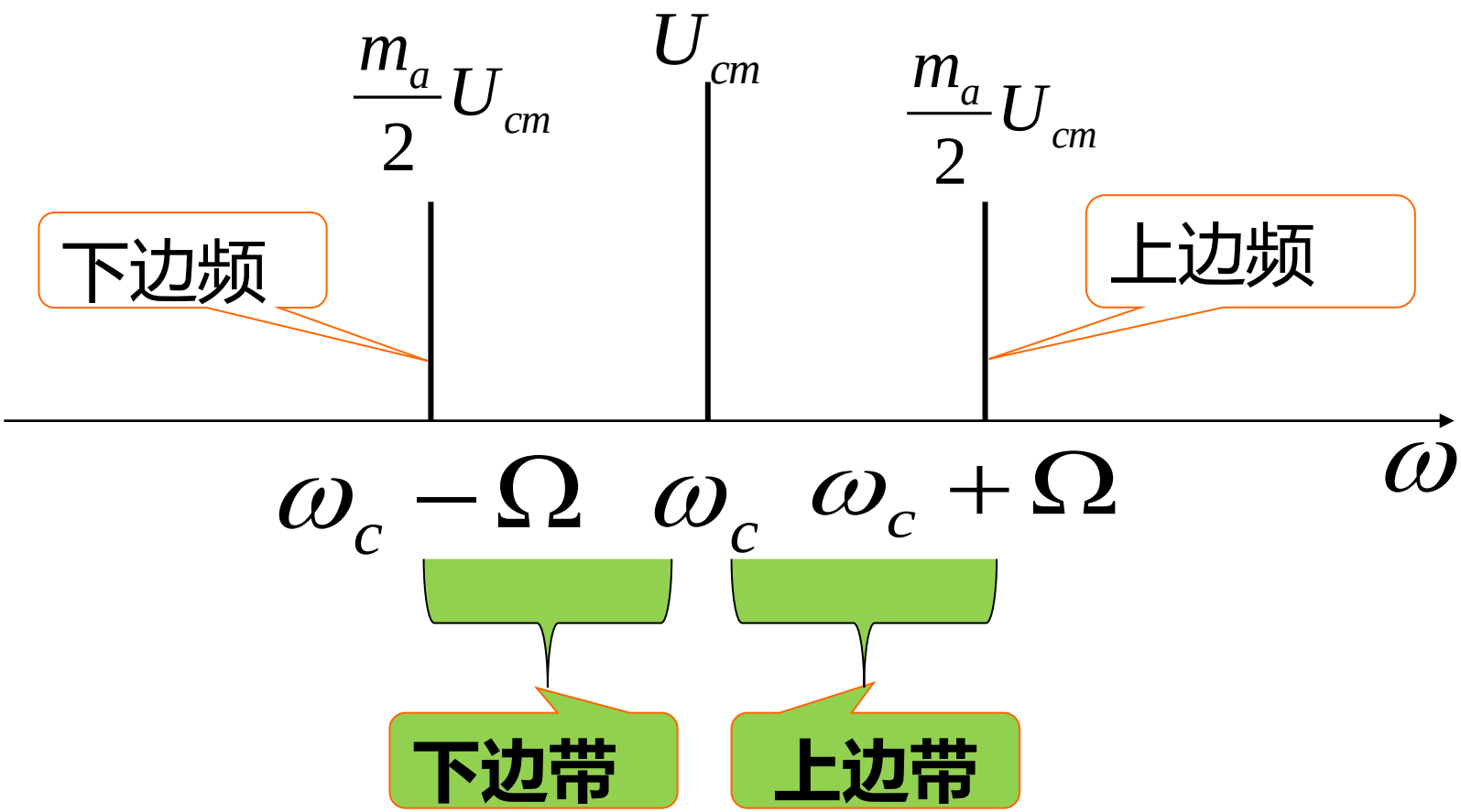
$$\therefore \quad m_a = \frac{\frac{1}{2}(U_{\max} - U_{\min})}{U_{cm}} = \frac{U_{cm} - U_{\min}}{U_{cm}}$$

3. AM信号的频谱的带宽

$$u(t) = U_{cm}(1 + m_a \cos \Omega t) \cos \omega_c t$$

$$= U_{cm} \cos \omega_c t + \frac{1}{2} m_a U_{cm} \cos(\omega_c + \Omega)t$$

$$+ \frac{1}{2} m_a U_{cm} \cos(\omega_c - \Omega)t$$



$$BW_{\mathrm{AM}}=2\Omega$$

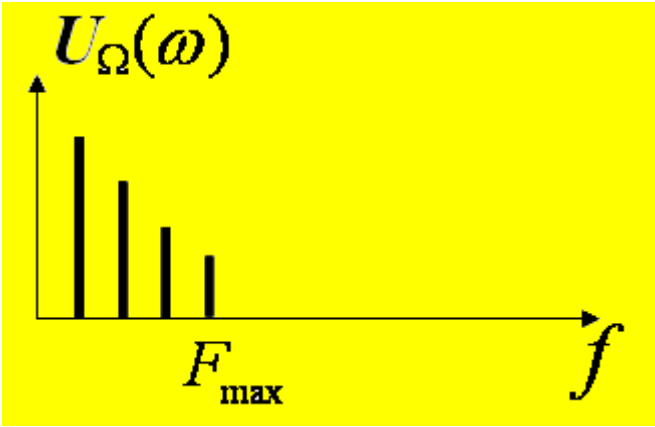
若 $u_\Omega(t)$ 为复杂音调制，即 $u_\Omega(t)=\sum_{n=1}^{n_{\max}} U_{\Omega mn}\cos n\Omega t$

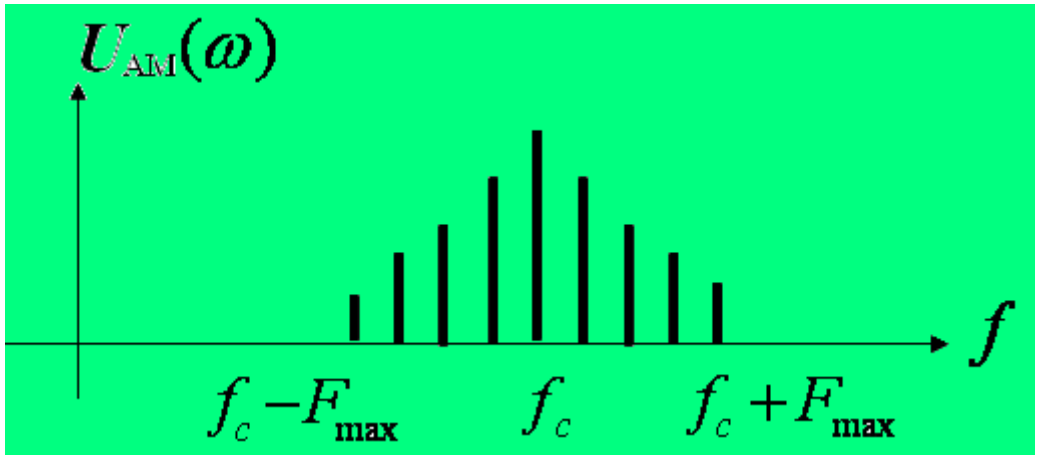
其中 $F_{\max}=\dfrac{\Omega_{\max}}{2\pi}$

$$n_{\max}=\Omega_{\max}/\Omega=F_{\max}/F$$

$$BW_{\mathrm{AM}}=2\Omega_{\max}$$

或 $BW_{\mathrm{AM}}=2F_{\max}$





4. 功率

载波 $$P_o = \frac{1}{2}\frac{U_{cm}^2}{R_L}$$

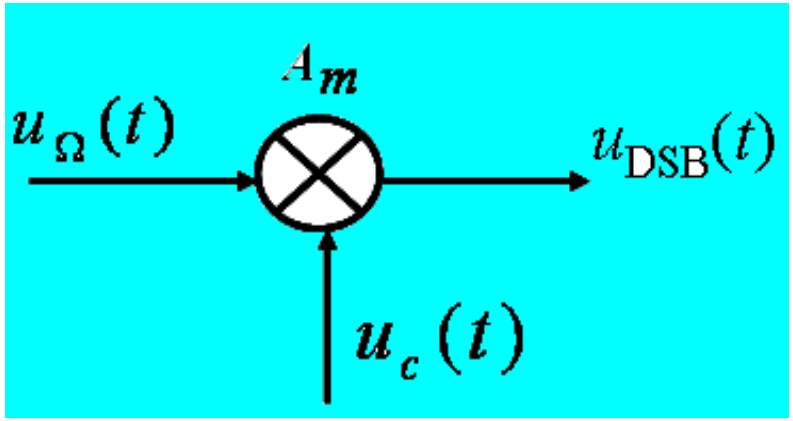
上边频或下边频 $$P_{\mathrm{SSB}} = (\frac{1}{2}m_a U_{cm})^2 \cdot \frac{1}{2R_L} = \frac{m_a^2}{4}P_o$$

$$P_{\mathrm{DSB}} = 2P_{SSB} = \frac{m_a^2}{2}P_o$$

$$P_{\mathrm{AM}} = P_o + P_{\mathrm{DSB}} = (1+\frac{m_a^2}{2})P_o$$

二、抑制载波的双边带调制

(Suppressed-Carrier Double Sideband Modulation. DSB)

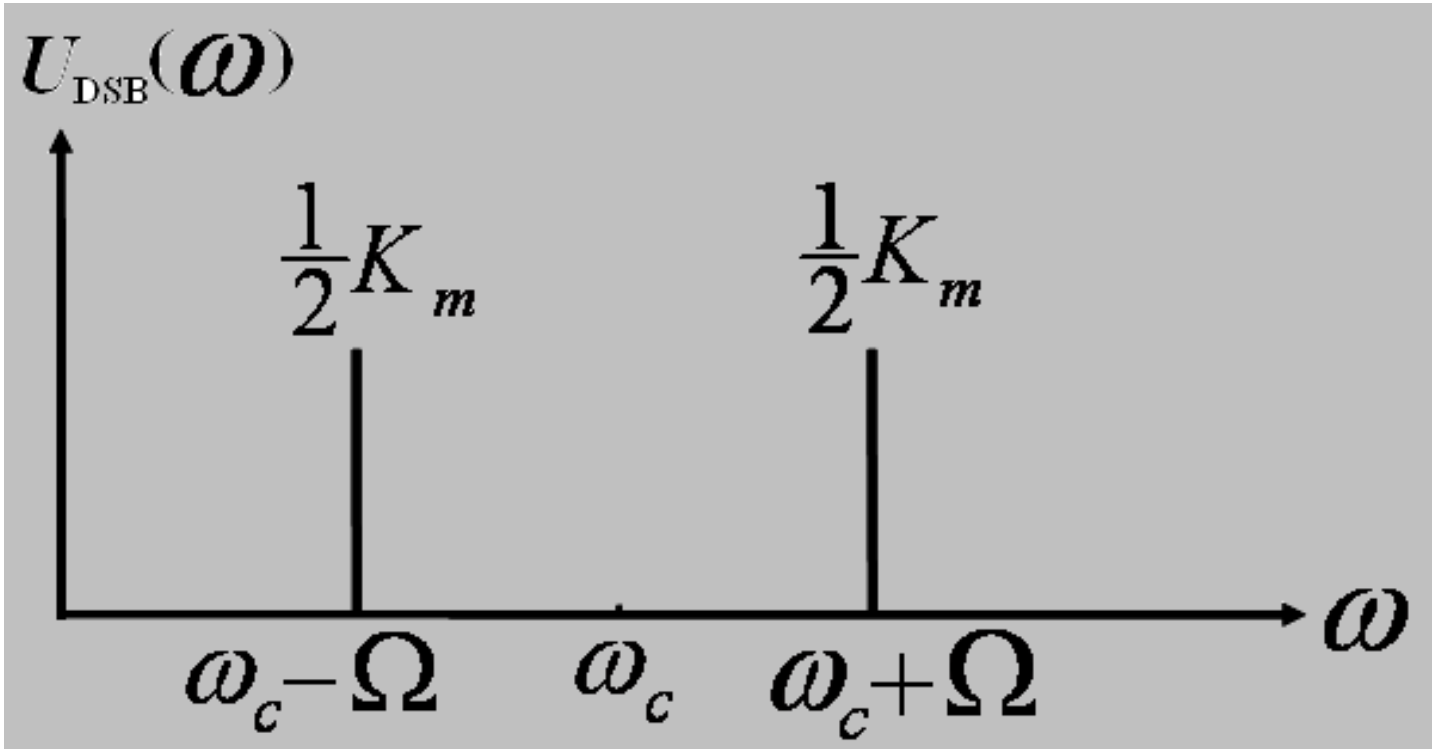


$$u_{DSB}(t) = A_m U_{cm} U_{\Omega m} \cos\Omega t \cdot \cos\omega_c t$$

$$= \frac{1}{2} A_m U_{cm} U_{\Omega m} \cos(\omega_c + \Omega)t + \frac{1}{2} A_m U_{cm} U_{\Omega m} \cos(\omega_c - \Omega)t$$

$$= \frac{1}{2} K_m \cos(\omega_c + \Omega)t + \frac{1}{2} K_m \cos(\omega_c - \Omega)t$$

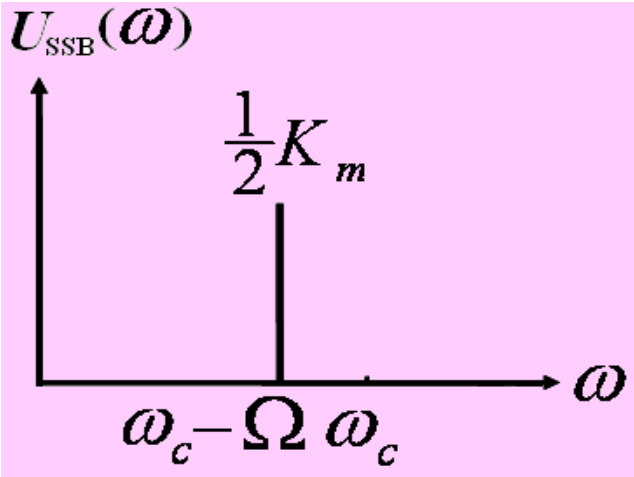
其中$K_m = A_m U_{cm} U_{\Omega m}$



$$BW_{DSB} = 2\Omega$$

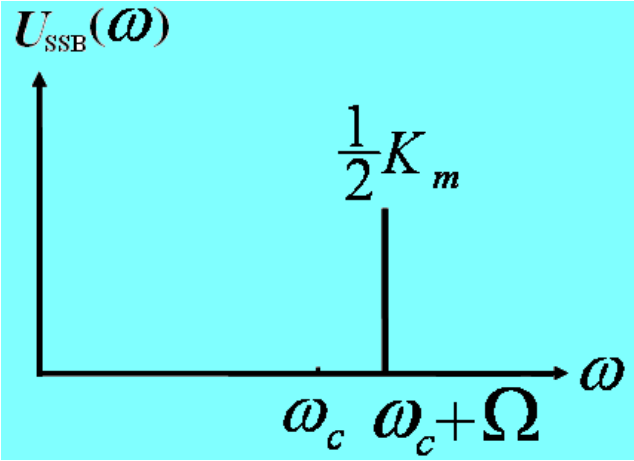
三、抑制载波的单边带调制

(Single Sideband Modulation. SSB)



下边带

$$u_{SSB}=\frac{K_m}{2}\cos(\omega_c-\Omega)t$$

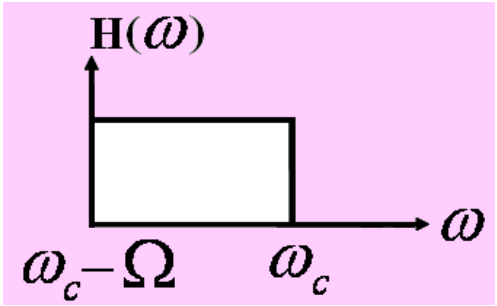


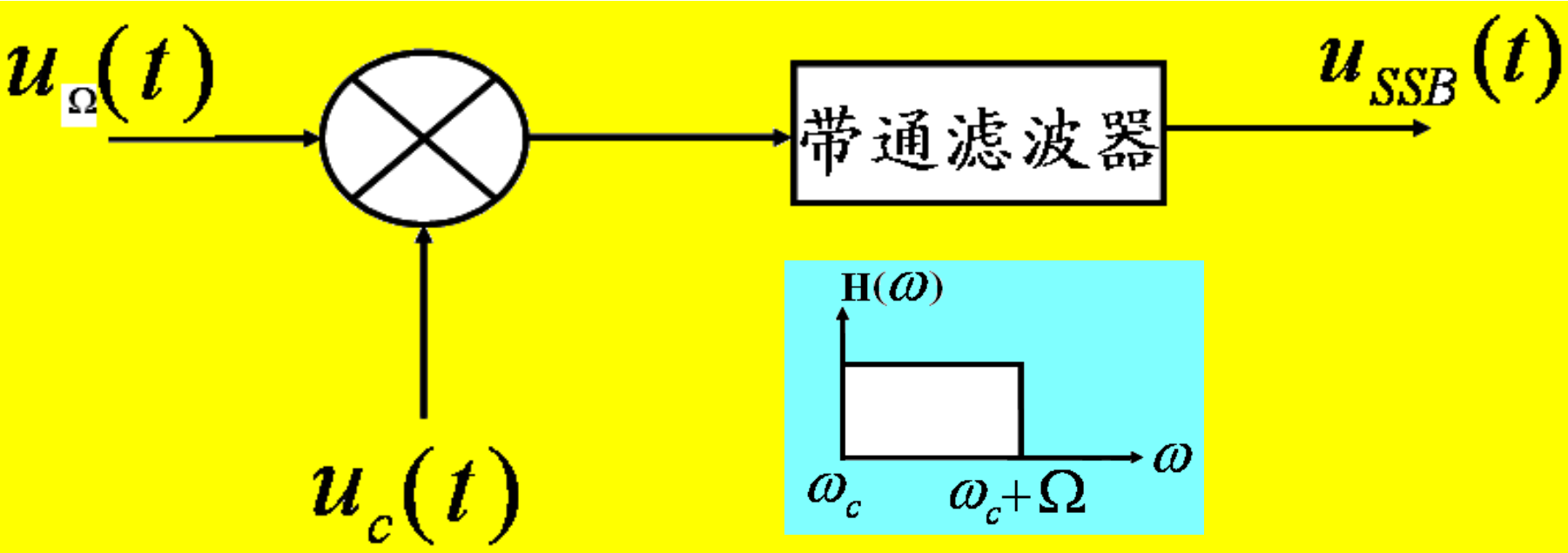
上边带

$$u_{SSB}=\frac{K_m}{2}\cos(\omega_c+\Omega)t$$

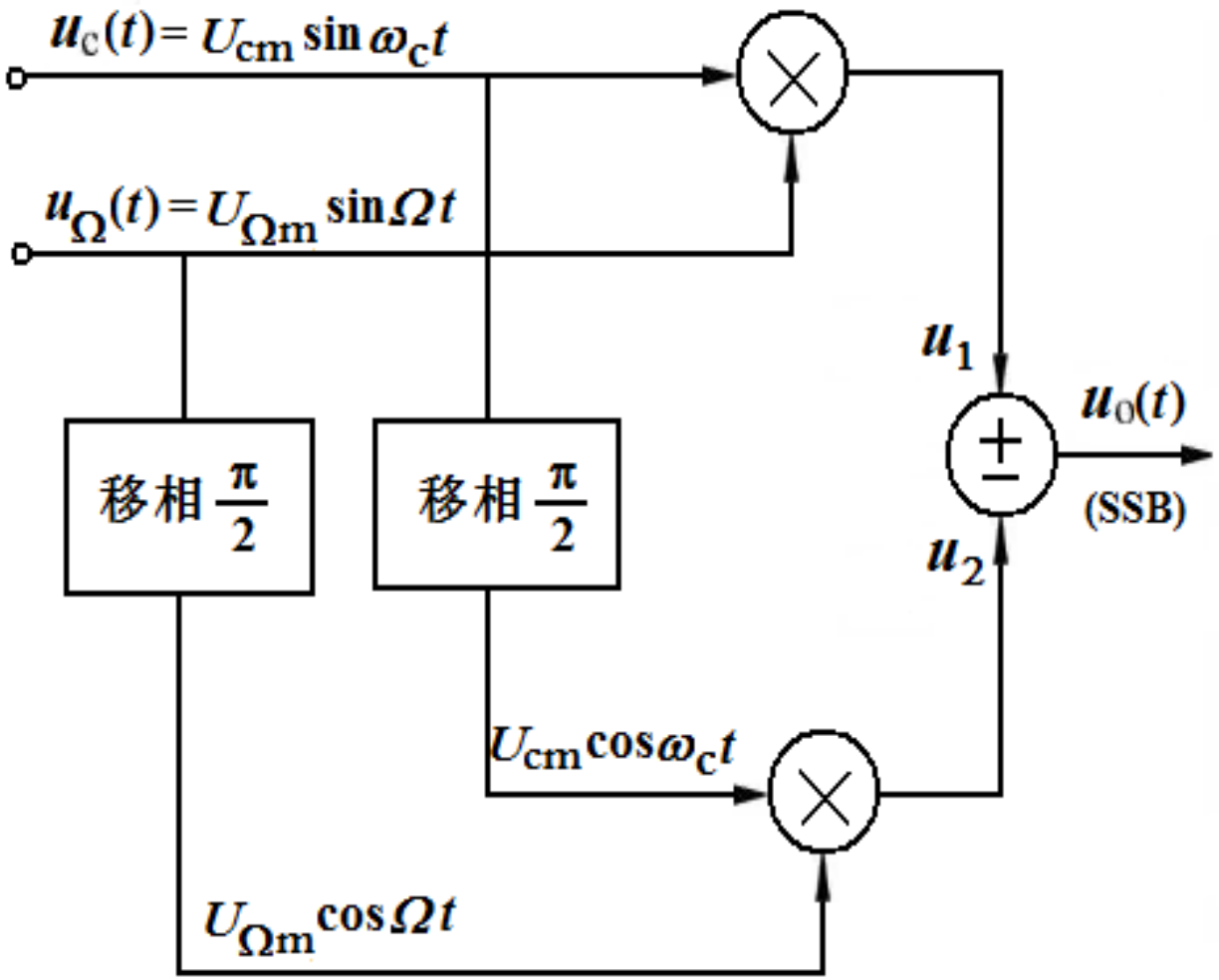
$$BW_{SSB}=\Omega$$

（1） 滤波法





（２） 移相法

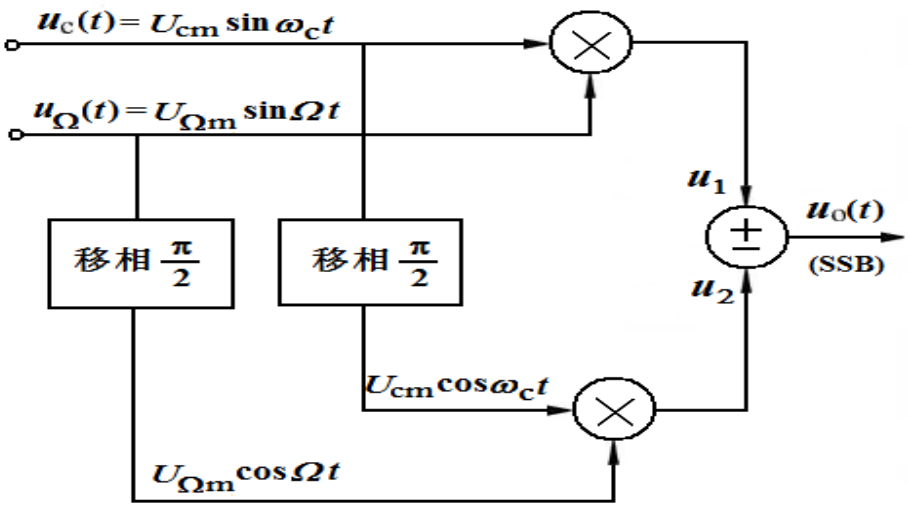

$u_c(t) = U_{cm}\sin\omega_c t$
$u_\Omega(t) = U_{\Omega m}\sin\Omega t$
移相 $\frac{\pi}{2}$
移相 $\frac{\pi}{2}$
u_1
u_2
$u_0(t)$
(SSB)
$U_{cm}\cos\omega_c t$
$U_{\Omega m}\cos\Omega t$

$$u_1(t) = A_m U_{\Omega m} U_{cm} \sin \Omega t \sin \omega_c t$$

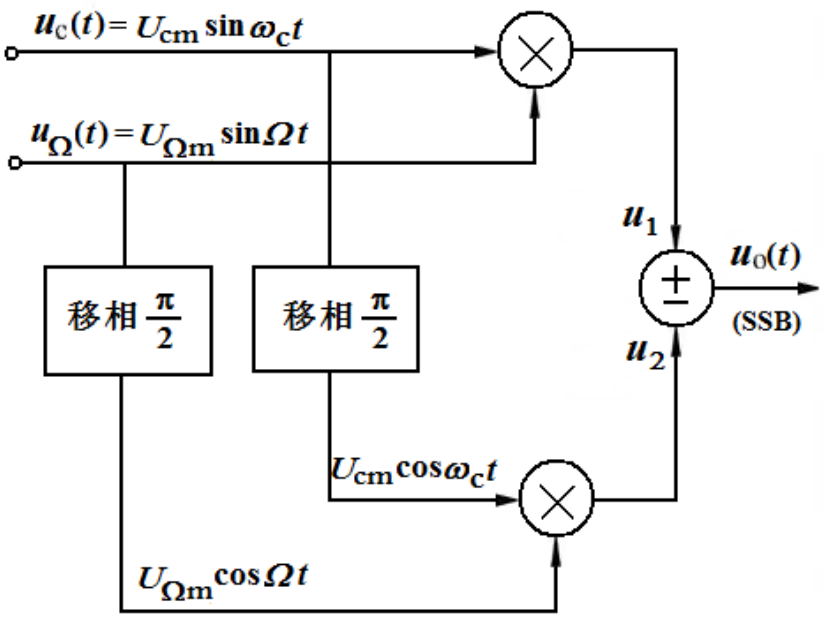
$$= \frac{1}{2} A_m U_{\Omega m} U_{cm} \left[\cos(\omega_c - \Omega)t - \cos(\omega_c + \Omega)t \right]$$

$$u_2(t) = A_m U_{\Omega m} U_{cm} \cos \Omega t \cos \omega_c t$$

$$= \frac{1}{2} A_m U_{\Omega m} U_{cm} \left[\cos(\omega_c - \Omega)t + \cos(\omega_c + \Omega)t \right]$$

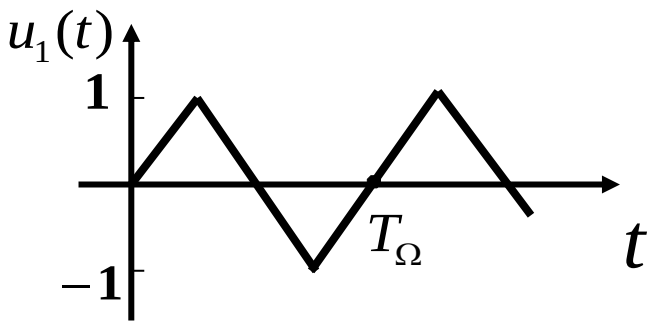


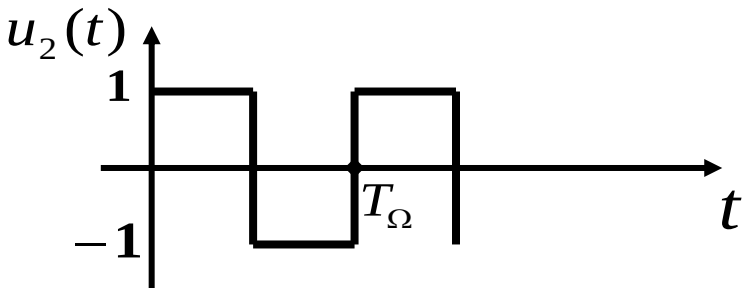
$$u_{SSB}(t)=\begin{cases} u_1(t)-u_2(t)=A_m U_{\Omega m} U_{cm}\cos(\omega_c+\Omega)t \\ u_1(t)+u_2(t)=A_m U_{\Omega m} U_{cm}\cos(\omega_c-\Omega)t \end{cases}$$



例:

已知载波信号为 $u_c(t)=U_{cm}\cos\omega_c t$,调制信号分别为 $u_1(t)$，$u_2(t)$，且 $f_c >> \dfrac{1}{T_\Omega}$ ，分别画出m_a=0.5及m_a=1 两种情况下AM波形以及DSB波形。

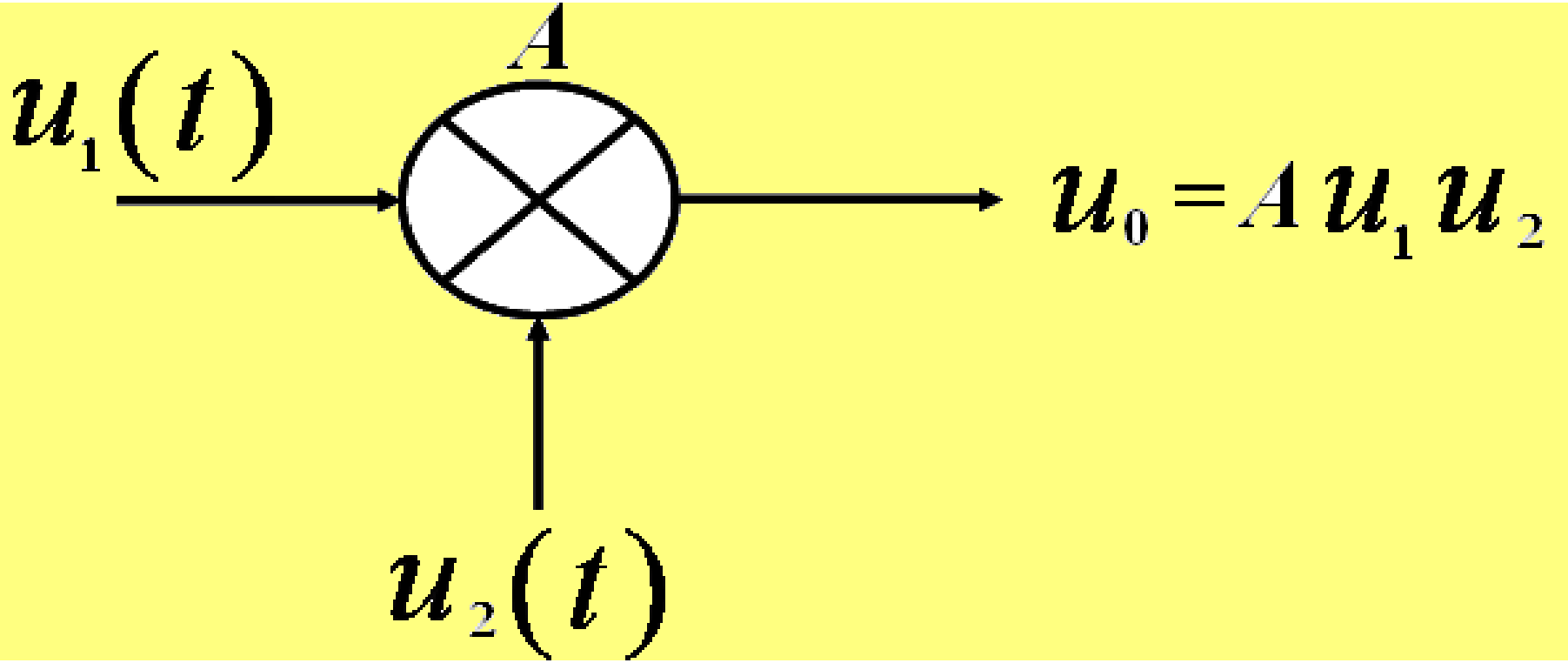




解法

解法

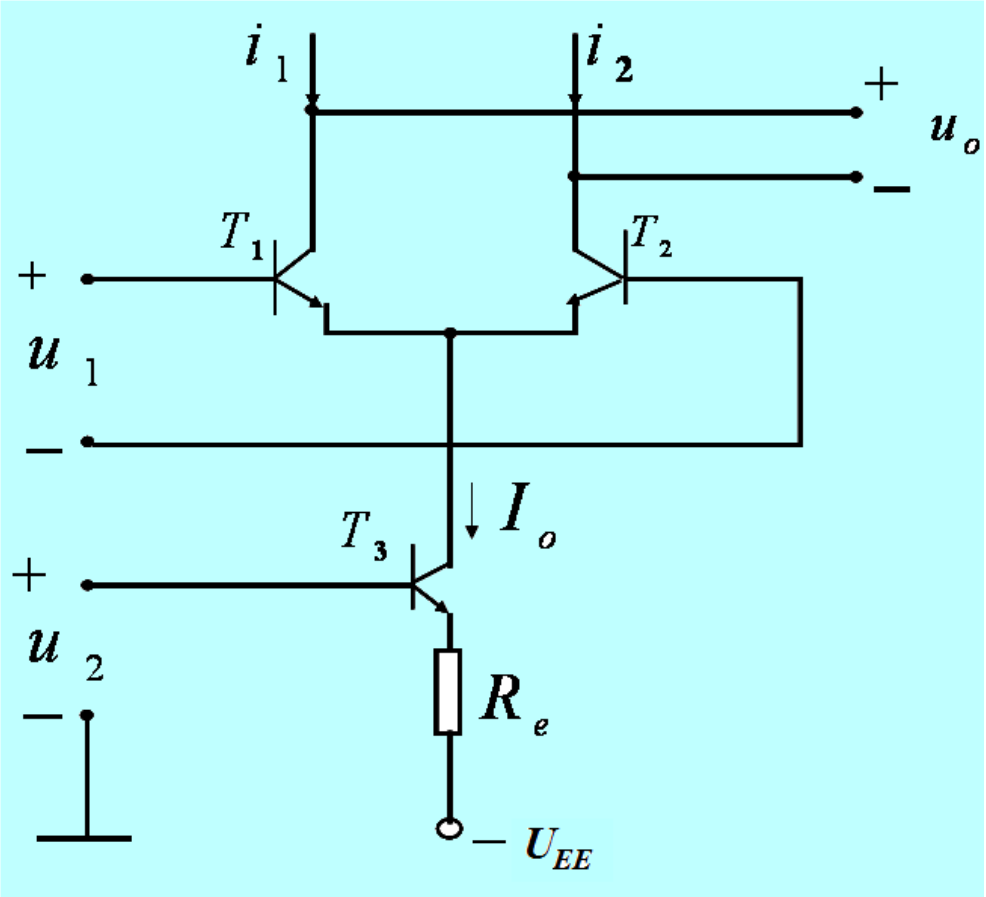
5.2 乘法器



1. 单差分对乘法电路

非线性通道

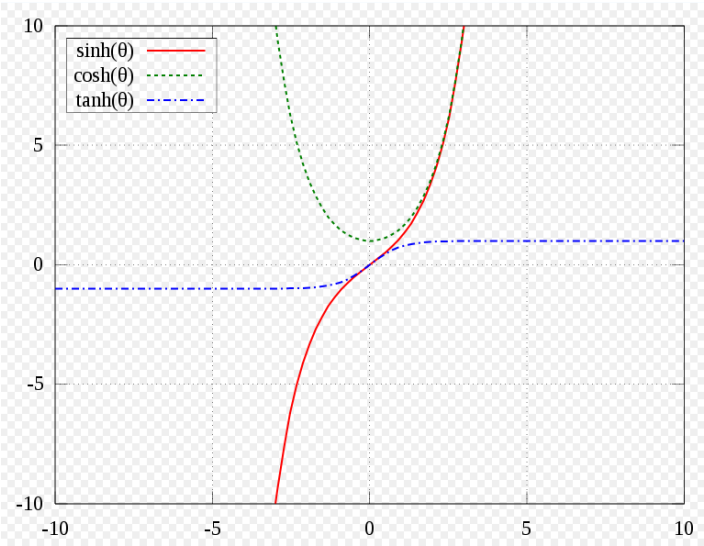
线性通道

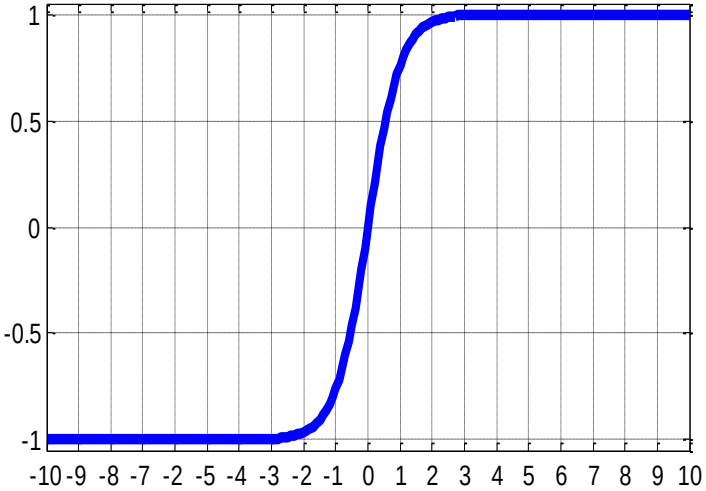


$$u_1 = U_{1m}\cos\omega_1 t \qquad u_2 = U_{2m}\cos\omega_2 t$$

$$\begin{cases} i_1 = i_{c1} = \dfrac{I_o}{2} + \dfrac{I_o}{2}\tanh\left(\dfrac{u_1}{2U_T}\right) \\ i_2 = i_{c2} = \dfrac{I_o}{2} - \dfrac{I_o}{2}\tanh\left(\dfrac{u_1}{2U_T}\right) \end{cases}$$

$$u_o = R_L(i_1 - i_2) = R_L\Delta i = R_L I_o \tanh\left(\frac{u_1}{2U_T}\right)$$


sinh(θ)
cosh(θ)
tanh(θ)
10
5
0
-5
-10
-10
-5
0
5
10


1
0.5
0
-0.5
-1
-10 -9 -8 -7 -2 -5 -4 -3 -2 -1 0 1 2 3 4 5 6 7 8 9 10

$$I_o = \frac{U_{EE} - U_{BE} + u_2}{R_e}$$

$$= \frac{U_{EE} - U_{BE}}{R_e} + \frac{u_2}{R_e} = A + Bu_2$$

与u_2呈线性变化

$$\Delta i = i_1 - i_2 = (A + Bu_2)\tanh(\frac{u_1}{2U_T})$$

与u_1呈非线性变化

（1） u_1 为小信号，$\tanh(\frac{x}{2})$，$x<1$

$$U_{1m} \le U_T = 26\ \text{mV}$$

$$\tanh(\frac{u_1}{2U_T}) \approx \frac{u_1}{2U_T}$$

$$\Delta i = (A + Bu_2) \cdot \frac{u_1}{2U_T}$$

$$= \frac{A}{2U_T} \cdot u_1 + \frac{B}{2U_T} \cdot u_1 u_2$$

（2） u_1 为中等信号 $U_T < U_{1m} < 10U_T$

$$\tanh\frac{u_1}{2U_T} = \tanh\left[\frac{U_{1m}}{2U_T}\cos\omega_1 t\right] = \tanh\left[\frac{x}{2}\cos\omega_1 t\right]$$

式中 $x = \dfrac{U_{1m}}{U_T}$ **再用傅里叶级数展开**

$$\tanh\left[\frac{x}{2}\cos\omega_1 t\right] = 2\beta_1(x)\cos\omega_1 t + 2\beta_3(x)\cos 3\omega_1 t + \cdots\cdots$$

$$= 2\sum_{n=1}^{\infty}\beta_{2n-1}(x)\cos(2n-1)\omega_1 t$$

$$\Delta i = 2(A + Bu_2)\sum_{n=1}^{\infty}\beta_{2n-1}(x)\cos(2n-1)\omega_1 t$$

（3） u_1 为大信号，$U_{1m} \geq 10U_T = 260\,\text{mV}$

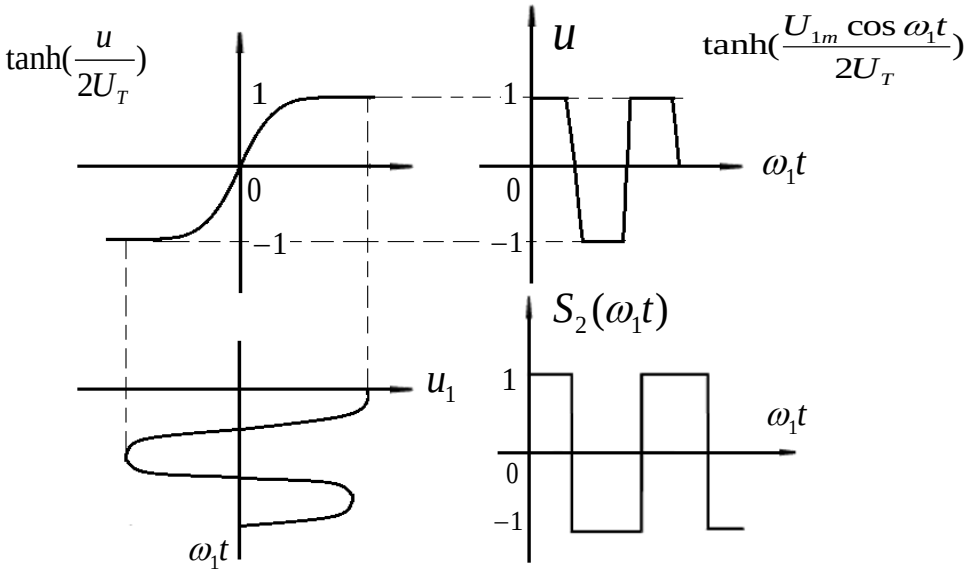
$$\tanh(\frac{x}{2}\cos\omega_1 t) \approx S_2(\omega_1 t)$$

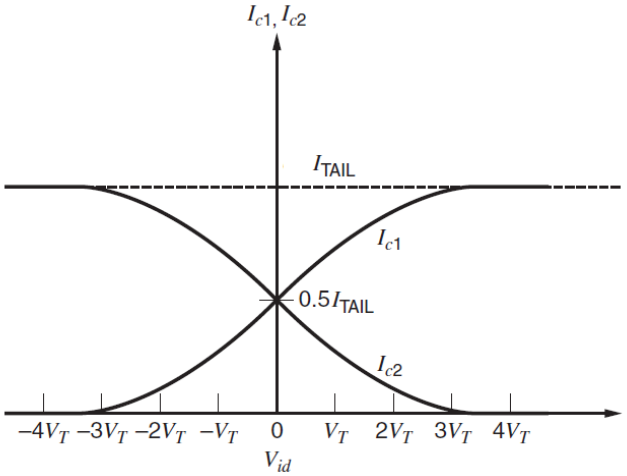
$$\Delta i = (A + Bu_2)\tanh(\frac{u_1}{2U_T})$$

$$= (A + Bu_2)S_2(\omega_1 t)$$

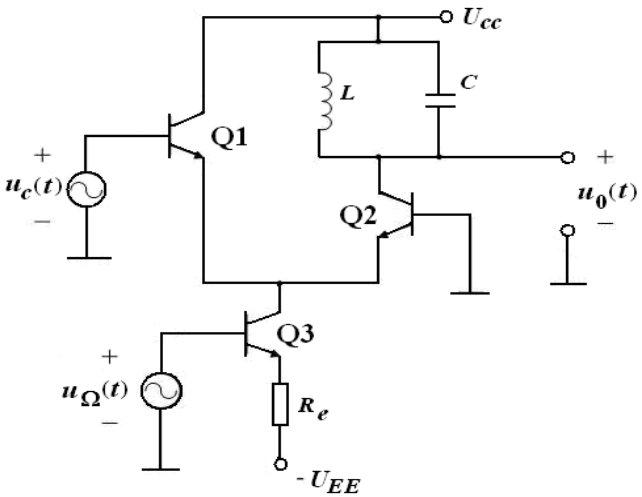
展开 $S_2(\omega_1 t)$

$$S_2(\omega_1 t) = \frac{4}{\pi}\cos\omega_1 t - \frac{4}{3\pi}\cos 3\omega_1 t + \frac{4}{5\pi}\cos 5\omega_1 t + \cdots$$


$\tanh(\frac{u}{2U_T})$
1
0
−1
u
$\tanh(\frac{U_{1m}\cos\omega_1 t}{2U_T})$
1
0
−1
$\omega_1 t$
u_1
$\omega_1 t$
$S_2(\omega_1 t)$
1
0
−1
$\omega_1 t$


I_{c1}, I_{c2}
I_{TAIL}
I_{c1}
$0.5I_{TAIL}$
I_{c2}
$-4V_T$ $-3V_T$ $-2V_T$ $-V_T$ 0 V_T $2V_T$ $3V_T$ $4V_T$
V_{id}

例 差分对调制器电路如图所示



(1) $\omega_c = 10^7 \ \mathrm{rad/s}$

并联谐振回路对 ω_c 谐振，

谐振电阻 $R_L = 5\ \mathrm{k\Omega}$ $\quad U_{CC} = U_{EE} = 10\ (\mathrm{V})$

$u_c(t) = 156\cos\omega_c t\ (\mathrm{mV})$ $\quad R_E = 5\ \mathrm{k\Omega}$

$u_\Omega(t) = 5.63\cos 10^4 t\ (\mathrm{V})$ 求 $u_o(t) = ?$

(2) 此电路能否得到双边带信号？为什么？

解

Q3管中

$$i_{c3} \approx i_{e3} = \frac{u_{\Omega}(t) - U_{be3} + U_{EE}}{R_e} \approx \frac{u_{\Omega}(t) + U_{EE}}{R_e}$$

$$= 2(1 + 0.563\cos 10^4 t) \quad （\mathbf{mA}）$$

Q1管中 $U_{cm} = 156\ \text{mV} > 26\ \text{mV}$

$$i_{c2} = \frac{I_o}{2}\left[1 - \tanh\left(\frac{u_c}{2U_T}\right)\right]$$

$$= \frac{i_{c3}}{2}\left[1 - \tanh\left(\frac{156}{2\times 26}\cos\omega_c t\right)\right]$$

$$= (1 + 0.563\cos 10^4 t)\left[1 - 2\beta_1(6)\cos\omega_c t + 2\beta_3(6)\cos 3\omega_c t + \cdots\right]$$

$$= (1 + 0.563\cos 10^4 t)\left[1 - 1.2\cos\omega_c t + 0.28\cos 3\omega_c t + \cdots\right]$$

i_{c2}中能够通过滤波器的分量 i_{c2}' 为

$$i_{c2}' = -\mathbf{1.2}(\mathbf{1}+\mathbf{0.563}\cos\mathbf{10^4}t)\cos\omega_c t$$

则输出电压为:

$$u_o(t) = U_{cc} - i_{c2}' R_L$$

$$= 10 + 6(1 + 0.563\cos 10^4 t)\cos\omega_c t \quad (\mathrm{V})$$

AM波

(2) 此电路不能得到DSB信号

单差分对电路要得到DSB信号的条件是：

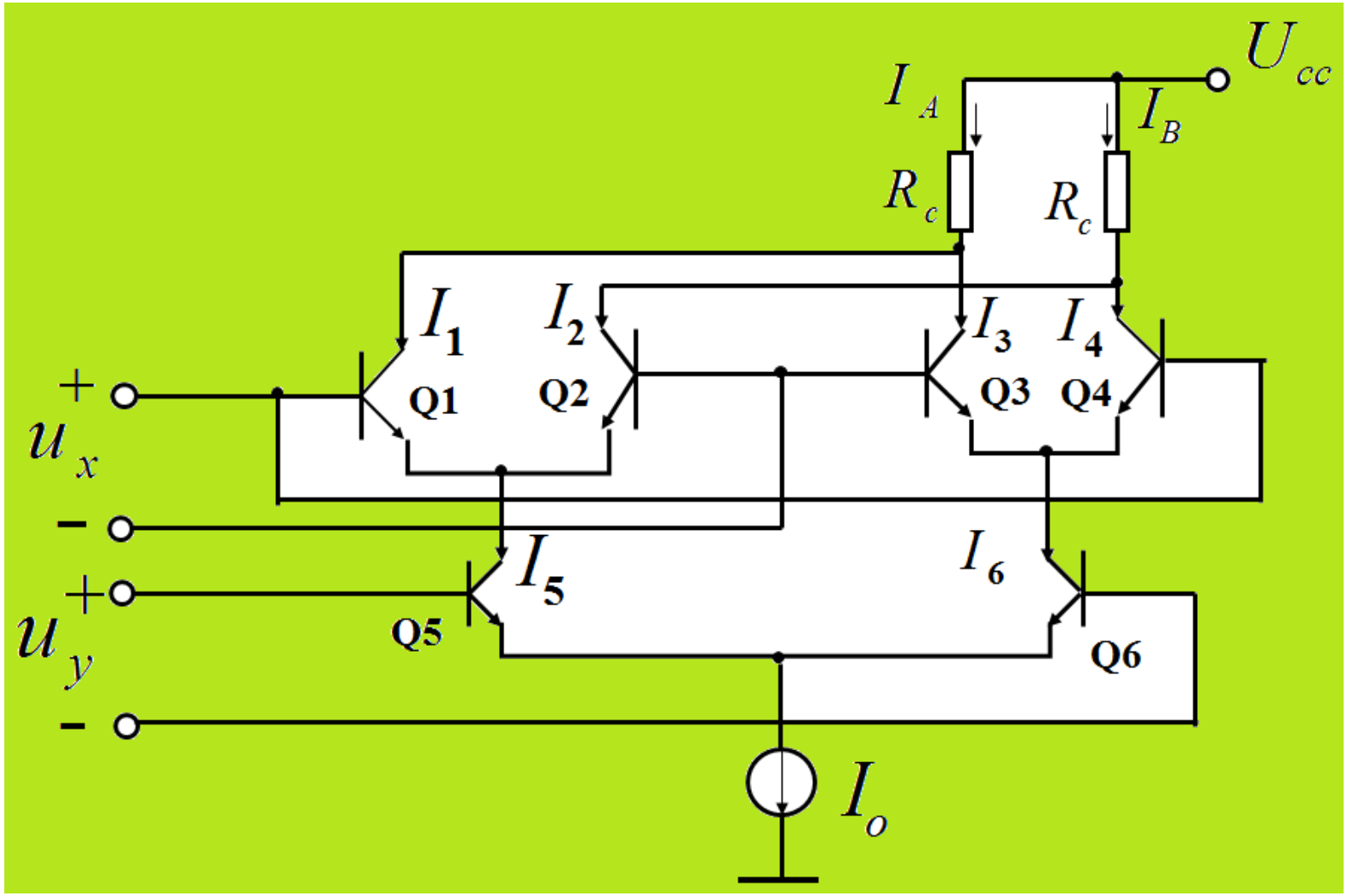
- ◆ u_Ω加在非线性信道

  u_c 加在线性信道。

- ◆双端输出

讨论：单差分对电路产生DSB信号与信号所加位置有关且必须双端输出，否则只能产生AM信号。

2. 双差分对乘法器（Gilbert）



双差分对乘法器

$$I_5 - I_6 = I_o \tanh(\frac{u_y}{2U_T})$$

$$I_1 - I_2 = I_5 \tanh(\frac{u_x}{2U_T})$$

$$I_4 - I_3 = I_6 \tanh(\frac{u_x}{2U_T})$$

双差分对乘法器

$$i = I_A - I_B = I_1 - I_2 - (I_4 - I_3)$$

$$= (I_5 - I_6)\tanh(\frac{u_x}{2U_T}) = I_o \tanh(\frac{u_y}{2U_T})\tanh(\frac{u_x}{2U_T})$$

$$\therefore u_o = iR_c = I_o R_c \tanh(\frac{u_y}{2U_T})\tanh(\frac{u_x}{2U_T})$$

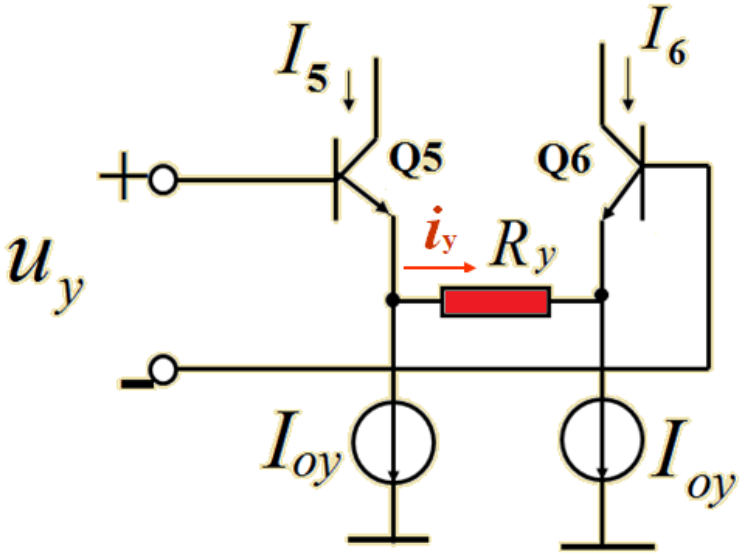
(1) 当 $u_x < U_T$ 时 $u_y < U_T$

$$u_o = \frac{I_o R_c}{4U_T^2} u_x u_y = A u_x u_y$$

(2) 当 $|u_y| < 26\ \text{mV}$ $|u_x| > 260\ \text{mV}$

$$u_o = \frac{I_o R_c}{2U_T} u_y S_2(\omega_1 t)$$

3.扩展 u_y 的动态范围



$$\left.\begin{aligned} R_y &>> 2r_{e5} = \frac{2U_T}{I_{oy}} \\ R_y &>> 2r_{e6} = \frac{2U_T}{I_{oy}} \end{aligned}\right\}$$ **共射极结电阻**

$$u_y = u_{BE5} - u_{BE6} + i_y R_y$$

3.扩展 u_y 的动态范围

$$u_y = u_{BE5} - u_{BE6} + i_y R_y$$

其中 $u_{BE5} - u_{BE6} = U_T \ln(\frac{I_5}{I_6})$

$$\therefore \quad u_y = U_T \ln(\frac{I_5}{I_6}) + i_y R_y$$

$$I_6 \approx I_{e6} = I_{oy} - i_y = (1 - \frac{i_y}{I_{oy}}) I_{oy}$$

$$I_5 \approx I_{e5} = I_{oy} + i_y = (1 + \frac{i_y}{I_{oy}}) I_{oy}$$

3.扩展 u_y 的动态范围

$$u_y = U_T \ln \frac{1 + i_y / I_{oy}}{1 - i_y / I_{oy}} + i_y R_y$$

$$= U_T \left[\ln(1 + i_y / I_{oy}) - \ln(1 - i_y / I_{oy}) \right] + i_y R_y$$

3.扩展 u_y 的动态范围

由 $$\ln(1+x)=x-\frac{1}{2}x^2+\frac{1}{3}x^3-\frac{1}{4}x^4+\cdots$$

$$\ln(1-x)=-x-\frac{1}{2}x^2-\frac{1}{3}x^3-\frac{1}{4}x^4-\cdots$$

当 $\dfrac{i_y}{I_{oy}}=x<0.5$ 时，忽略三次方以上项

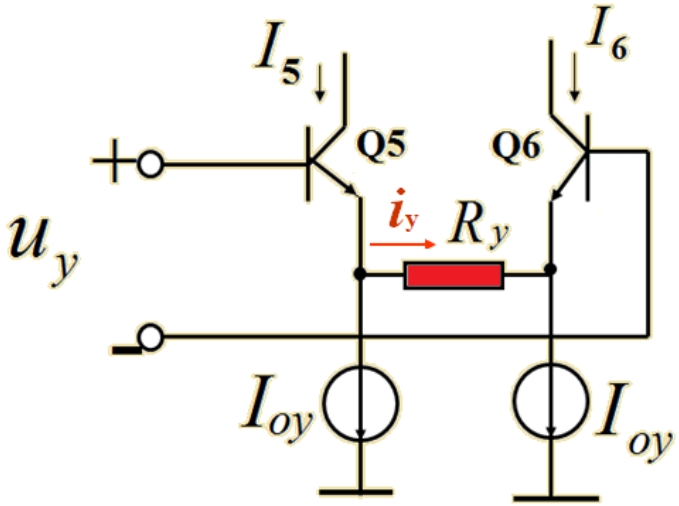
3.扩展 u_y 的动态范围

$$u_y \approx U_T \frac{2i_y}{I_{oy}} + i_y R_y$$

（其中 $\frac{2U_T}{I_{oy}} = 2r_{e5} << R_y$）

$$= \frac{2U_T}{I_{oy}} i_y + i_y R_y \approx i_y R_y$$

$$I_5 - I_6 = 2i_y = \frac{2u_y}{R_y}$$



3.扩展 u_y 的动态范围

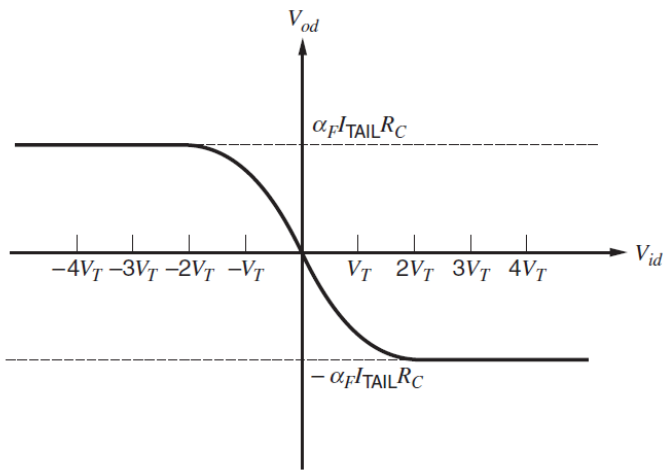
$$\therefore \quad i = I_A - I_B = (I_5 - I_6)\tanh(\frac{u_x}{2U_T})$$

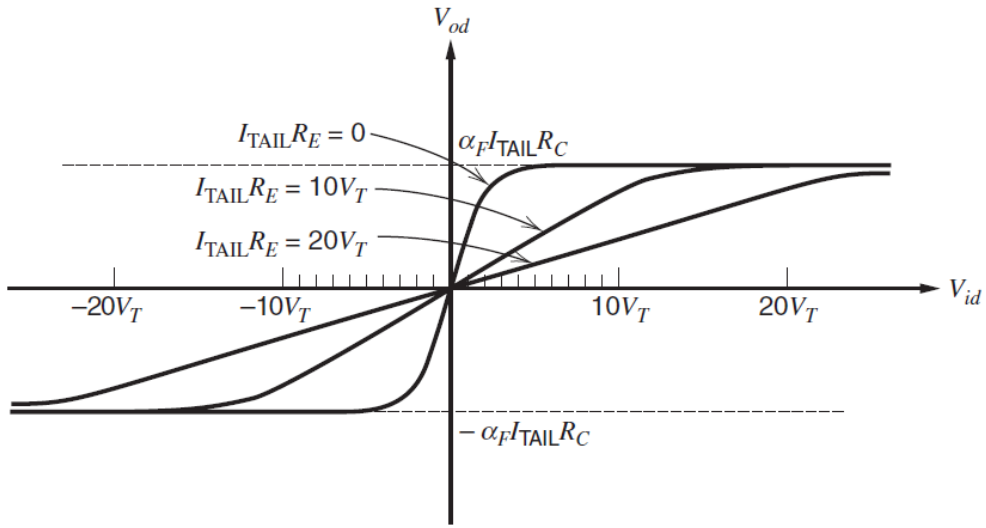
$$= \frac{2u_y}{R_y}\tanh(\frac{u_x}{2U_T})$$

$$u_o = R_c \times i = \frac{2R_c}{R_y}u_y \tanh(\frac{u_x}{2U_T})$$

(1) u_x 为小信号 $$u_o = \frac{R_e}{R_y U_T} u_x u_y = A u_x u_y$$

(2) u_x 为大信号 $$u_o = \frac{2R_c}{U_T} u_y S_2(\omega_1 t)$$





3.扩展 u_y 的动态范围

u_y 的动态范围（在 $I_{oy} = 0.5\,\mathrm{mA}$，$R_y = 1\,\mathrm{k}\Omega$ 时）

$$(-276\,\mathrm{mV}\ ,\ +276\,\mathrm{mV})$$

比不接 R_y 时扩大了10倍

集成电路MC1496/MC1596 (XFC1596 , F1596)

MC1596

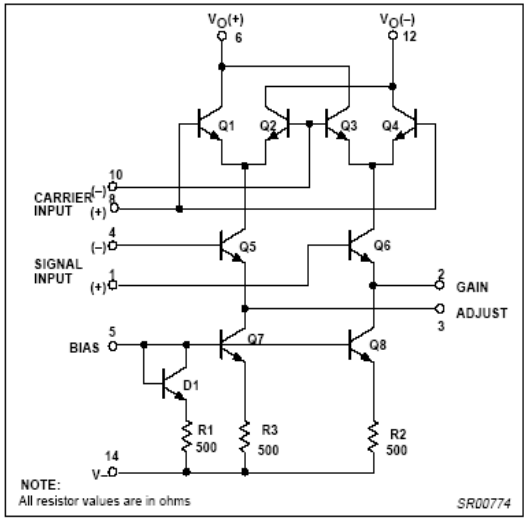


Figure 1. Balanced Modulator Schematic

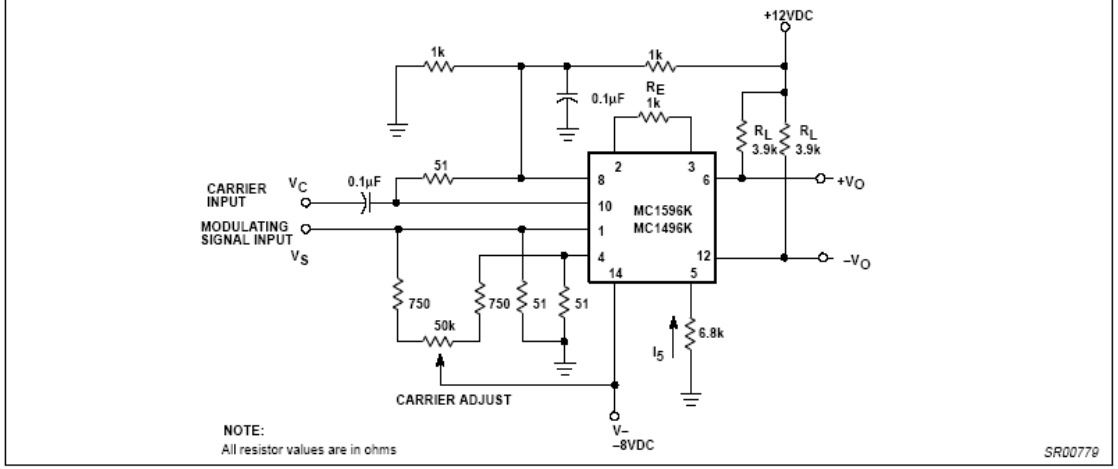


Figure 6. AM Modulator

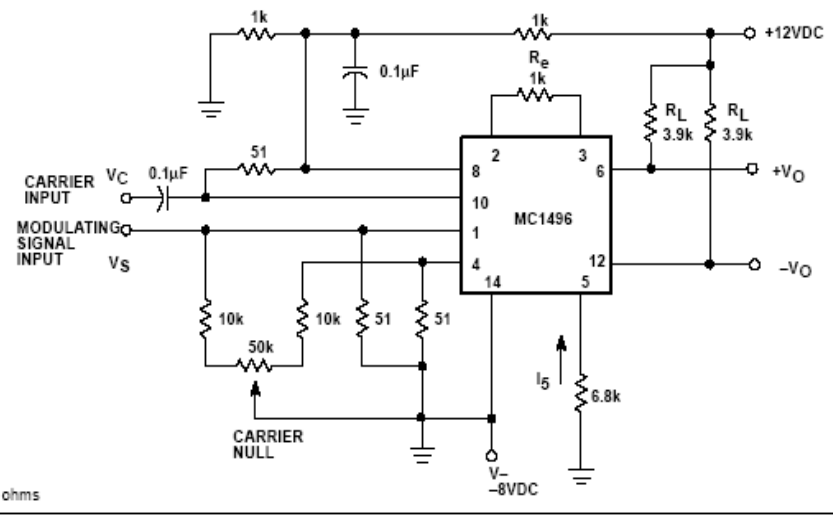
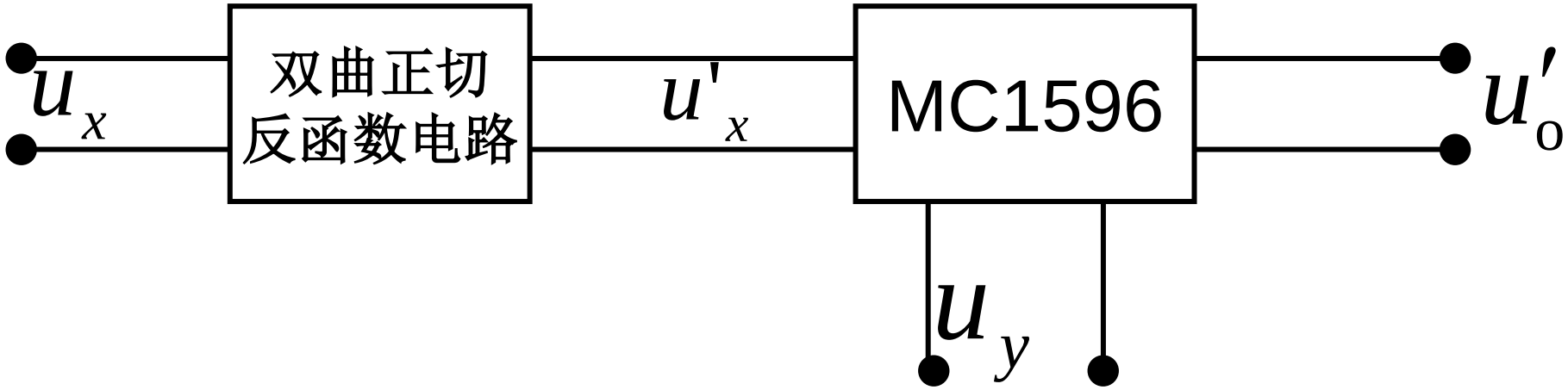


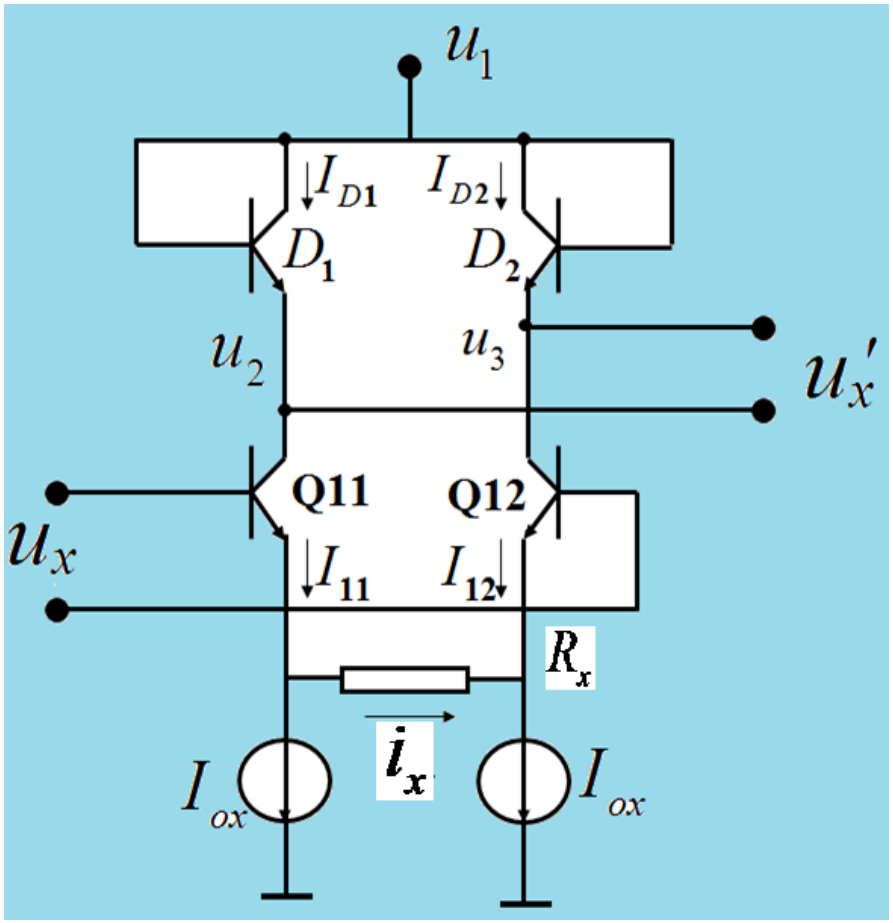
Figure 5. Double Suppressed Carrier Modulator

4. 线性化双平衡模拟乘法器

MC1596利用 R_y 扩大了 u_y 的线性动态范围, 但 u_x 的动态范围很小



双曲正切反函数电路



$I-U$ 变换

$U-I$ 变换

$$i_x \approx \frac{u_x}{R_x} \qquad I_{12} = I_{ox} - \frac{u_x}{R_x} \qquad I_{11} = I_{ox} + \frac{u_x}{R_x}$$

二级管 D_1, D_2 **上的电流关系为:** $u_D = U_T \ln \frac{I_D}{I_S}$

$$u'_x = u_3 - u_2 = u_{D1} - u_{D2} = U_T \ln \frac{I_{D1}}{I_{D2}}$$

$$= U_T \ln \frac{I_{11}}{I_{12}} = U_T \ln \frac{1 + \frac{u_x}{I_{ox} R_x}}{1 - \frac{u_x}{I_{ox} R_x}}$$

由 $\frac{1}{2}\ln\frac{1+x}{1-x}=\tanh^{-1}x$

$$\therefore \quad u'_x = 2U_T \tanh^{-1}\frac{u_x}{I_{ox}R_x}$$

$$u'_o = \frac{2R_c}{R_y}u_y \tanh\frac{u'_x}{2U_T} = \frac{2R_c}{I_{ox}R_xR_y}u_xu_y$$

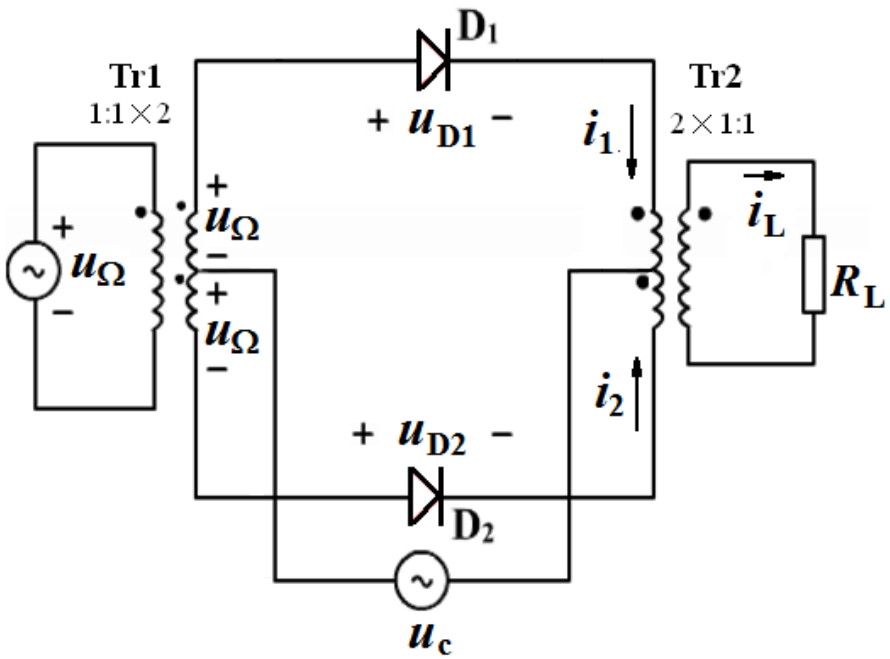
得到DSB信号

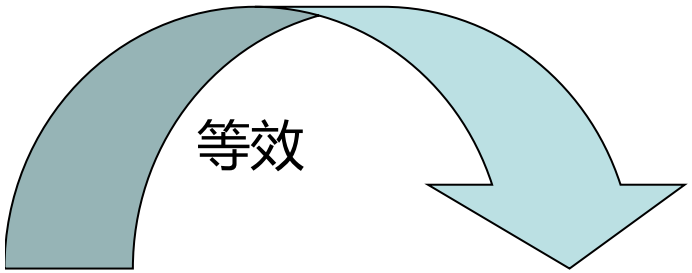
u_x, u_y 的线性动态范围可达±10V左右

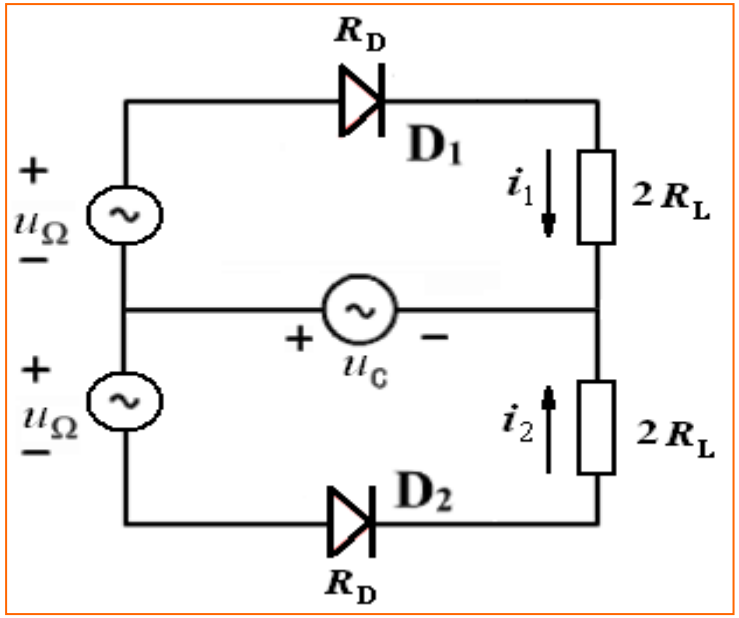
MC1595 (双差分对平衡调制器)

5.3 平衡调制器

1. 平衡调制器(用于产生AM波、DSB波、SSB波)







$U_{cm} > 0.5V$

$U_{cm} >> U_{\Omega m}$

$$u_{D1} = u_c + u_\Omega \qquad u_{D2} = u_c - u_\Omega$$

$$i_1 = g_D u_{D1} S(t) \qquad i_2 = g_D u_{D2} S(t)$$

$$i_L = i_1 - i_2 = 2 g_D u_\Omega S(t)$$

$$= 2 g_D U_{\Omega m} \cos \Omega t \left(\frac{1}{2} + \frac{2}{\pi} \cos \omega_c t - \frac{2}{3\pi} \cos 3\omega_c t + \cdots\right)$$

5-5.exe

$$= g_D U_{\Omega m} \cos \Omega t + \underline{\frac{2 g_D}{\pi} U_{\Omega m} \left[\cos(\omega_c + \Omega) t + \cos(\omega_c - \Omega) t\right]}$$

$$- \frac{2 g_D}{3\pi} U_{\Omega m} \left[\cos(3\omega_c + \Omega) t + \cos(3\omega_c - \Omega) t\right] + \ldots$$

经滤波，得到

$$i_L' = \frac{2g_D}{\pi} U_{\Omega m}\left[\cos(\omega_c + \Omega)t + \cos(\omega_c - \Omega)t\right]$$

5-5.exe

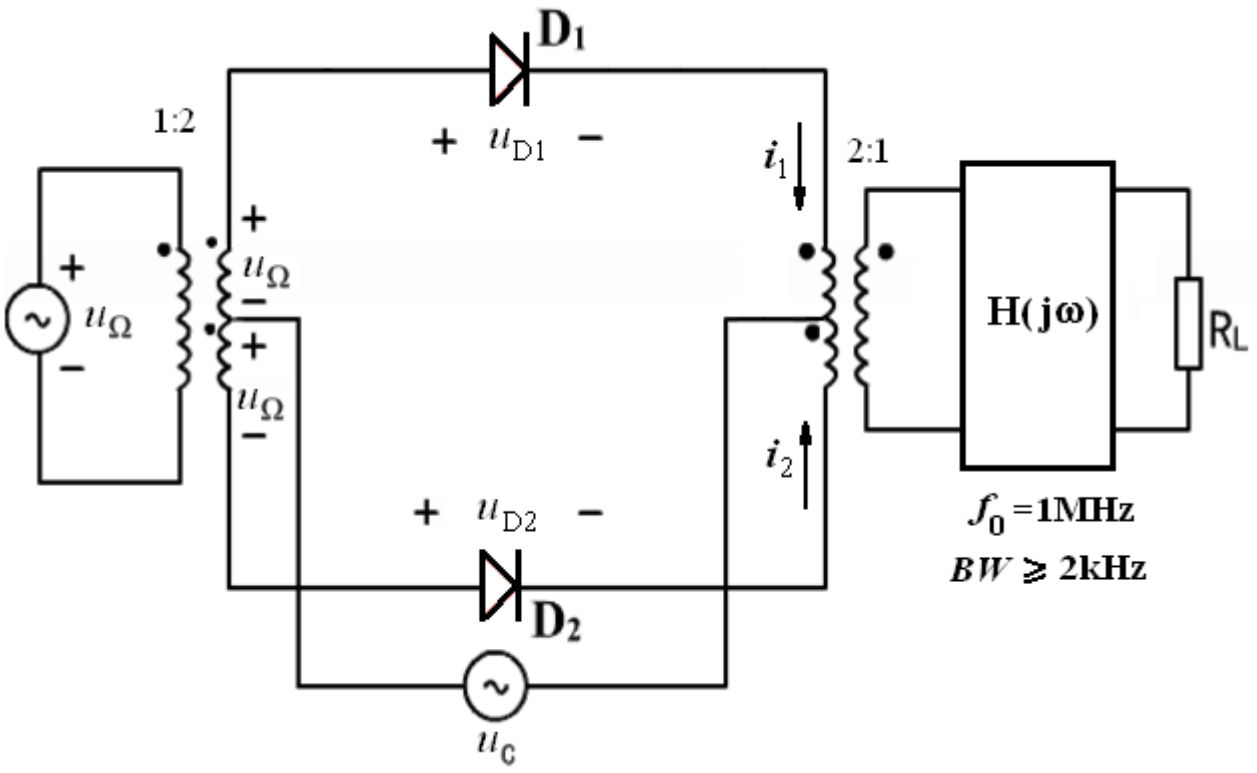
例：　二极管平衡调制器如图所示，已知

$$u_{\Omega} = U_{\Omega m} \cos 2\pi \times 10^{3} t \ \ (\mathrm{V})$$

$$u_{c} = U_{cm} \cos 2\pi \times 10^{6} t \ \ (\mathrm{V})$$

$$U_{cm} >> U_{\Omega m} \quad U_{cm} > 0.5 \ \ (\mathrm{V})$$

- 绘出 $u_o(t) =$ ？
- 若 u_c 与 u_{Ω} 对调位置 $u_o(t) =$ ？
- 若将此电路产生SSB信号，电路应如何变化?


D1
1:2
+ uD1 −
i1
2:1
+
uΩ
−
+
uΩ
−
H(jω)
RL
i2
+ uD2 −
f0 =1MHz
BW ≥ 2kHz
D2
uC

解

$$i = i_1 - i_2 = 2g_D U_{\Omega m}\cos\Omega t\left(\frac{1}{2} + \frac{2}{\pi}\cos\omega_c t - \frac{2}{3\pi}\cos 3\omega_c t + \cdots\right)$$

$$u_0(t) = \frac{4}{\pi} g_D U_{\Omega m} R_L \cos\Omega t \cos\omega_c t$$

经滤波后，输出电压 $u_o(t)$ 为

$$u_o(t) = i' R_L$$

$$= \frac{4}{\pi} g_D U_{\Omega m} R_L \cos(2\pi \times 10^3 t)\cos(2\pi \times 10^6 t) \ \ (\mathrm{V})$$

它是DSB信号

(2) $u_{D1} = u_\Omega + u_c \qquad u_{D2} = -(-u_\Omega + u_c) = -u_c + u_\Omega$

$$i_{D1} = g_D u_{D1} S(t) \quad i_{D2} = g_D u_{D2} S(t + \frac{T_c}{2})$$

$$i' = i_{D1} - i_{D2}$$

$$= g_D u_c \left[S(t) + S(t + \frac{T_c}{2}) \right] + g_D u_\Omega \left[S(t) - S(t + \frac{T_c}{2}) \right]$$

$$= g_D U_{cm} \cos \omega_c t + g_D U_{\Omega m} \cos \Omega t \left[\frac{4}{\pi} \cos \omega_c t - \frac{4}{3\pi} \cos 3\omega_c t + \cdots \right]$$

经滤波：

$$u_o(t) = i'R_L$$

$$= g_D R_L U_{cm} \cos\omega_c t + g_D R_L \frac{4}{\pi} U_{\Omega m} \cos\Omega t \cos\omega_c t$$

$$= U_{om}(1 + m_a \cos\Omega t)\cos\omega_c t$$

其中 $U_{om} = g_D R_L U_{cm}$ $\quad m_a = \dfrac{4U_{\Omega m}}{\pi U_{cm}}$

AM 信号

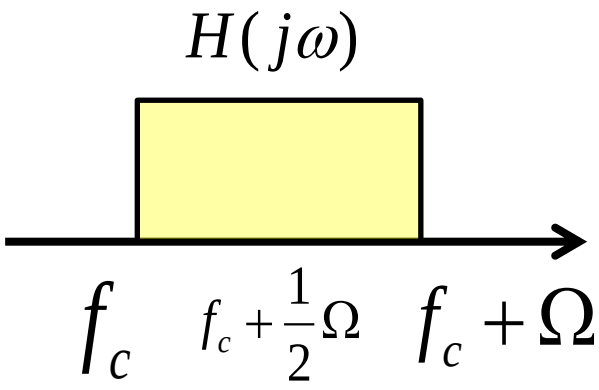
(3) 通过滤波器产生SSB信号。

上边频：$f_c + \Omega$

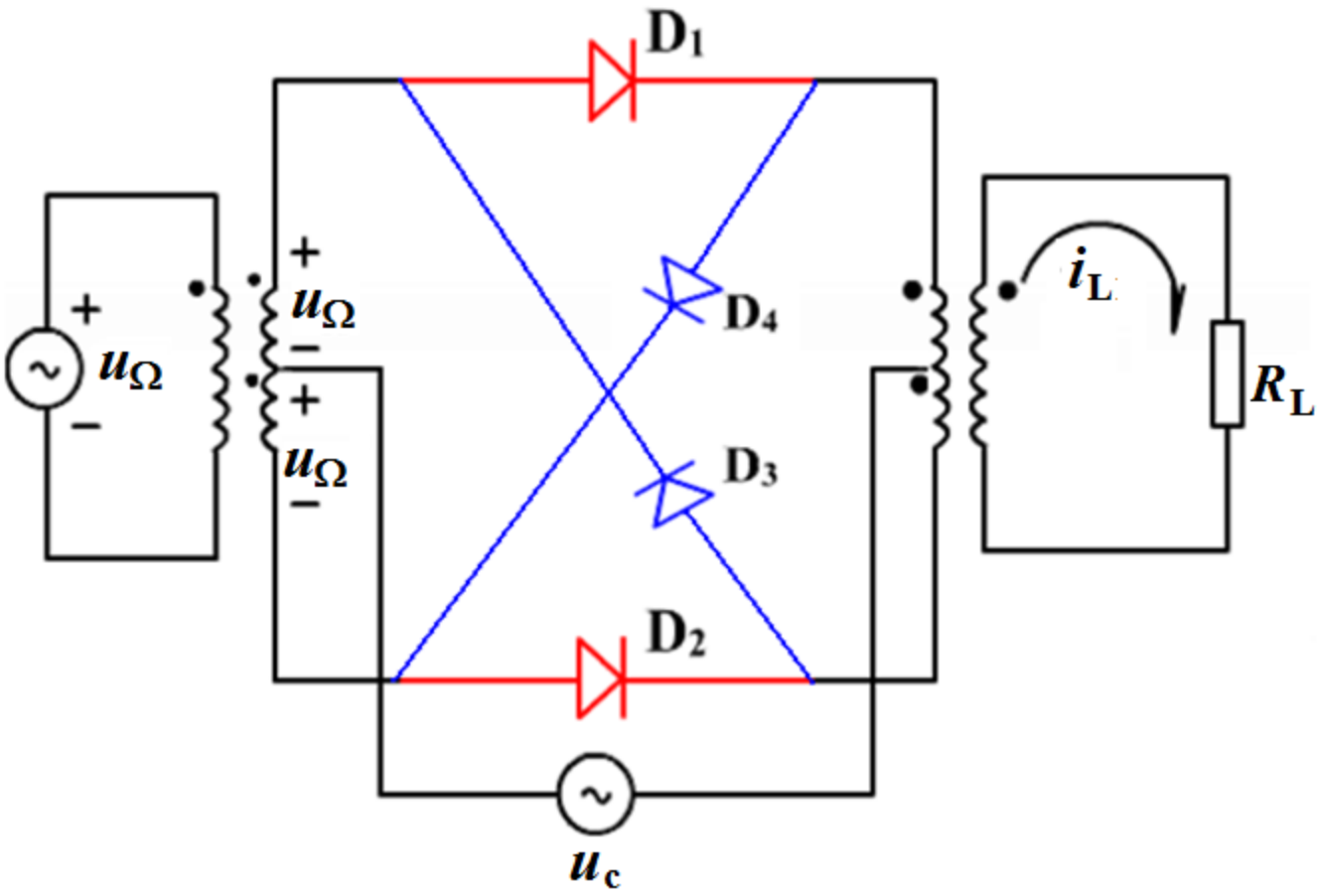
下边频：f_c

中心频率：$f_c + \frac{1}{2}\Omega$

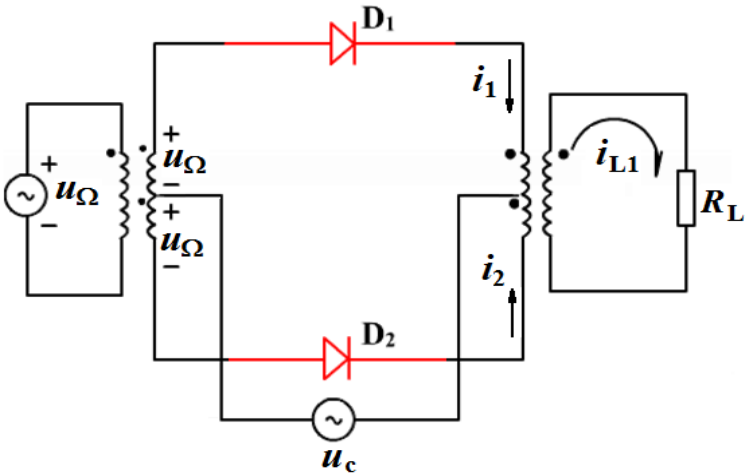
带宽：Ω

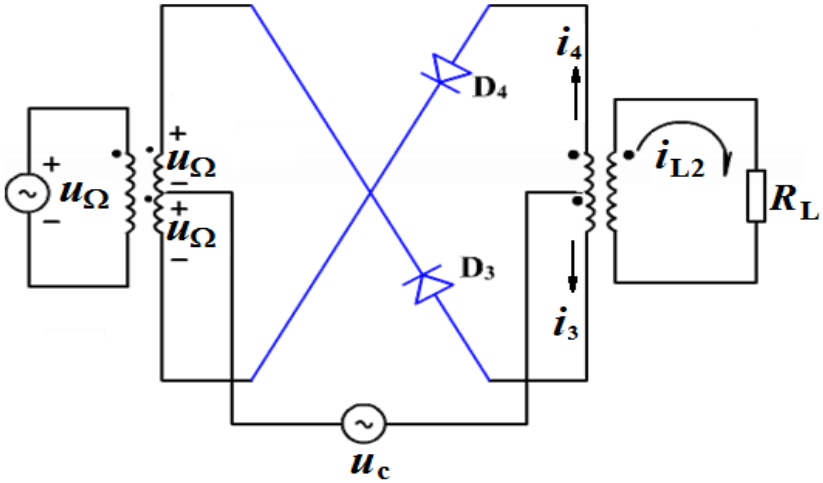


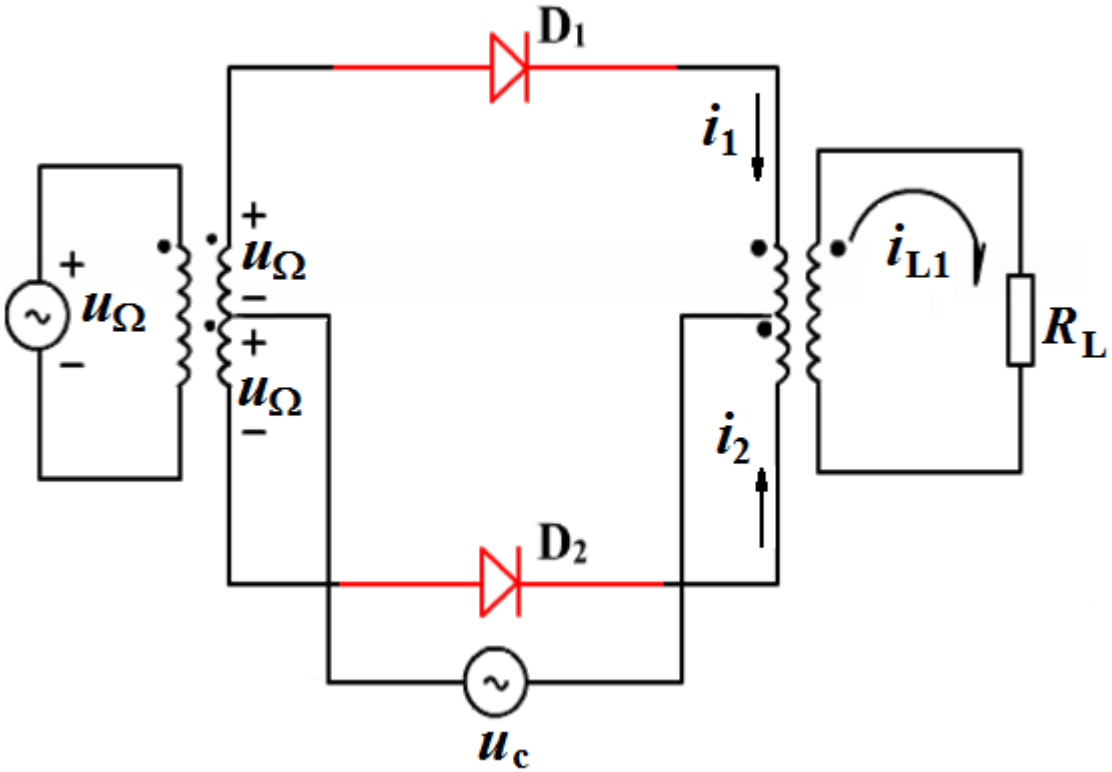
2. 环形调制器（双平衡调制器）



等效为

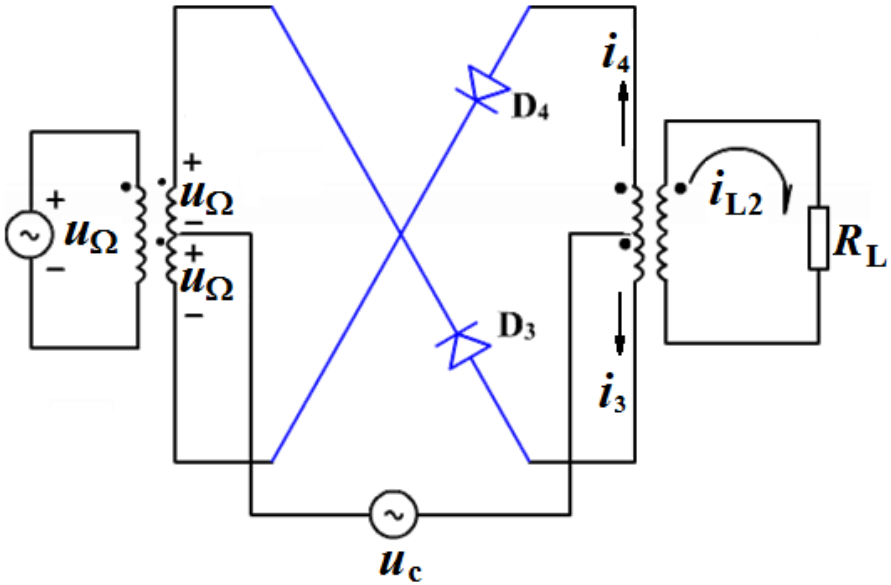






$$i_{L1} = i_1 - i_2 = 2g_D u_\Omega S(t)$$

$$u_{D3} = -u_{\Omega} - u_c \qquad u_{D4} = u_{\Omega} - u_c$$



$$i_{L2} = i_3 - i_4 = -2g_D u_{\Omega} S(t + \frac{T_c}{2})$$

$$i_L = i_{L1} + i_{L2} = 2g_D u_\Omega \left[S(t) - S(t + \frac{T_c}{2}) \right]$$

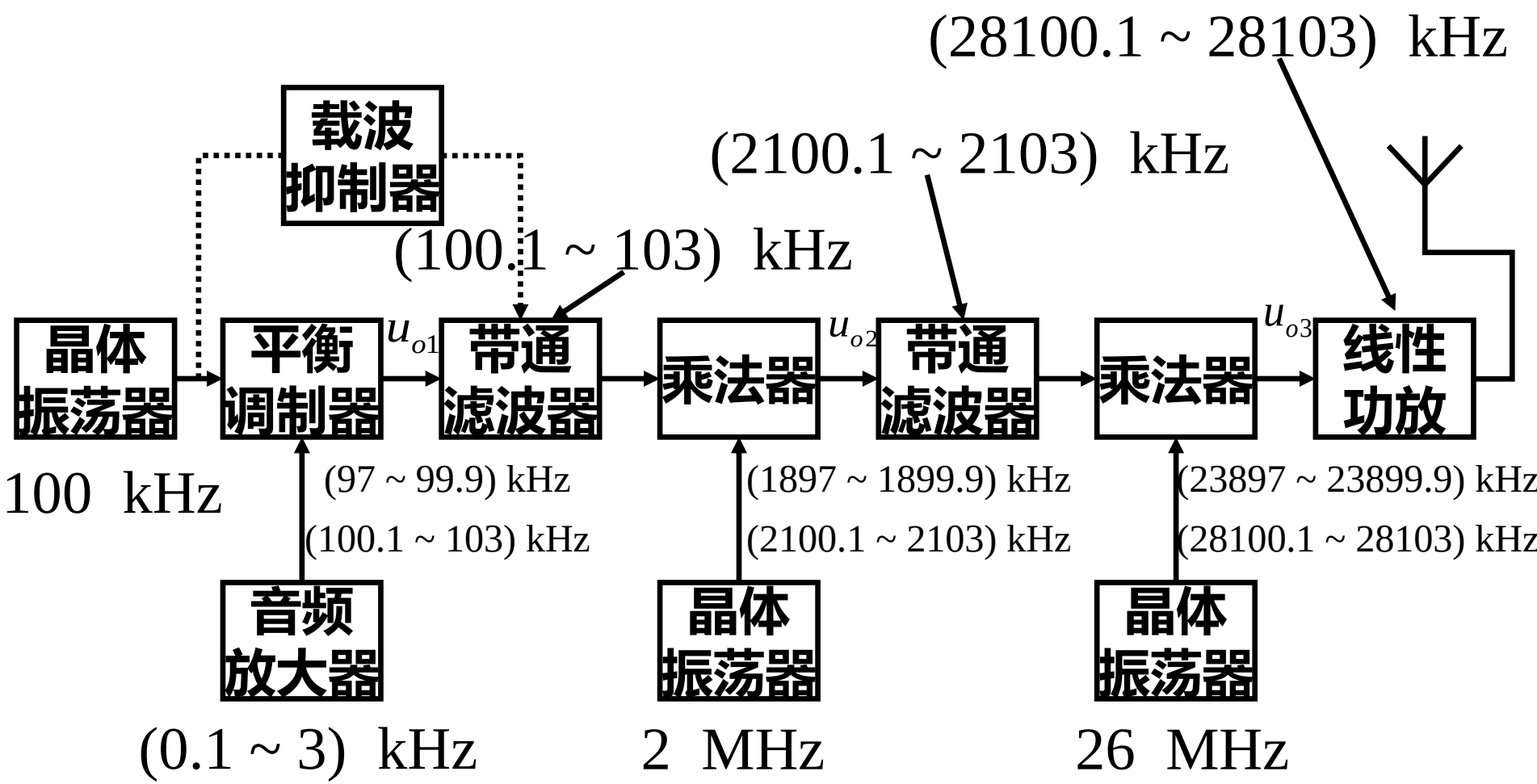
$$= 2g_D u_\Omega \left[\frac{4}{\pi} \cos \omega_c t - \frac{4}{3\pi} \cos 3\omega_c t + \cdots \right]$$

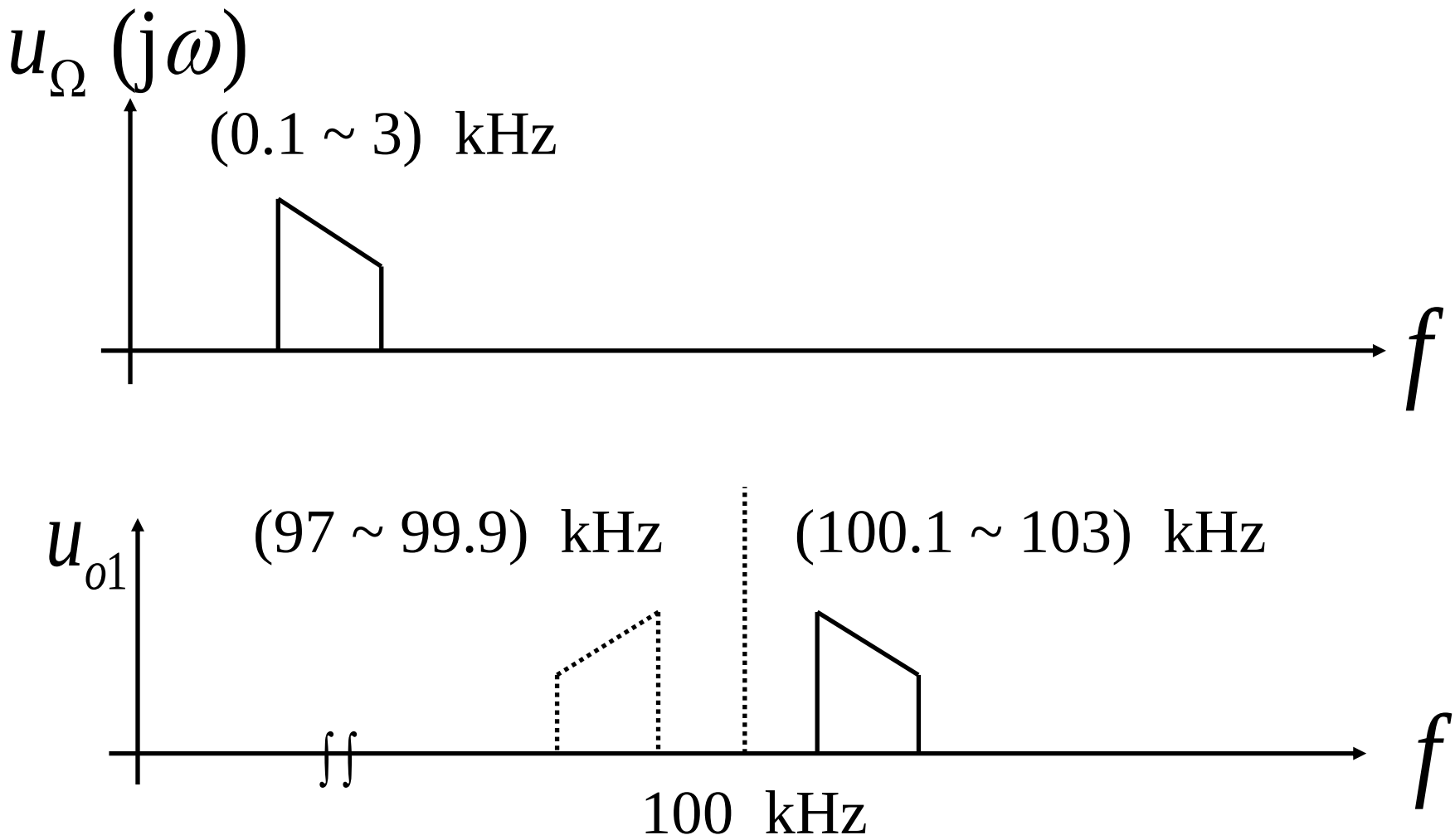
$$= \underline{g_D u_{\Omega m} \frac{4}{\pi} [\cos(\omega_c + \Omega)t + \cos(\omega_c - \Omega)t]}$$

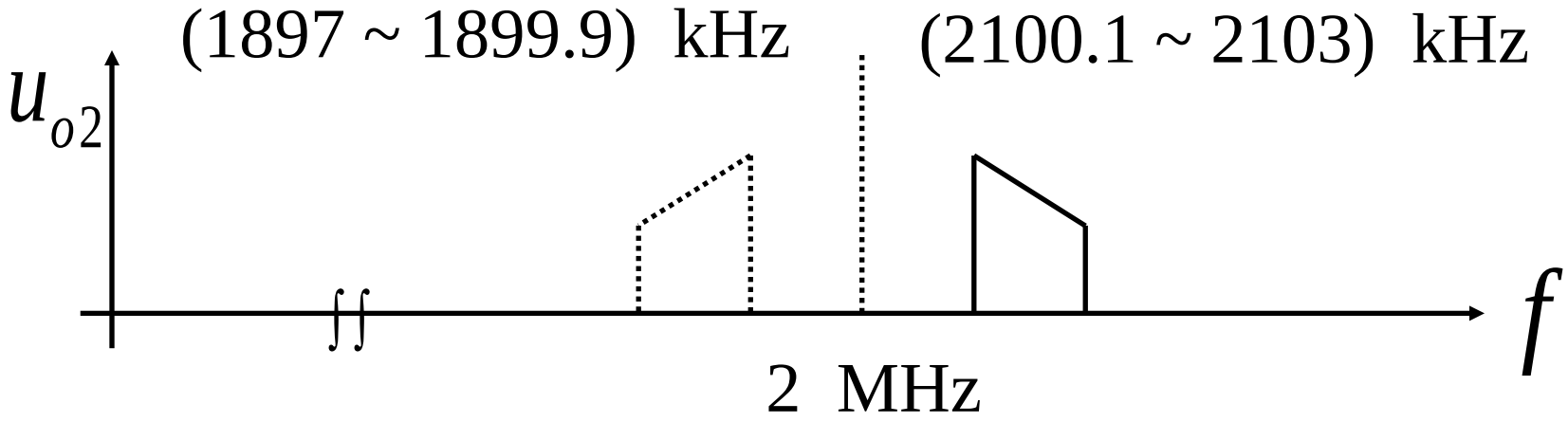
$$- g_D u_{\Omega m} \frac{4}{3\pi} [\cos(3\omega_c + \Omega)t + \cos(3\omega_c - \Omega)t] + \cdots$$

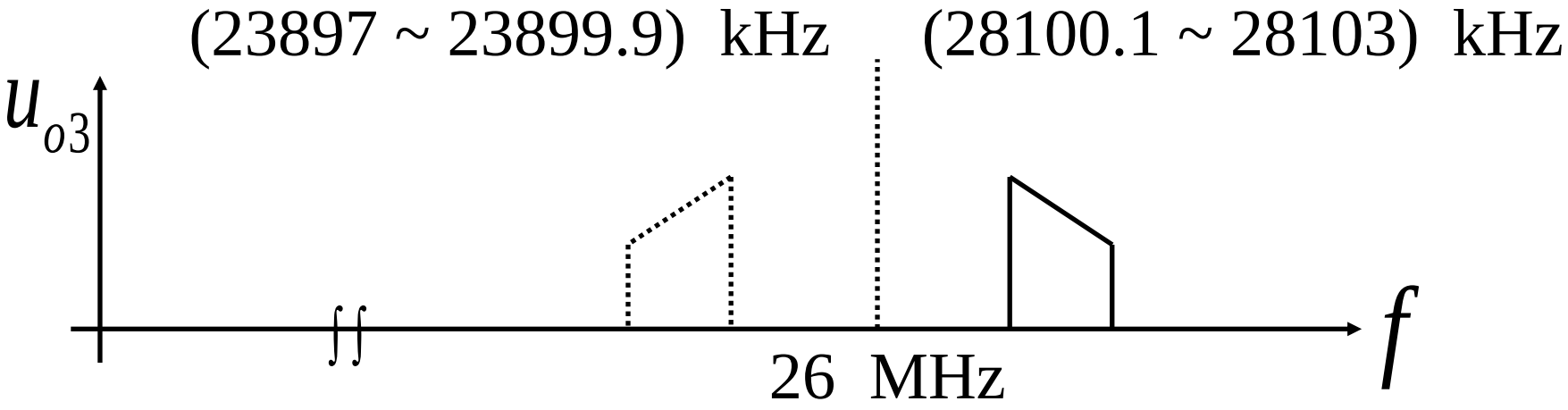
输出DSB

3.单边带发射机　(采用滤波法)

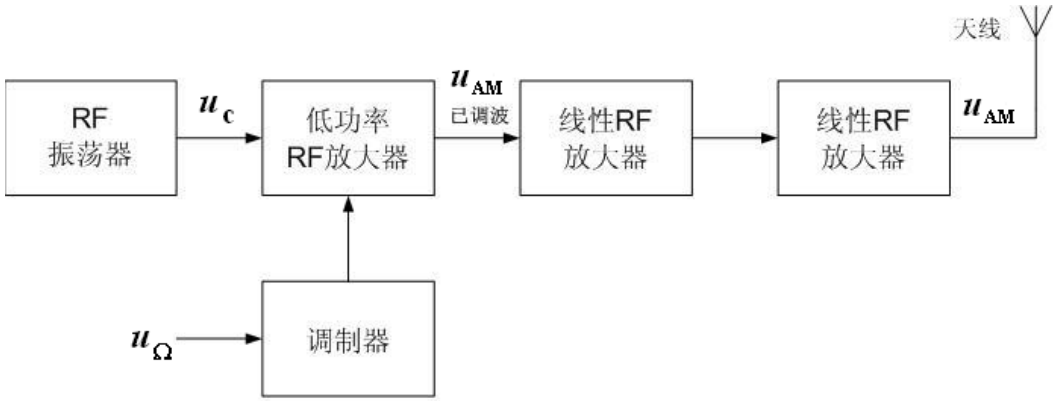



$u_\Omega(j\omega)$
(0.1 ~ 3) kHz
f
u_{o1}
(97 ~ 99.9) kHz
(100.1 ~ 103) kHz
f
100 kHz

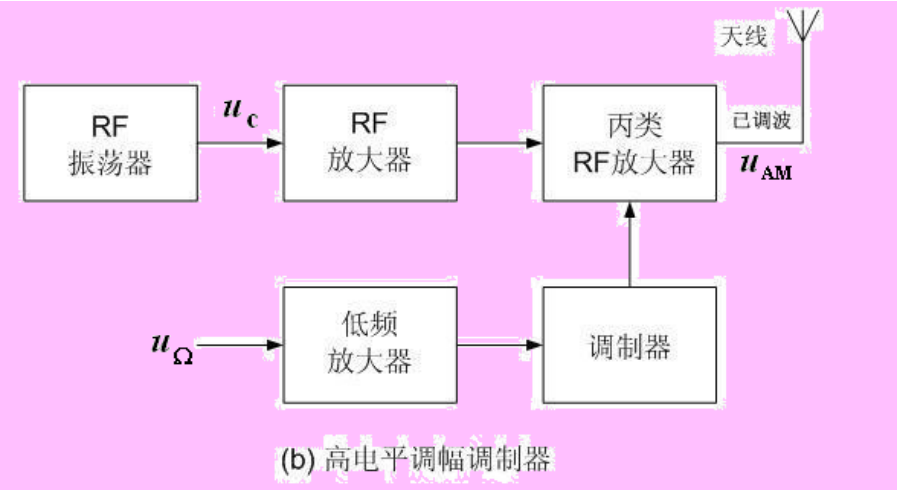

(1897 ~ 1899.9) kHz
(2100.1 ~ 2103) kHz
u_{o2}
f
2 MHz


(23897 ~ 23899.9) kHz
(28100.1 ~ 28103) kHz
u_{o3}
f
26 MHz

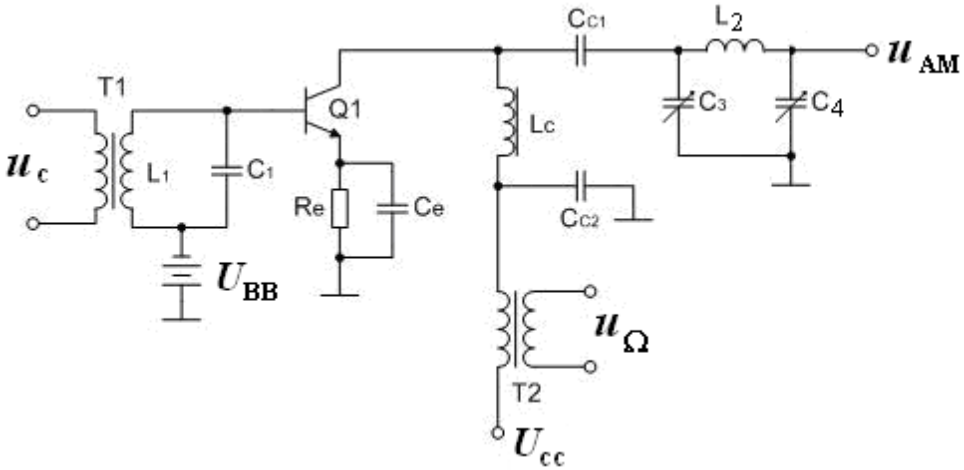
5.4　　高电平调幅电路

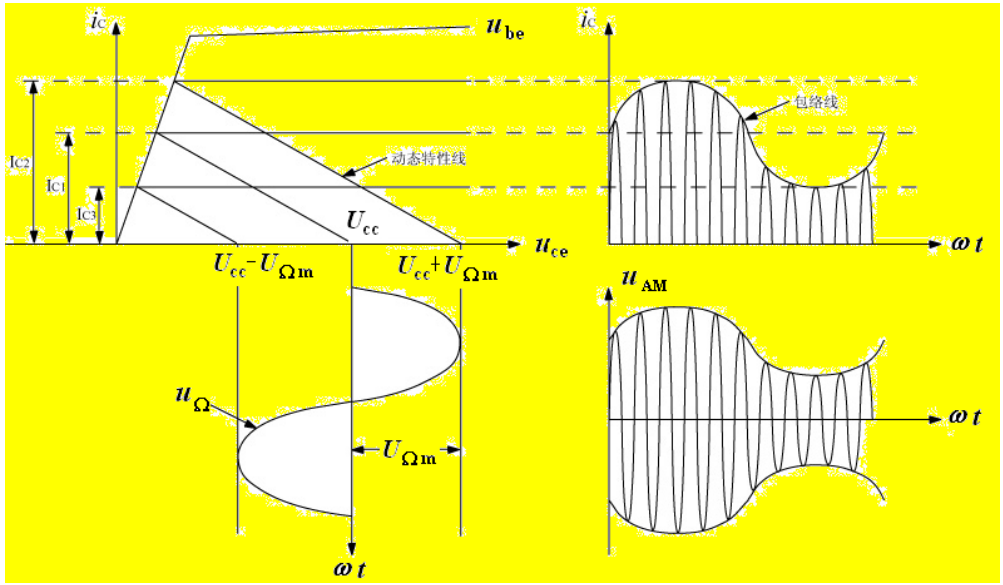


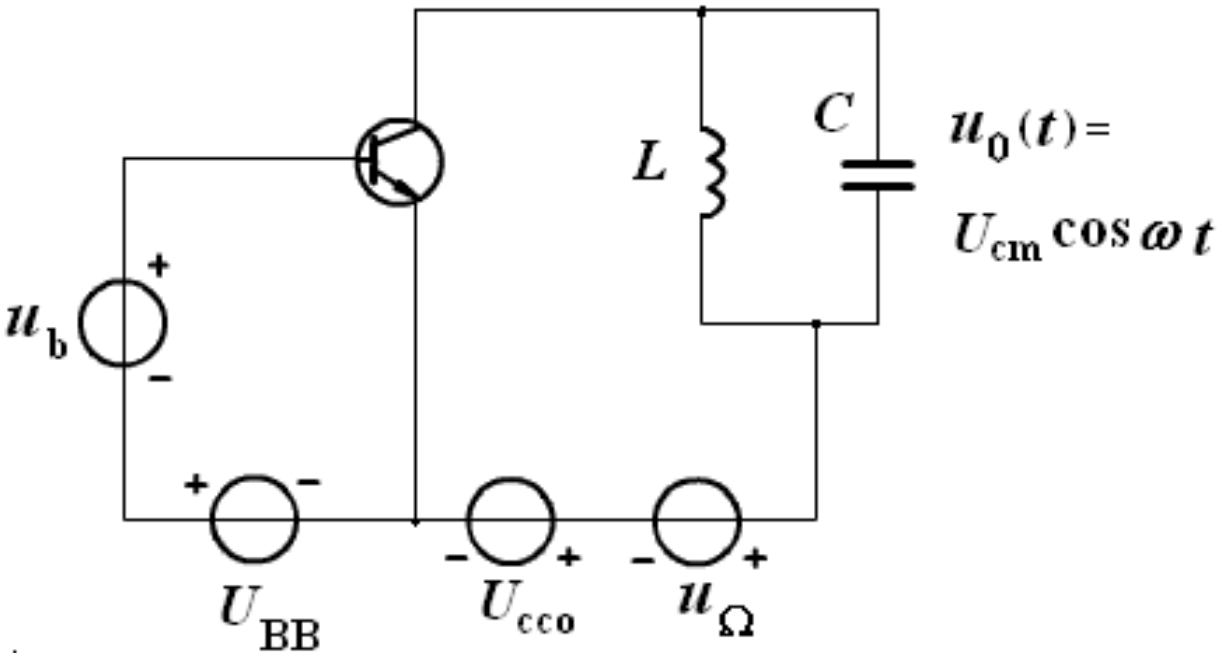
(a) 低电平调幅调制器



(b) 高电平调幅调制器

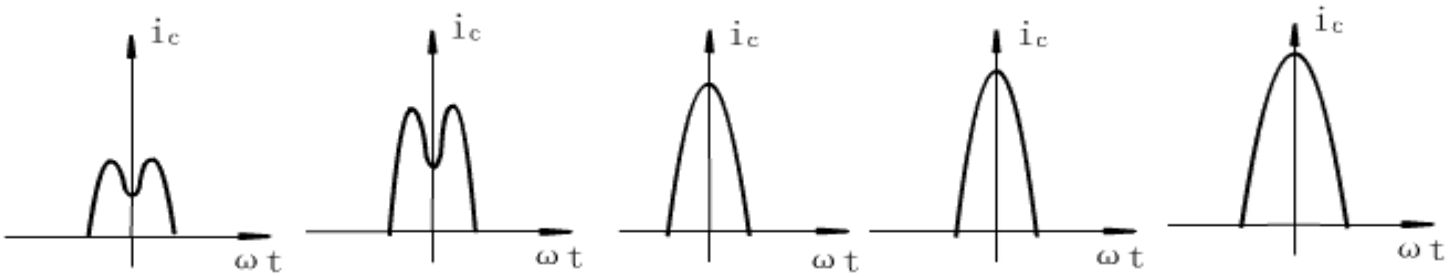

T1
u_c
L1
C1
Q1
Re
Ce
U_{BB}
Cc1
L2
u_{AM}
C3
C4
Lc
Cc2
u_Ω
T2
U_{cc}


i_c
u_{be}
Ic2
Ic1
Ic3
动态特性线
U_{cc}
u_{ce}
$U_{cc}-U_{\Omega m}$
$U_{cc}+U_{\Omega m}$
u_Ω
$U_{\Omega m}$
ωt
i_c
包络线
ωt
u_{AM}
ωt

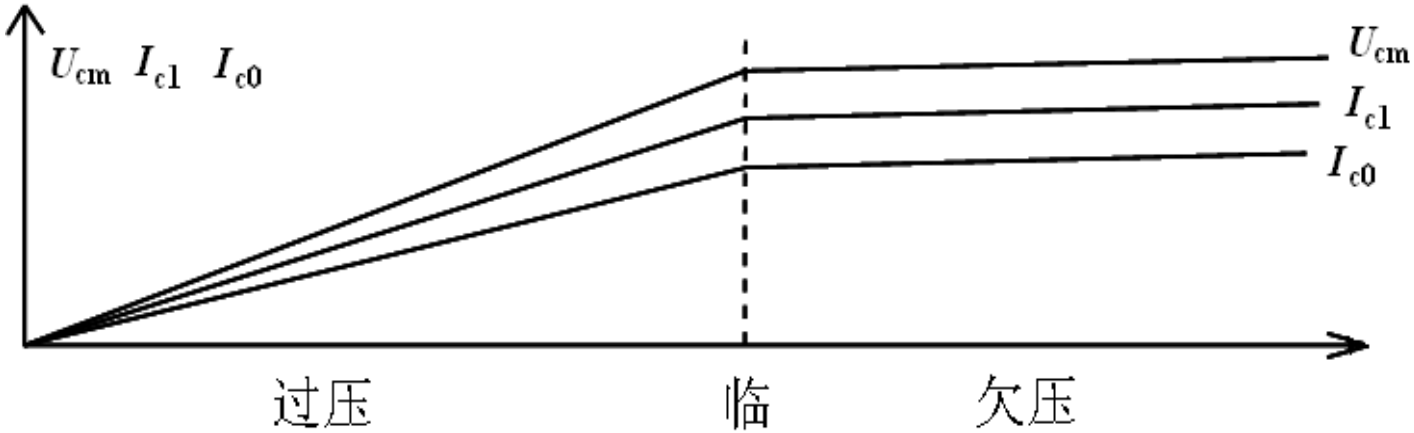

L
C
$u_0(t) =$
$U_{cm} \cos \omega t$
u_b
+
−
+
−
U_{BB}
−
+
U_{cco}
−
+
u_Ω

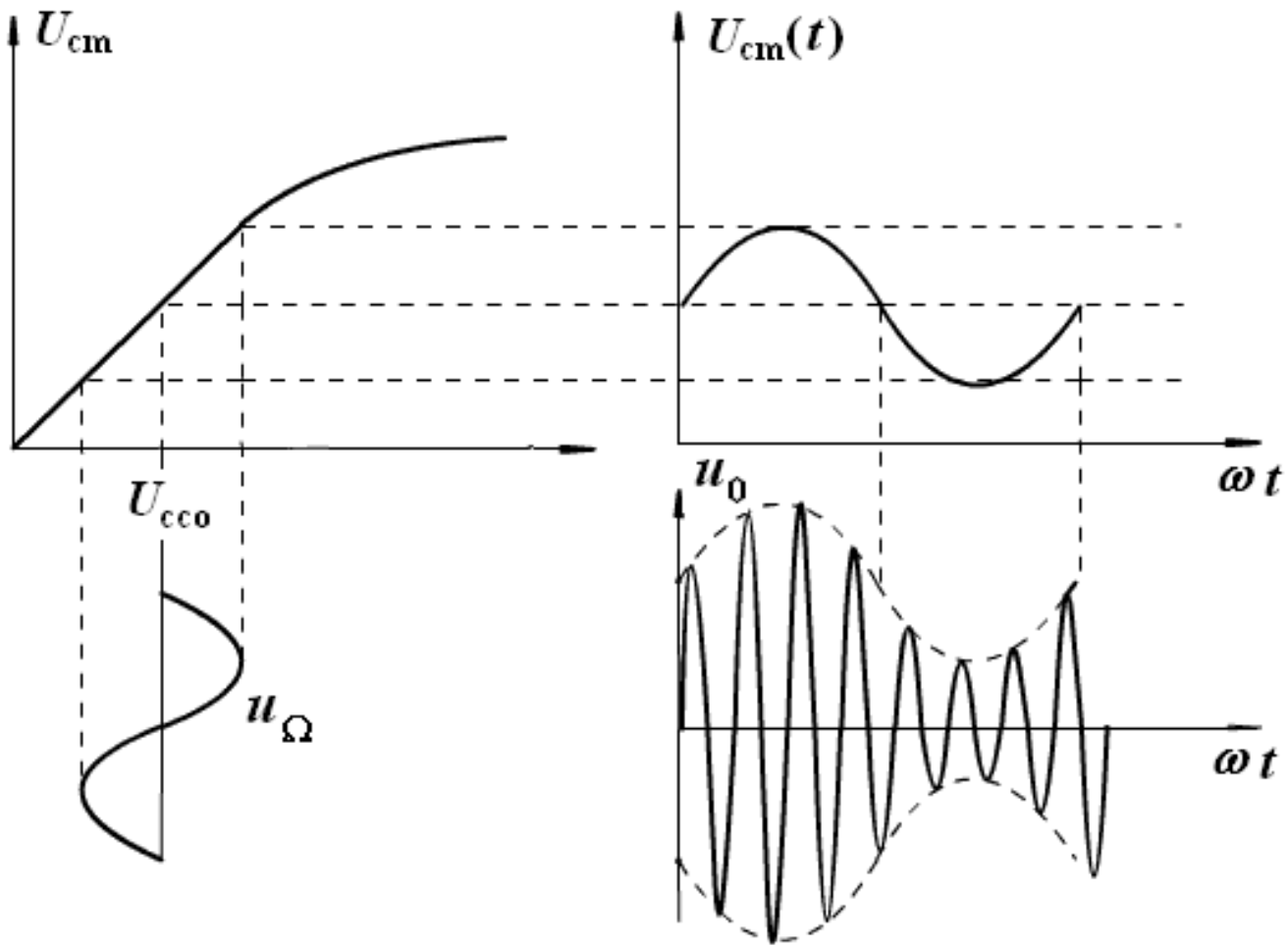
丙类功率放大器的原理应用 --------- 调制特性

集电极调制： U_{BB} ，U_{bm} ，R_e **一定。** U_{CC} **变化对性能的影响**

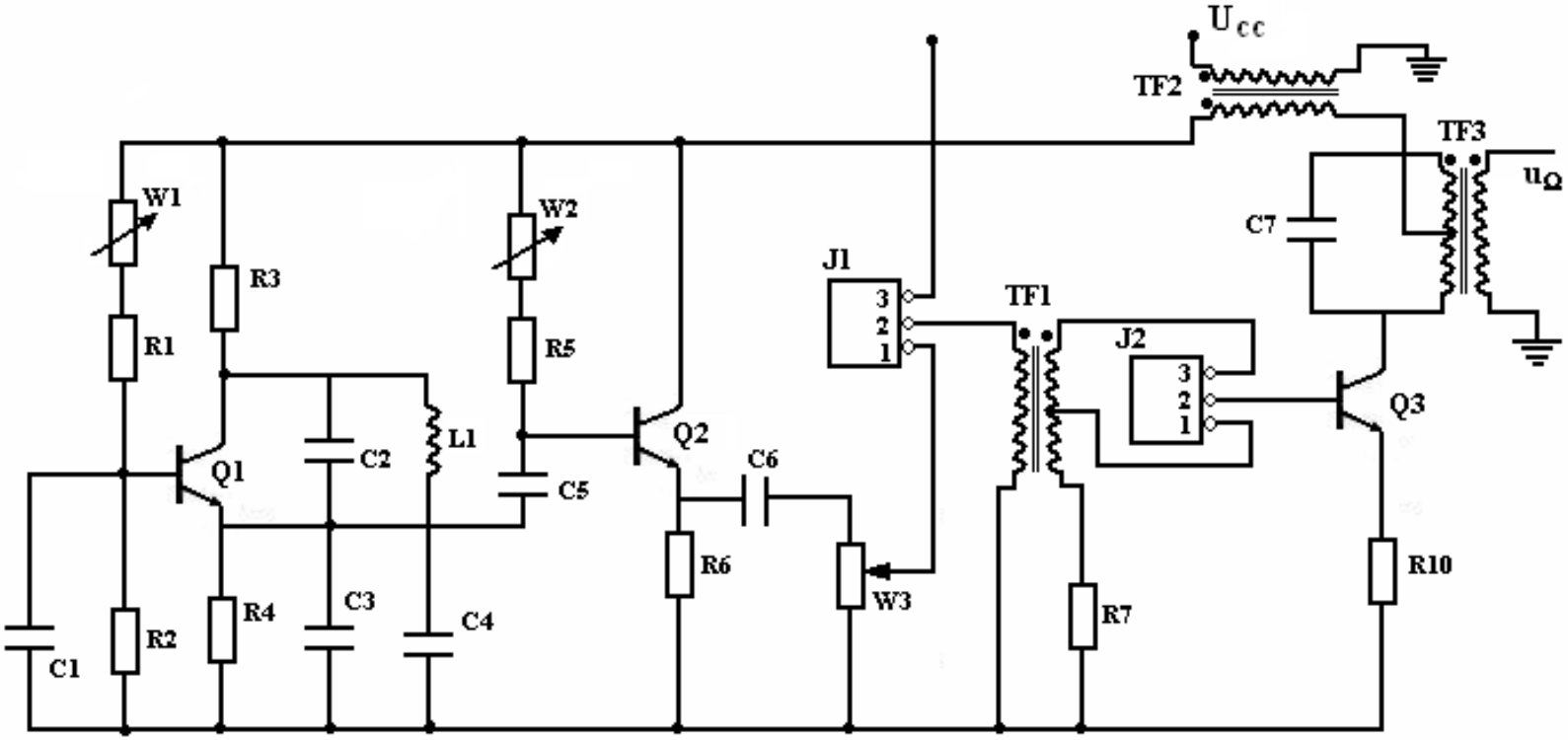


Ucc、增大




U_{cm}
$U_{cm}(t)$
U_{cco}
u_{Ω}
u_0
ωt
ωt

实验教学电路


Ucc
TF2
TF3
uΩ
C7
W1
R3
R1
W2
R5
J1
3
2
1
TF1
J2
3
2
1
Q3
Q1
C2
L1
C5
Q2
C6
R6
W3
R7
R10
C1
R2
R4
C3
C4

思考题

1.改变U_{cc}：记录数据，分析工作原理

2.改变U_{bm}：记录数据，分析工作原理

3.改变$U_{\Omega m}$：记录数据，分析工作原理

5.5 包络检波

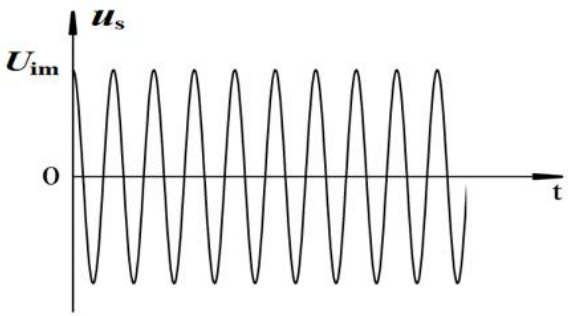
检波： 从已调波中提取，恢复调制信号，这一过程称为解调或检波。完成检波作用的电路称为检波器。

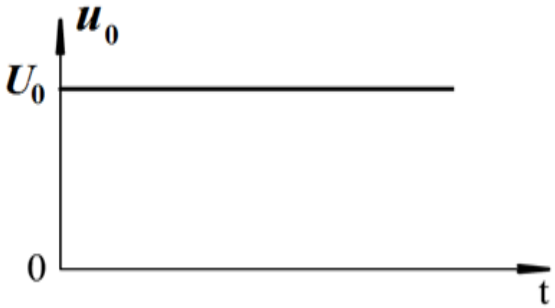
检波方式：

1. **峰值包络检波**
2. **平方律检波**
3. **平均值包络检波**

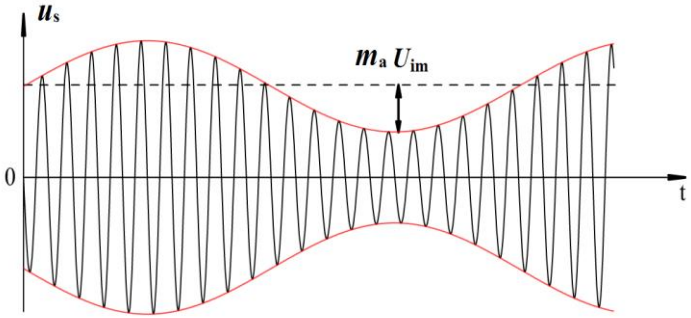
} **解调AM波**

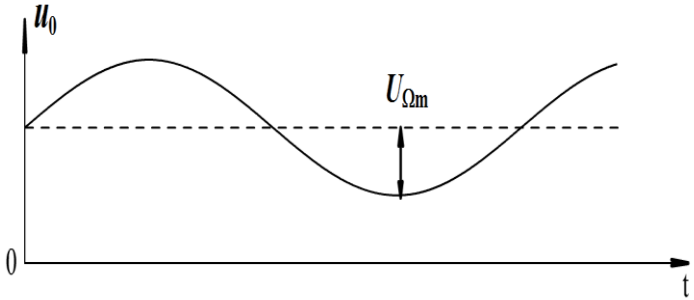
4. **同步检波** **解调AM, DSB, SSB波**





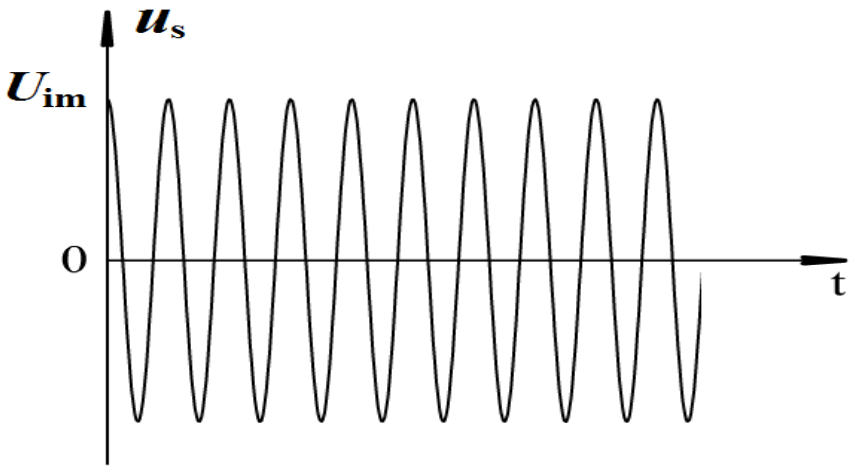
（已调波）



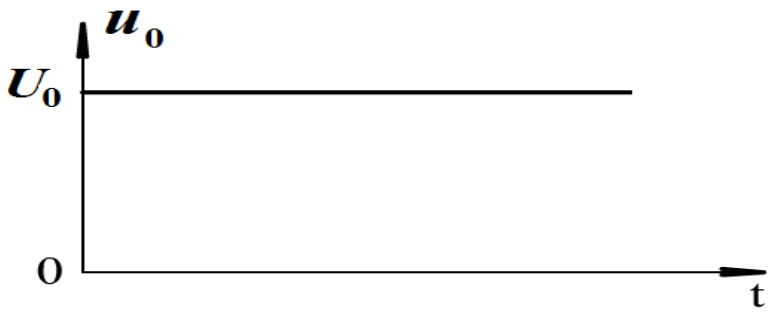


检波器的性能：

1. 检波效率



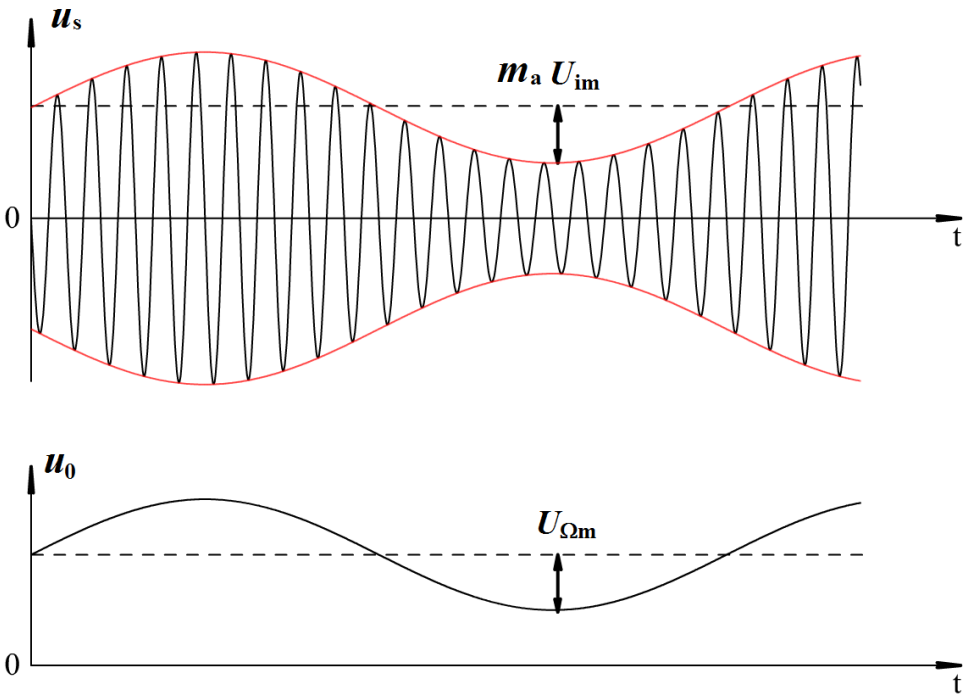
$$K_d = \frac{U_o}{U_{im}}$$



K_d **越大越好** $K_d \to 1$

检波器的性能：

1. 检波效率

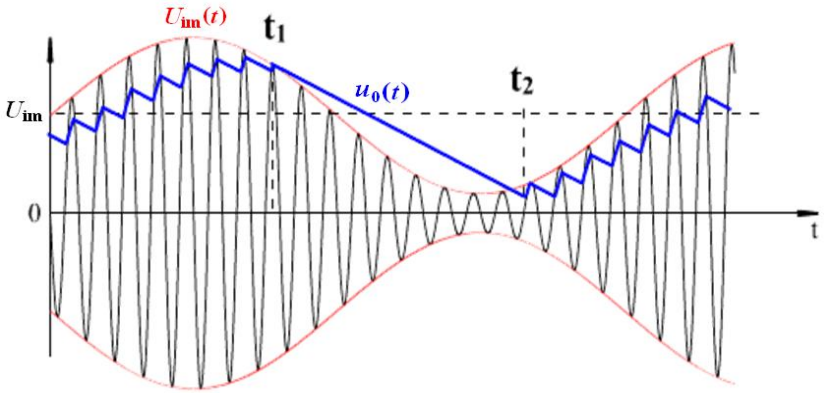


$$K_d = \frac{U_{\Omega m}}{m_a U_{im}}$$

K_d **越大越好** $K_d \rightarrow 1$

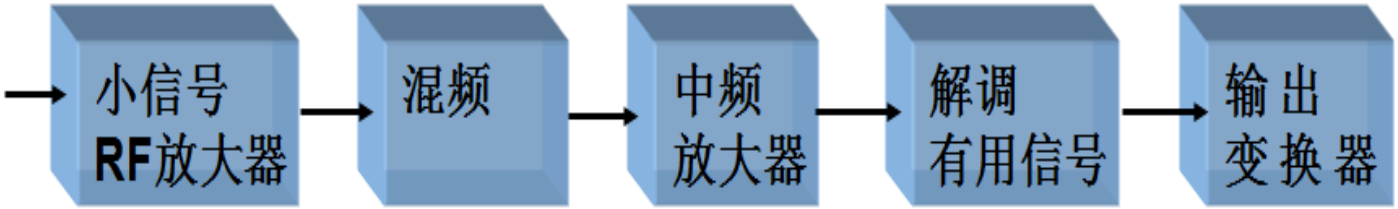
检波器的性能：

1. 检波效率



2. 失真小

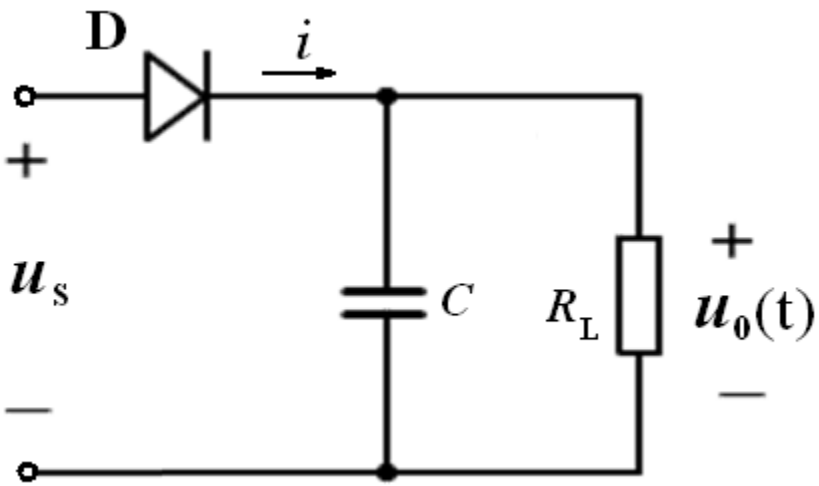
3. 输入阻抗要高



一. 大信号二极管包络检波

1. 工作原理： u_s 为AM波

$$u_s = U_{im}(1 + m_a \cos \Omega t) \cos \omega_c t$$ 且 $U_{im} > 0.5$ (V)

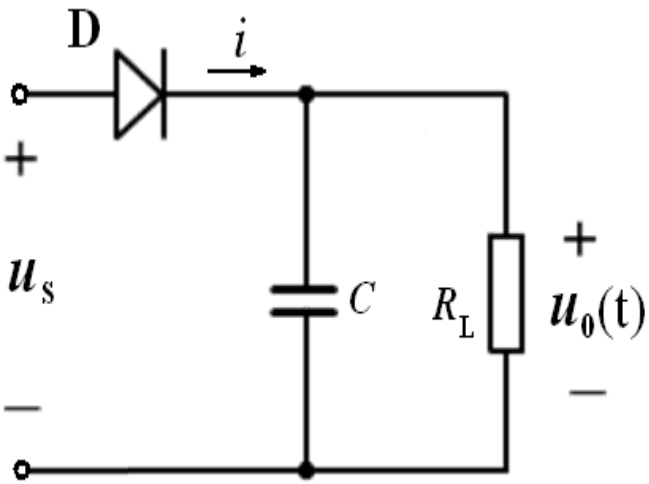


➢二极管导通：

充电时间常数 $R_D C \quad (R_D << R_L)$

充电时间常数越小，充电越快。

充电很快，很短时间就冲到峰值

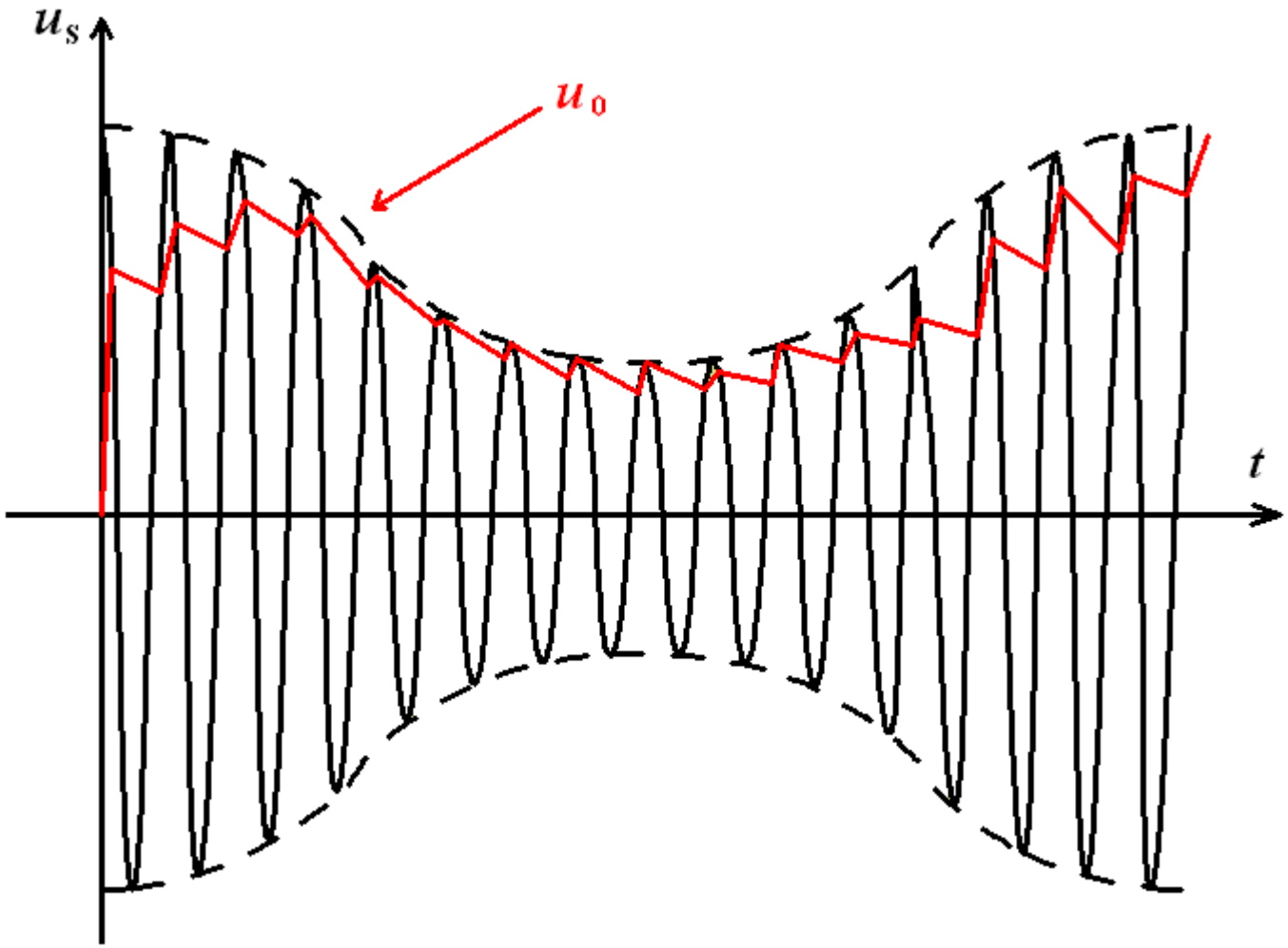


➢二极管截止：

放电时间常数 $R_L C$

放电时间常数越大，放电越慢。

放电较慢，时间较长


u_S
u_0
t

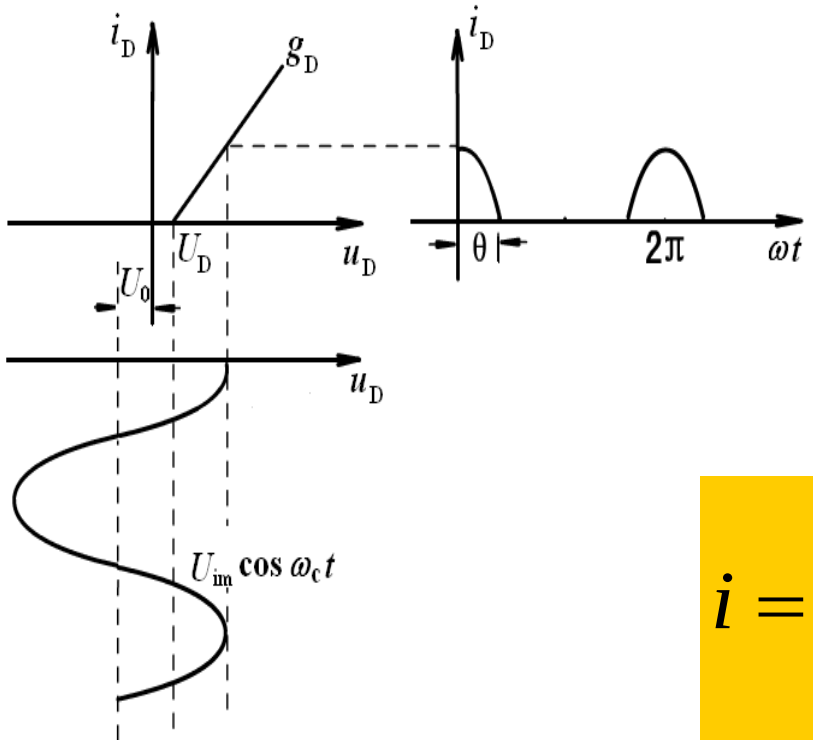
❖对Ω信号，C 相当开路

$$\frac{1}{\Omega C} >> R_L \rightarrow R_L C << \frac{1}{\Omega}$$

❖对ω_c信号，C 相当短路

$$\frac{1}{\omega_c C} << R_L \rightarrow R_L C >> \frac{1}{\omega_c}$$

由于U_{im}足够大，二极管的伏安特性折线近似。



$$i=\begin{cases} g_D(u_D-U_D) & u_D>U_D \\ 0 & u_D\le U_D \end{cases}$$

1） 若输入为载波，输出为直流

$$u_D = u_s - U_o = U_{im}\cos\omega_c t - U_o$$

$$i = g_D(u_D - U_D)$$

$$= g_D(U_{im}\cos\omega_c t - U_o - U_D)$$

当$\omega t = \theta$ $\quad i = \mathbf{0}$

$$\cos\theta = \frac{U_o + U_D}{U_{im}} \approx \frac{U_o}{U_{im}} \text{ 或 } U_o = U_{im}\cos\theta \quad ①$$

式(1)说明：输出的直流电压与导通角有关。

2）θ角仅由电路决定，与外加信号无关

（参考方程式1）

当$\omega t = 0$时，

$$I_m = g_D(U_{im} - U_o - U_D)$$
$$= g_D U_{im}(1 - \cos\theta)$$

直流量 $I_o = \alpha_o I_m = \dfrac{I_m(\sin\theta - \theta\cos\theta)}{\pi(1-\cos\theta)}$

$$= \frac{g_D U_{im}}{\pi}(\sin\theta - \theta\cos\theta) \qquad (2)$$

输出平均电压 $U_o = R_L I_o$ (3)

式 ① ② 代入式 ③

$$U_{im}\cos\theta = R_L \cdot \frac{g_D U_{im}}{\pi}(\sin\theta - \theta\cos\theta)$$

$$\frac{tg\theta - \theta}{\pi} = \frac{1}{g_D R_L}$$

$g_D R_L$ 很大，近似： $tg\theta \approx \theta + \frac{\theta^3}{3}$

得 $$\theta \approx \sqrt[3]{\frac{3\pi}{g_D R_L}}$$

2. 性能指标：

❖检波效率： $$K_d = \frac{U_o}{U_{im}} \approx \cos\theta$$

❖输入阻抗： $$\frac{1}{2} \cdot \frac{U_{im}^2}{R_i} = \frac{U_o^2}{R_L}$$ (载波功率近似等于输出功率)

$$\therefore \ R_i = \frac{1}{2} R_L$$

若输入为调幅波(AM)

$$u_s = U_{im}(1 + m_a \cos\Omega t)\cos\omega_c t$$

则输出 u_o**正比于**$U_{im}(1^{+} m_a \cos\Omega t)$

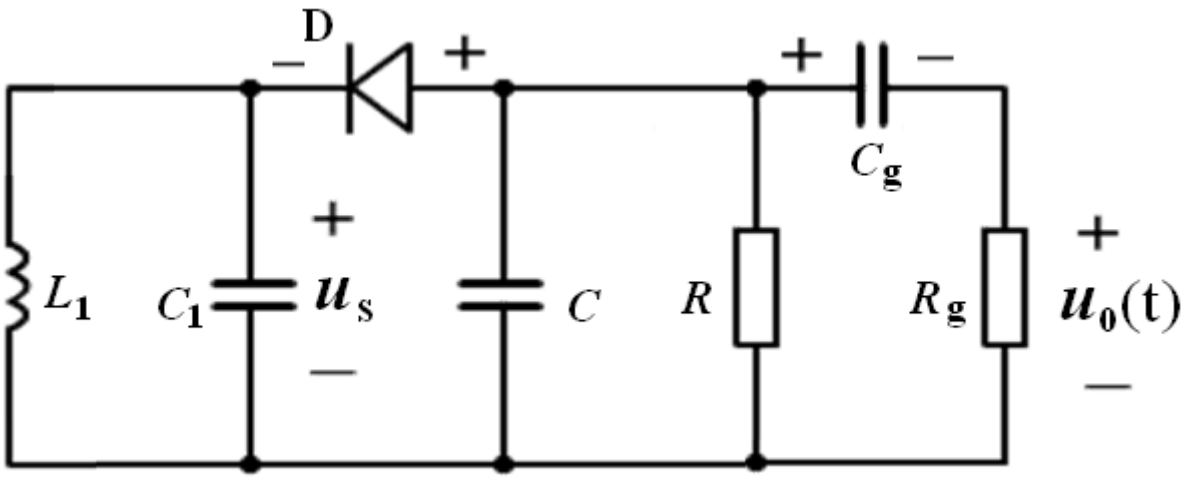
$$u_0 = K_d U_{im}(1 + m_a \cos\Omega t)$$

5-9.EXE

例1　　检波电路如图所示，

$$u_s = U_{sm}(1 + m\cos\Omega t)\cos\omega_c t \ ,\ \ U_{sm} > 0.5\ \text{V}$$

画出RC两端，R_g两端，C_g两端，二极管两端的电压波形。



解

$u_D = u_c - u_s$ 二极管 u_s 在负半周导通，因此检波应是调幅波 u_s 的下包络。

$$u_c = K_d \times (u_s\text{下包络})$$

$$= K_d U_{sm}(1 + m_a \cos \Omega t)$$

$$u_D = u_c - u_s$$

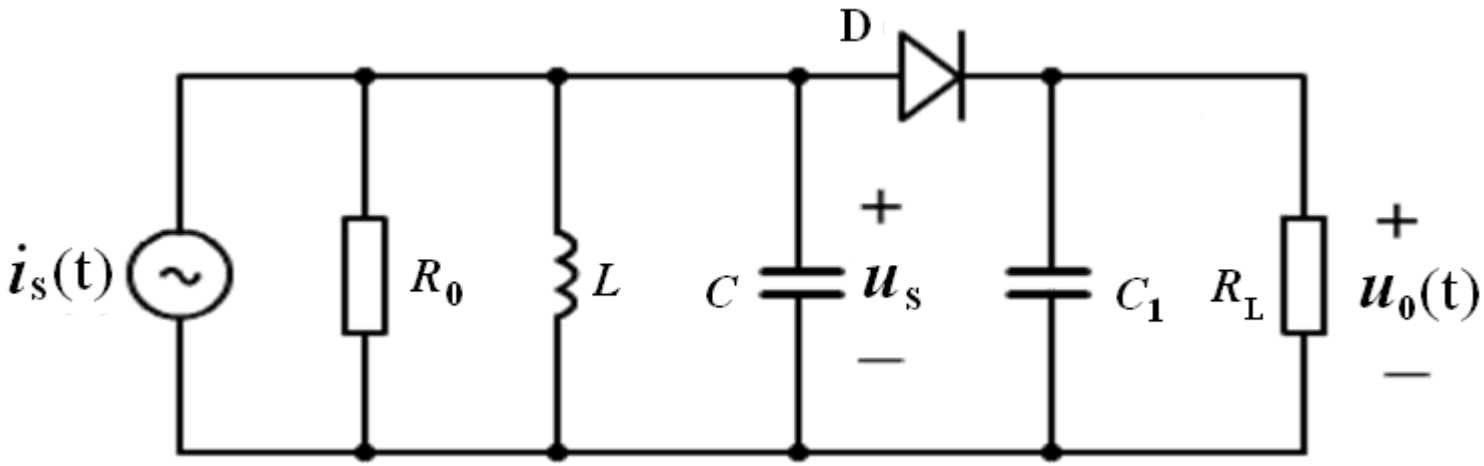
图解

例2 在题图所示的包络检波电路中，输入回路为并联谐振电路，其谐振频率 $f_o = 10^6$ Hz，谐振电阻 $R_o = 10\ \text{k}\Omega$，检波负载为 $R_D = 100\ \Omega$，$C_1 = 0.01\ \mu\text{F}$

(1) 若 $i_s = 0.5\cos 2\pi \times 10^6 t$ (mA)，求检波器输入电压 $u_i(t)$ 及检波器输出电压 $u_o(t)$ 的表达式。

（2） 若 $i_s = 0.5(1+0.5\cos 2\pi\times 10^3 t)\cos 2\pi\times 10^6 t$ (mA)，求 $u_o(t)$ 的表达式

(1) 若 $i_s = 0.5\cos 2\pi \times 10^6 t$ (mA)，求检波器输入电压 $u_s(t)$ 及检波器输出电压 $u_o(t)$ 的表达式。



（2） 若 $i_s = 0.5(1+0.5\cos 2\pi \times 10^3 t)\cos 2\pi \times 10^6 t$ (mA)

求 $u_o(t)$ 的表达式

解

（1） $\theta = \sqrt[3]{\dfrac{3\pi R_D}{R_L}} = \sqrt[3]{\dfrac{3\pi \times 100}{10 \times 10^3}} = 0.455\,\text{rad} = 26^\circ$

$$K_d = \cos\theta = \mathbf{0.9} \qquad R_i = \frac{R_L}{2} = 5\ \text{k}\Omega$$

$$u_s = (R_o // R_i) \cdot i_s = (20 // 5) \cdot i_s$$

$$= 4 \times 0.5 \cos(2\pi \times 10^6)t = 2\cos(2\pi \times 10^6)t\ \ (\text{V})$$

$$u_o(t) = K_d \cdot u_{sm}(t) = 1.8\ \text{V}$$

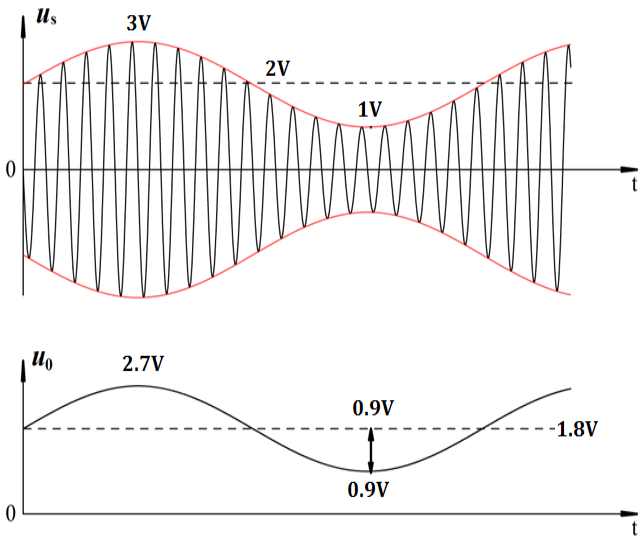
（2）

$$u_s = 4\times 0.5\left[1+0.5\cos(2\pi\times 10^3)t\right]\cos(2\pi\times 10^6)t$$

$$= 2\left[1+0.5\cos(2\pi\times 10^3)t\right]\cos(2\pi\times 10^6)t \quad \text{(V)}$$

$$u_o(t) = K_d U_{sm}(t)$$

$$= 1.8\left[1+0.5\cos(2\pi\times 10^3)t\right] \quad \text{(V)}$$



3. 检波失真

为了保证检波的输出电压不失真地反映输入调幅波的包络变化，输入调幅波足够大。

$$u_S = U_{im}(1 + m_a \cos \Omega t) \cos \omega_c t$$

则包络最小值：

$U_{im}(1 - m_a)$ 应大于大信号检波所需的电压值

$$\therefore\ U_{im}(1 - m_a) \geq 0.5 \quad \mathrm{V}$$

若输入的调制信号为复杂波时，最高频率为 $F_{\max}$，则低通滤波器的带宽应大于 $R_L C$。 $F_{\max}$

除此之外，如果电路参数选择不当，还产生以下两种失真。

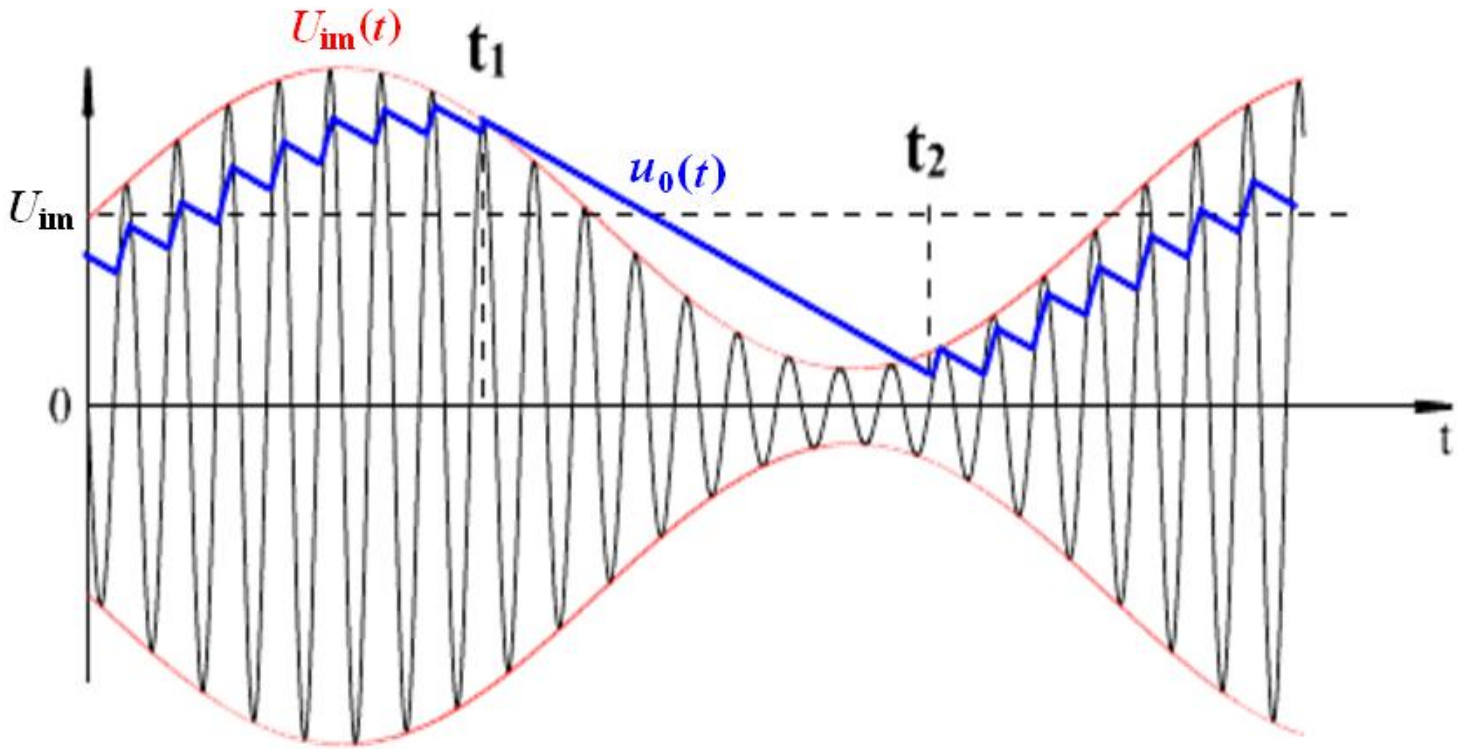
- 惰性失真
- 负峰切割失真

（1） 惰性失真

从充放电的观点来看，增大 R_L 和 C 值，可以提高检波效率和滤波能力。

但 R_L 、C 过大，二极管放电速度跟不上调幅波包络的下降速度，输出电压就会产生失真。通常将这种失真称为惰性失真。

（Failure-Follow Distortion）



$$u_S(t) = U_{im}(1 + m_a \cos \Omega t) \cos \omega_c t$$

可见为了避免产生惰性失真。C 通过 R_L 放电速度必须大于或等于包络的下降速度。

$$\left|\frac{\partial u_0}{\partial t}\right|_{t=t_1} \geq \left|\frac{\partial U_{im}(t)}{\partial t}\right|_{t=t_1}$$

$$U_{im}(t) = U_{im}(1 + m_a \cos \Omega t)$$

$$\left.\frac{\partial U_{im}(t)}{\partial t}\right|_{t=t_1} = -m_a U_{im} \Omega \sin \Omega t_1$$

C **通过** R_L **放电规律**

$$u_0 = U_{01} e^{-\frac{t-t_1}{R_L C}} \qquad U_{01} \approx U_{im}(1 + m_a \cos \Omega t_1)$$

$$\left.\frac{\partial u_0}{\partial t}\right|_{t=t_1} = -U_{01}\cdot\frac{1}{R_L C}\cdot e^{-\frac{t-t_1}{R_L C}} = -\frac{U_{im}(1+m_a\cos\Omega t_1)}{R_L C}$$

令：

$$A = \frac{\left|\frac{\partial U_{im}(t)}{\partial t}\right|_{t=t_1}}{\left|\frac{\partial u_0}{\partial t}\right|_{t=t_1}}$$

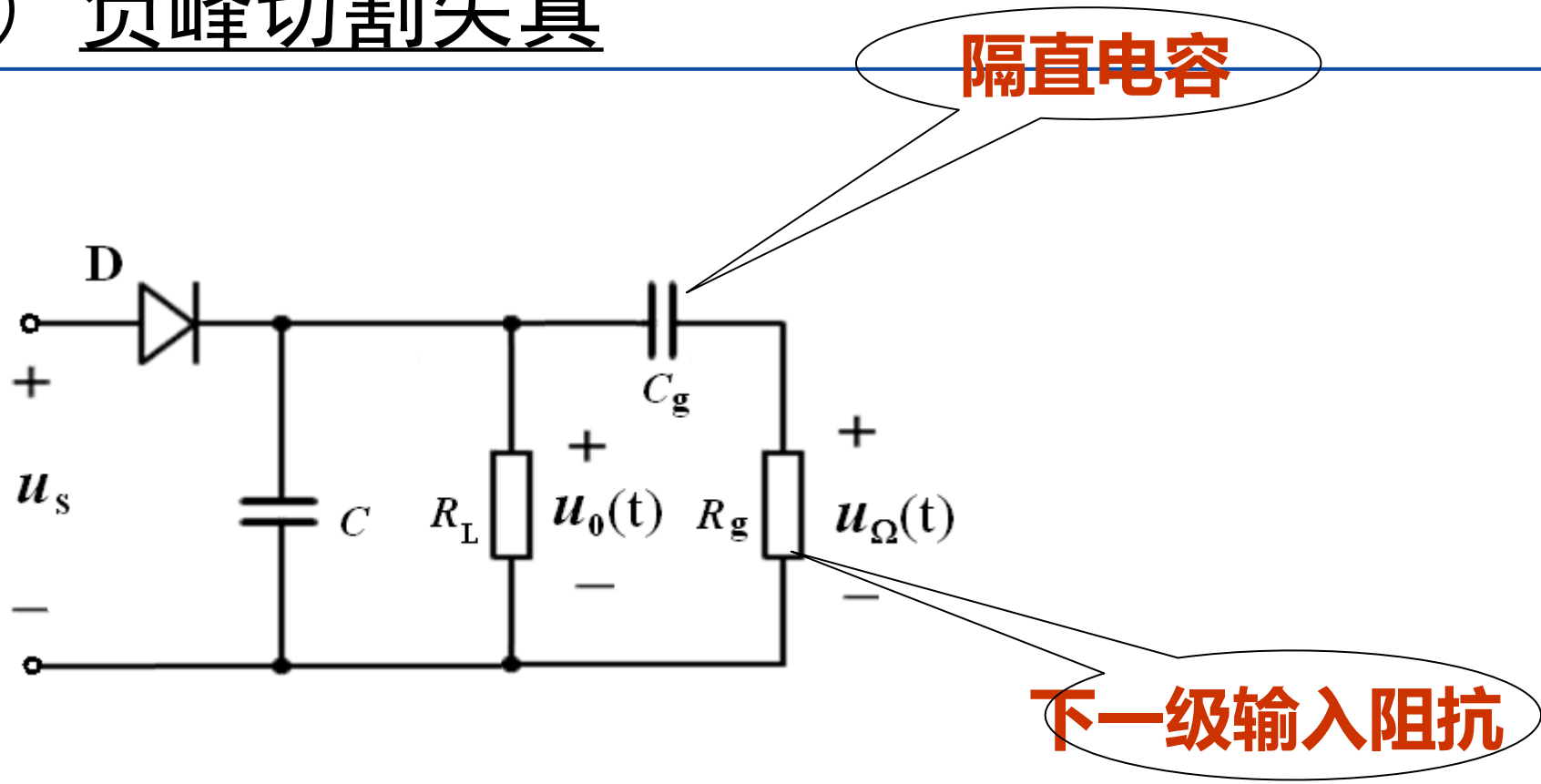
$$= \Omega R_L C\left|\frac{m_a\sin\Omega t_1}{1+m_a\cos\Omega t_1}\right| \le 1$$

A 出现最大值仍要保证其值小于或等于1，为此，取A 对t_1的导数，并令它等于零，求A 呈最大值时刻为 $\cos\Omega t_1 = -m_a$ ，所以得到单音调制时不产生惰性失真的充要条件为：

$$R_L C \le \frac{\sqrt{1-m_a^2}}{\Omega m_a}$$

在设计中，应将 Ω_{max} ，$m_{a\,max}$ 代入上式，检验有无惰性失真。

(2) 负峰切割失真



当考虑到检波器和下级放大器连接时，采用的电路

当考虑到检波器和下级放大器连接时，采用上图所示电路。C_g 为耦合电容，以避免 $u_0(t)$ 中的直流分量影响下级放大器的静态工作点。C_g 为隔直电容，它对 Ω 呈交流短路，R_g 的下一级输入阻抗由图可知，检波器的交流负载 $Z_L(j\Omega)$ 和直流负载 $Z_L(0)$ 分别为：

$$Z_L(0) = R_L \qquad Z_L(j\Omega) = R_L // R_g$$

$$\therefore Z_L(j\Omega) < Z_L(0)$$

在不产生负峰切割失真的情况下：

当输入AM波： $u_s = U_{im}(1 + m_a \cos \Omega t) \cos \omega_c t$

输出电压： $u_0(t) = U_0 + U_{\Omega m} \cos \Omega t$ **，**

因而输出电流的平均值为

$$i_{AV} = I_o + I_{\Omega m} \cos \Omega t$$

其中 $I_o = \dfrac{U_o}{Z_L(0)} = \dfrac{U_o}{R_L}$

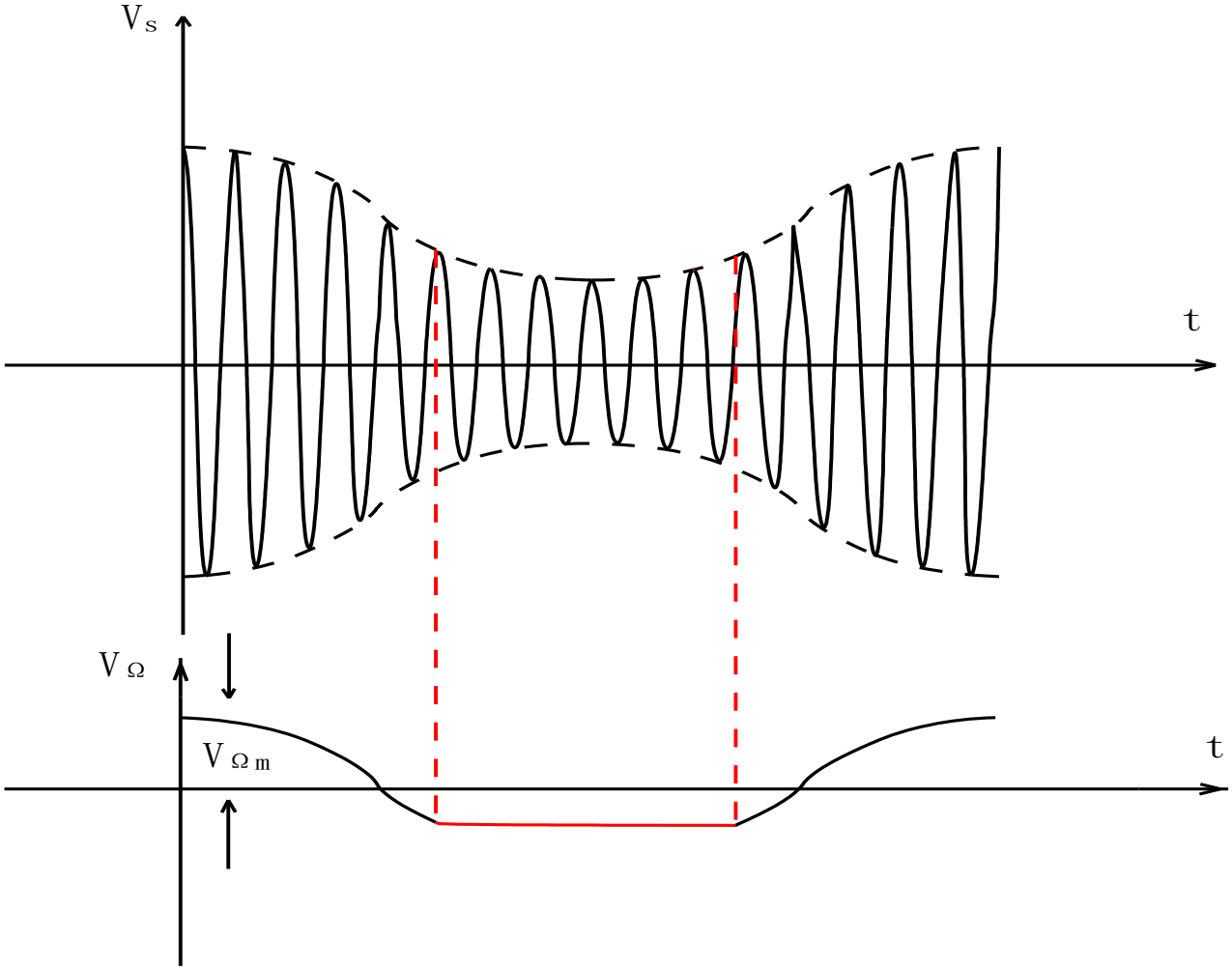
$$I_{\Omega m} = \frac{U_{\Omega m}}{Z_L(j\Omega)} = \frac{U_{\Omega m}}{R_L // R_g}$$

由于二极管单向导电性，i_{AV} 必须为正值，即

$$I_{\Omega m} \le I_o$$

反之，如果 $I_{\Omega m} > I_{AV}$ 则 i_{AV} 在一段时间内变为负值，二极管截止，检波器不能正常充放电，不能跟随包络变化，从而产生负峰值切割失真。

5-11.exe


Vs
t
VΩ
VΩm
t

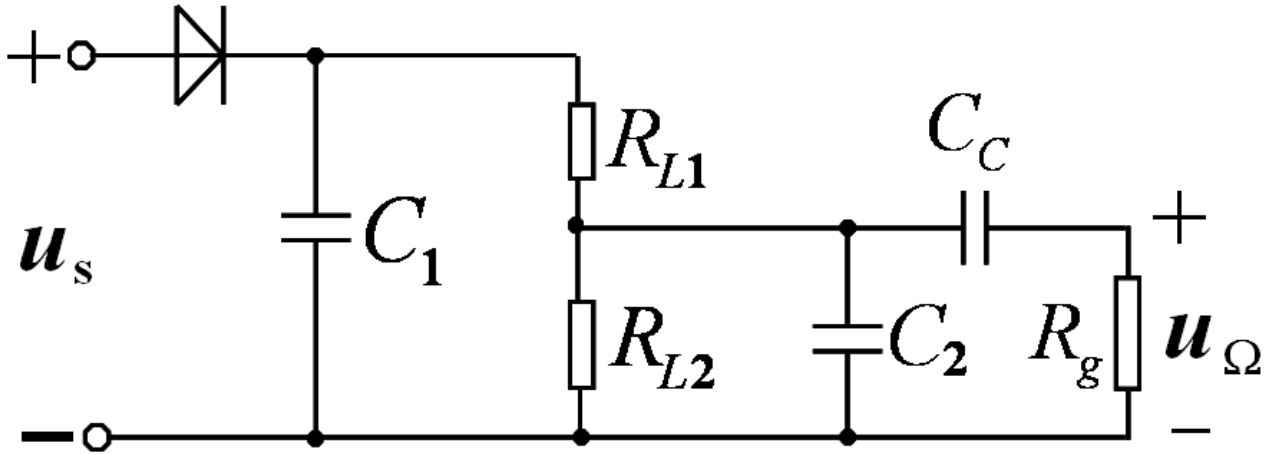
可见要使检波器不产生负峰切割失真，必须使 $I_{\Omega m} \le I_o$

即 $$\frac{U_{\Omega m}}{Z_L(\Omega)} \le \frac{U_o}{Z_L(o)}$$

设 $K_d \to 1$ $\quad U_0 \approx U_{im} \quad U_{\Omega m} \approx m_a U_{im}$

$$m_a \le \frac{Z_L(\Omega)}{Z_L(0)} = \frac{R_g}{R_L + R_g}$$

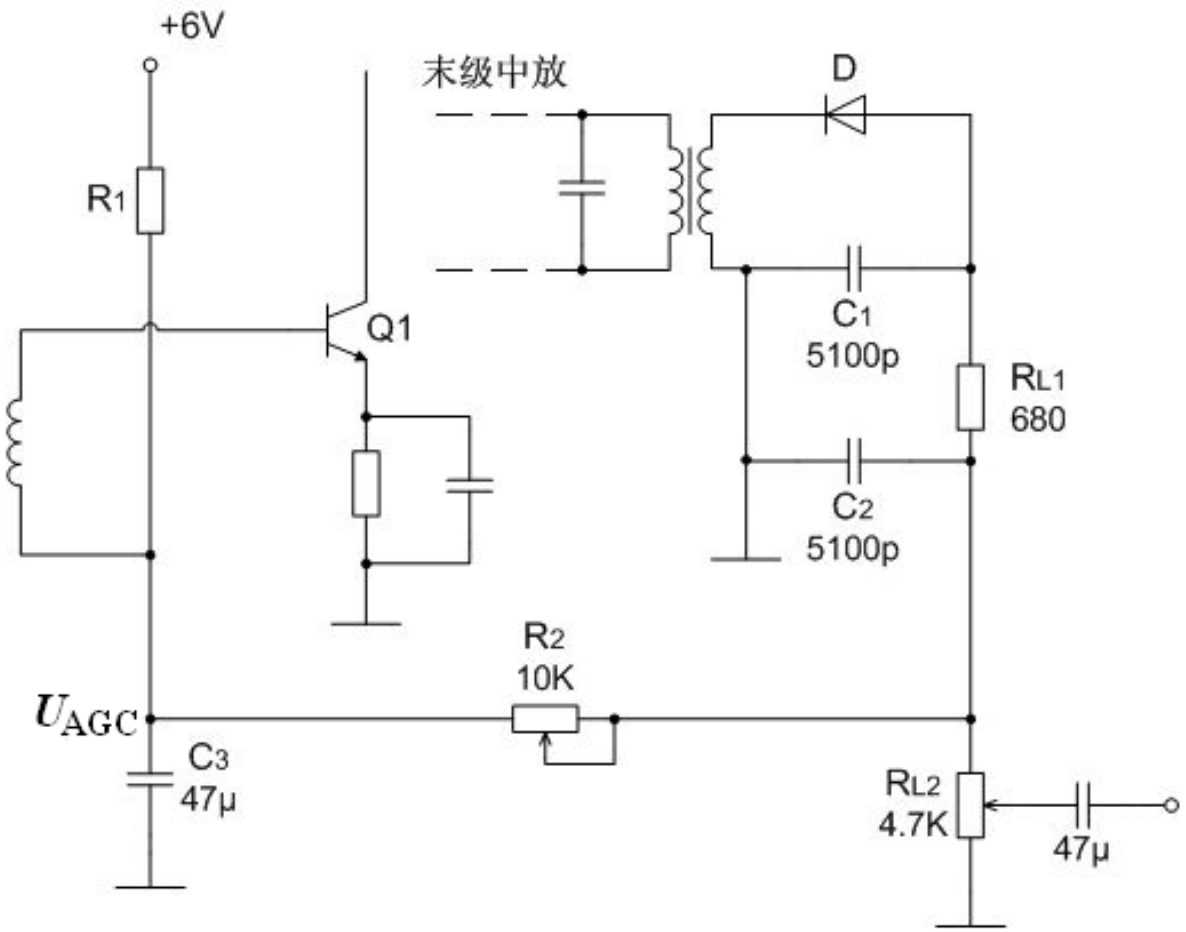
在实际电路中，可以采取各种措施来减少交、直流阻抗值的差别。



交 $Z_L(\Omega) = R_{L1} + R_{L2} // R_g$

直 $Z_L(0) = R_{L1} + R_{L2}$

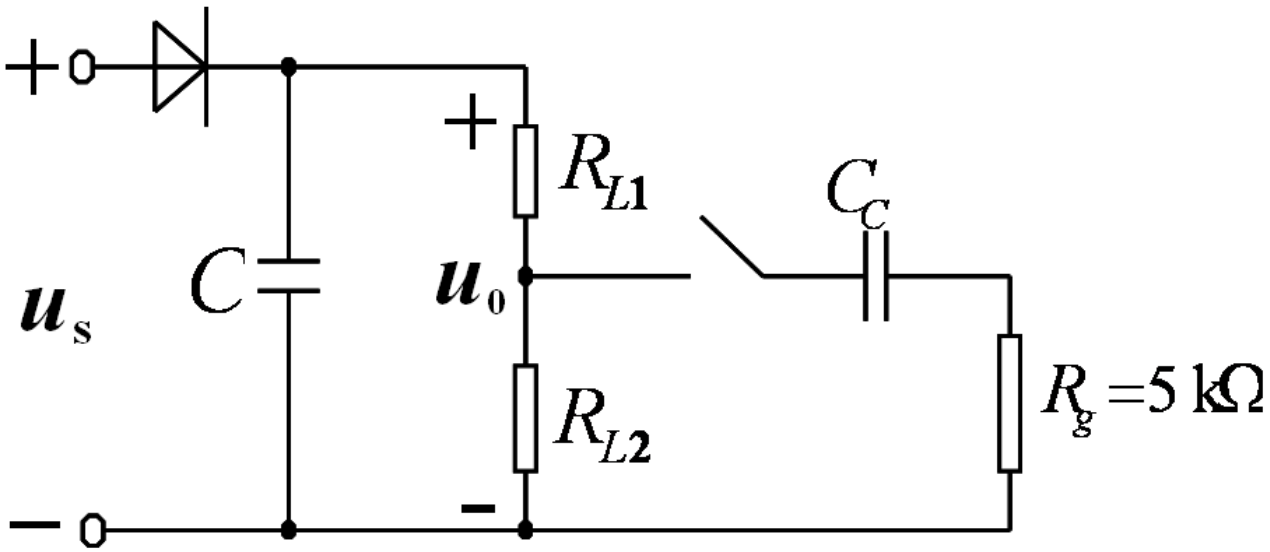
广播收音机中检波器的实用电路



例3　如图大信号二极管包络检波器，二极管导通时电阻 $R_D = \mathbf{100}\ \Omega$　$R_{L1} + R_{L2} = 5\ \mathrm{k}\Omega$

$$u_s = 2\left[1 + 0.6\cos(2\pi \times 10^4)t\right]\cos(2\pi \times 10^7)t$$

求开关断开时，(1) 检波器的输入阻抗 R_i ,和电压传输系数 K_d ,检波器的输出电压 u_0 的表达式。(2) 为了不产生惰性失真，电容 C 的最大值。 (3)合上开关后， 为了不产生负峰切割失真，R_{L1} 至少应为多少?


+
R_{L1}
C_C
u_s
C
u_0
$R_g = 5\ \mathrm{k}\Omega$
R_{L2}
−

解：(1)

$$R_i = \frac{1}{2}(R_{L1} + R_{L2}) = 2.5\ \text{k}\Omega$$

$$R_L = R_{L\mathbf{1}} + R_{L\mathbf{2}}$$

$$\theta = \sqrt[3]{\frac{3\pi R_D}{R_L}} = \sqrt[3]{\frac{3\pi \times 100}{5\times 10^3}} = 0.573\ \ rad = 32.86^\circ$$

$$K_d = \cos\theta \approx \mathbf{0.84}$$

$$u_o(t) = K_d U_{sm}(t) = 1.68\left[1 + 0.6\cos(2\pi \times 10^4)t\right]$$

(2) $$R_L C \le \frac{\sqrt{1-m_a^2}}{\Omega \cdot m_a}$$

$$\therefore\ C_{\max} = \frac{\sqrt{1-m_a^2}}{R_L \Omega m_a}$$

$$= \frac{\sqrt{1-0.6^2}}{5\times10^3\times2\pi\times10^4\times0.6}$$

$$= 4246\ \text{pF}$$

(3)

$$Z_L(\Omega) = R_{L1} + R_{L2} // R_g$$

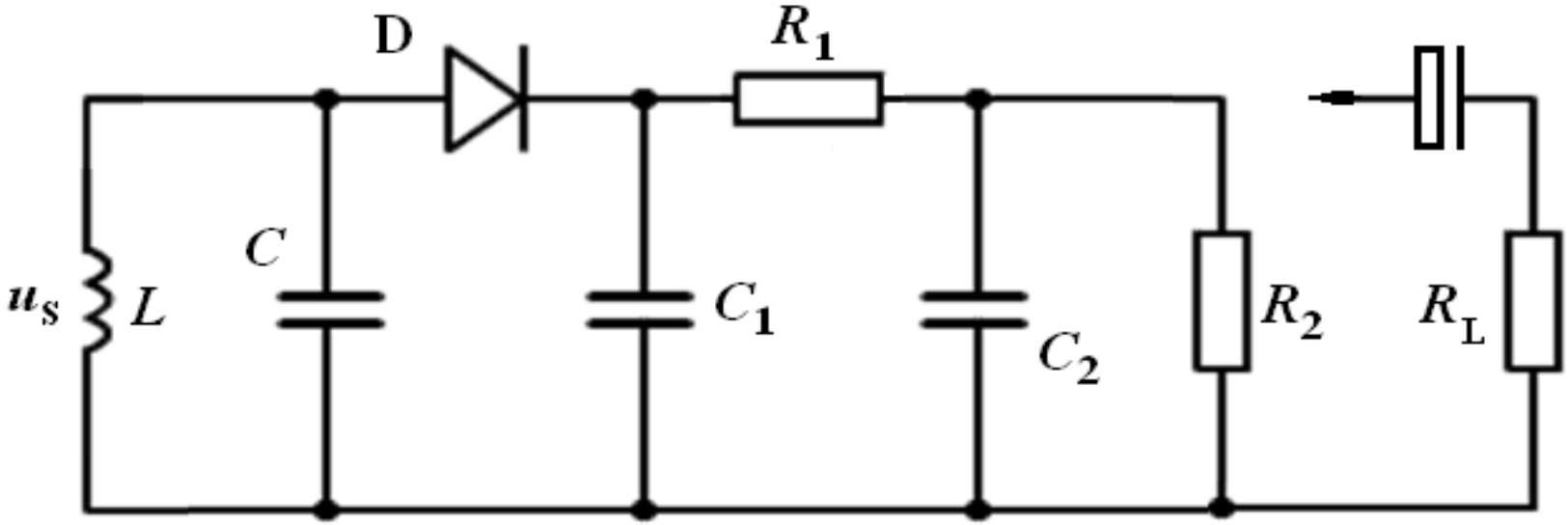
$$= R_{L1} + \frac{(5 - R_{L1})R_g}{5 - R_{L1} + 5}$$

$$Z_L(\mathbf{0}) = R_{L\mathbf{1}} + R_{L\mathbf{2}} = \mathbf{5}$$

$$\frac{Z_L(\Omega)}{Z_L(\mathbf{0})} \geq m_a \quad \Longrightarrow \quad R_{L1}^2 - 8R_{L1} + 5 \leq 0$$

$$R_{L1} \geq 0.683 \ \text{k}\Omega$$

例4 如图二极管大信号峰值包络检波电路，已知输入电路的空载Q值：$Q_0 = \mathbf{100}$， 载波频率$f_0 = 465\ \mathrm{kHz}$ $L = 560\ \mu\mathrm{H}$ $R_L = 5\ \mathrm{k\Omega}$ 若要求输入回路的通频带$BW = 30\ \mathrm{kHz}$不产生惰性失真，并且再接上交流负载后不产生负峰切割失真，试确定元件 R_1，R_2，C_1，C_2 的值。



$$u_s = U_{sm}(1+0.5\cos 2\pi\times 10^3 t)\cos 2\pi f_0 t$$

解

$$R_p = Q_0\omega_0 L = 100\times 2\pi\times 465\times 10^3\times 560\times 10^{-6}$$

$$= 163.53 \quad k\Omega$$

$$\frac{R_1+R_2}{2} // R_p = Q_L\omega_0 L = \frac{f_0}{\mathrm{BW}}\cdot\omega_0 L$$

$$= \frac{465\times 10^3}{30\times 10^3}\times 2\pi\times 465\times 10^3\times 560\times 10^{-6}$$

$$= 25 \ \mathrm{k\Omega}$$

设 $\dfrac{R_1+R_2}{2} = R_i \quad \Rightarrow \quad R_i \approx 29 \ \mathrm{k\Omega}$

$$\therefore \quad R_1 + R_2 = 2R_i = 58 \ \mathrm{k\Omega}$$

不产生惰性失真:

$$(R_1+R_2)(C_1+C_2)\le\frac{\sqrt{1-m_a^2}}{\Omega\cdot m_a}=\frac{\sqrt{1-0.5^2}}{2\pi\times10^3\times0.5}$$

$$\therefore\quad C_1+C_2\le4.755\ \mu\mathrm{F}$$

取 $C_1+C_2=4\ \mu\mathrm{F}$

$$C_1=C_2=\frac{C_1+C_2}{2}=2\ \mu\mathrm{F}$$

不产生负峰切割失真：

$$\frac{R_2 // R_g + R_1}{R_2 + R_1} \geq m_a = \mathbf{0.5}$$

$$\Rightarrow \begin{array}{l} R_1 \geq 24.66\ \text{k}\Omega \\ R_2 \leq 33.34\ \text{k}\Omega \end{array}$$

5.6　同步检波　又称相干检波
（Synchronous Detector）

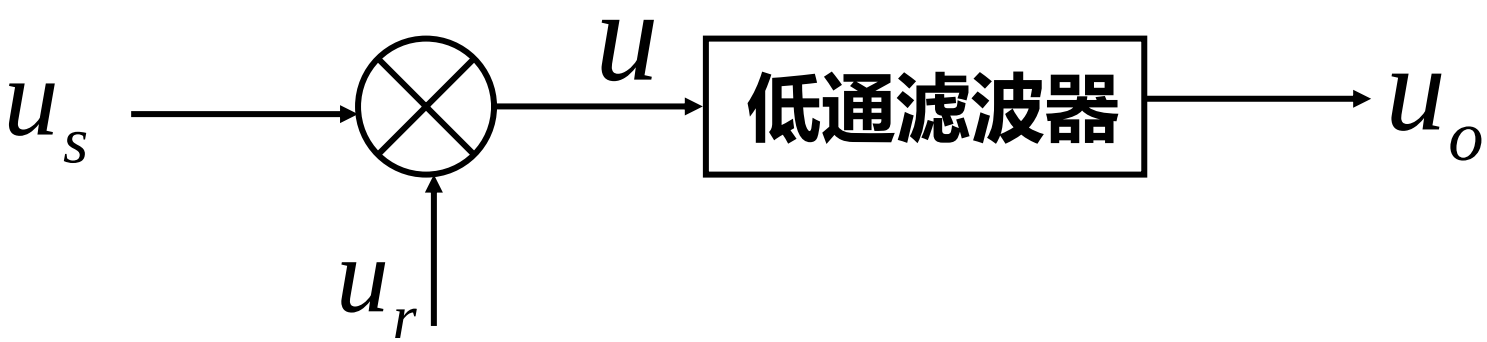
主要用来解调双边带和单边带调幅 (DSB，SSB)

也可解调AM

有两种实现电路：

1. **乘积型**
2. **叠加型**

一. 乘积检波电路



u_r **同步信号** $u_r(t) = U_{rm}\cos\omega_c t$

1. $u_s = U_{sm}\cos\Omega t\cos\omega_c t$ **(DSB)**

$$u = u_r \cdot u_s = U_{rm}U_{sm}\cos\Omega t\cos^2\omega_c t$$

$$= \frac{1}{2}U_{rm}U_{sm}\cos\Omega t(1+\cos 2\omega_c t)$$

经低通： $u_o = \frac{1}{2}KU_{rm}U_{sm}\cos\Omega t$

2. $u_s = U_{sm}\cos(\Omega + \omega_c)t$ **(SSB)**

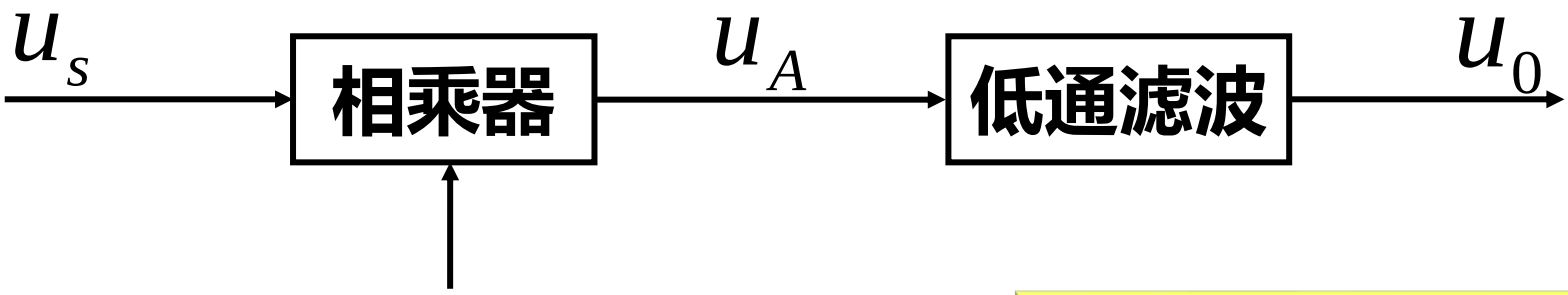
则 $u = u_s \cdot u_r = U_{sm}U_{rm}\cos(\Omega + \omega_c)t\cos\omega_c t$

$$= \frac{1}{2}U_{sm}U_{rm}\left[\cos\Omega t + \cos(2\omega_c + \Omega)t\right]$$

经低通： $u_0 = \frac{1}{2}KU_{rm}U_{sm}\cos\Omega t$

同步信号 u_r **必须与输入信号保持严格同步**

思考题:



$$u_r = U_{rm}\cos(\omega_c t + \varphi)$$

同频不同相

若 (1) $u_s = U_{sm}\cos\Omega t\cos\omega_c t$

(2) $u_s = U_{sm}\cos(\omega_c + \Omega)t$

时，输出电压 $u_0 =$ ？ 是否失真?

解

$$(1)\ u_o = \frac{1}{2}U_{rm}U_{sm}\cos\varphi\cos\Omega t$$

多了一个 $\cos\varphi$ 因子，是一个衰减因子

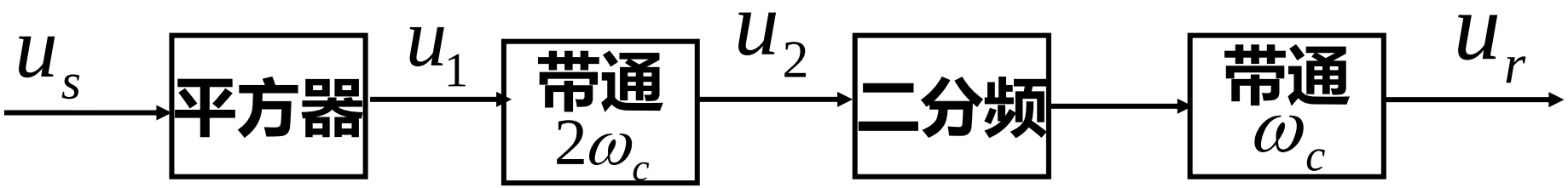
当 $\varphi = \frac{\pi}{2}$，$\cos\varphi = 0$

当 $\varphi = \varphi(t)$ 则输出导致相位失真

$$(2)\ u_o = \frac{1}{2}U_{rm}U_{sm}\cos(\Omega t - \varphi)$$

导致相位失真

3. 同步信号的产生



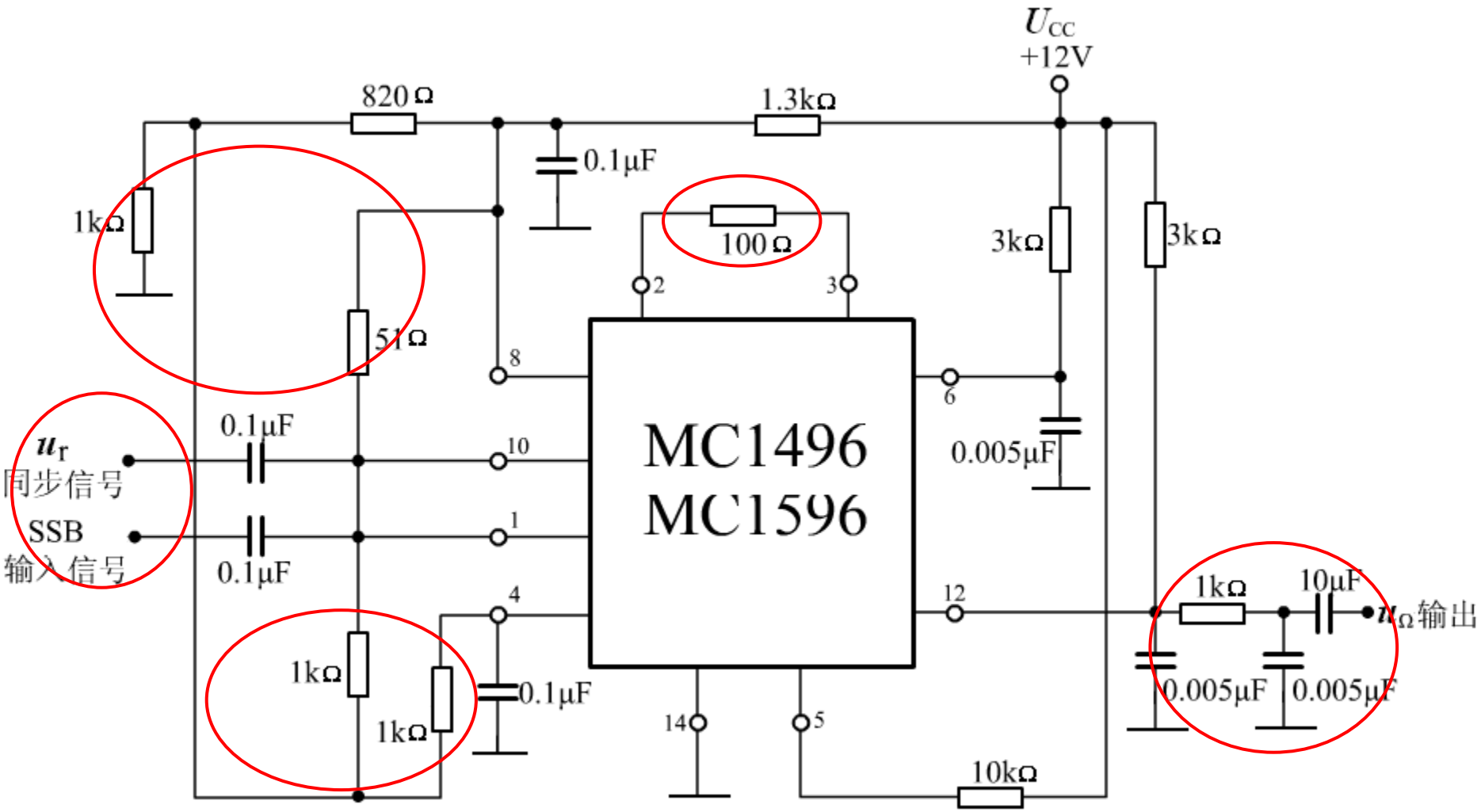
其中：

$$u_1 = K(U_{\mathrm{sm}} \cos \Omega t \cos \omega_{\mathrm{c}} t)^2$$

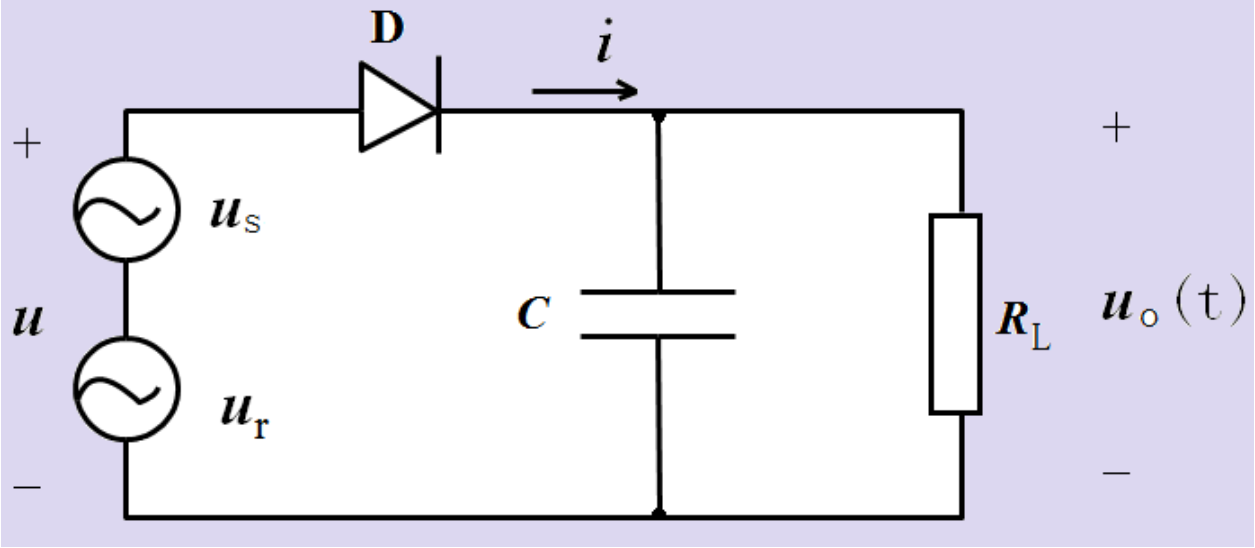
$$= KU_{\mathrm{sm}}^2 \cos^2 \Omega t \cos^2 \omega_{\mathrm{c}} t$$

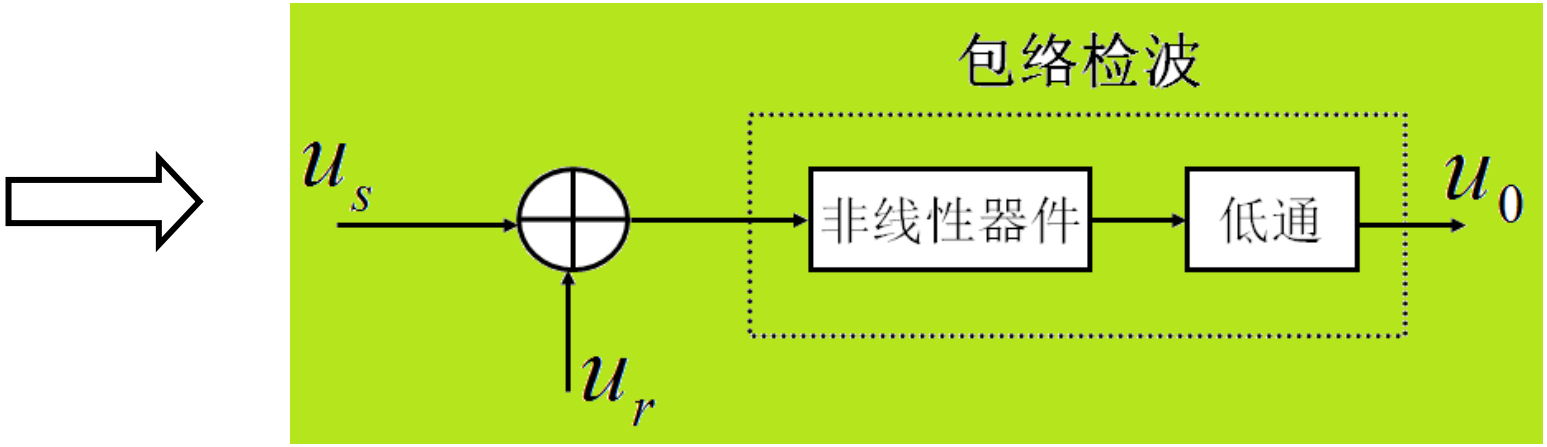
$$= KU_{\mathrm{sm}}^2 \cos^2 \Omega t \left[\frac{1}{2} + \frac{1}{2} \cos 2\omega_{\mathrm{c}} t \right]$$

MC1596集成同步检波器



二.叠加型同步检波 (平衡同步检波)





1. u_s 为DSB

$$u_s = U_{sm}\cos\Omega t\cos\omega_c t$$

$$u_r = U_{rm}\cos\omega_c t$$

$$u = u_s + u_r = U_{rm}(1+\frac{U_{sm}}{U_{rm}}\cos\Omega t)\cos\omega_c t$$

$$m_a = \frac{U_{sm}}{U_{rm}} < 1$$

2. u_s为SSB

$$u_s = U_{sm}\cos(\Omega + \omega_c)t$$

$$\begin{aligned} u(t) &= u_r(t) + u_s(t) = U_{rm}\cos\omega_c t + U_{sm}\cos(\omega_c + \Omega)t \\ &= (U_{rm} + U_{sm}\cos\Omega t)\cos\omega_c t - U_{sm}\sin\Omega t\sin\omega_c t \\ &= U_m\cos(\omega_c t + \varphi) \end{aligned}$$

$$U_{\mathrm{m}} = \sqrt{(U_{\mathrm{rm}} + U_{\mathrm{sm}} \cos \Omega t)^2 + (U_{\mathrm{sm}} \sin \Omega t)^2}$$

$$\varphi = -\mathrm{tg}^{-1} \frac{U_{\mathrm{sm}} \sin \Omega t}{U_{\mathrm{rm}} + U_{\mathrm{sm}} \cos \Omega t}$$

若 $U_{\mathrm{sm}} << U_{\mathrm{rm}}$ ， $(\frac{U_{\mathrm{sm}}}{U_{\mathrm{rm}}})^2 \approx 0$

$$U_{\mathrm{m}} = U_{\mathrm{rm}} \sqrt{1 + (\frac{U_{\mathrm{sm}}}{U_{\mathrm{rm}}})^2 + 2(\frac{U_{\mathrm{sm}}}{U_{\mathrm{rm}}}) \cos \Omega t}$$

则 $U_{sm} \approx U_{rm}\left[1+2\frac{U_{sm}}{U_{rm}}\cos\Omega t\right]^{\frac{1}{2}}$

$$=U_{rm}\left[1+\frac{U_{sm}}{U_{rm}}\cos\Omega t-\frac{1}{2}(\frac{U_{sm}}{U_{rm}})^2\cos^2\Omega t+\cdots\right]$$

进一步忽略三次方以上项

$$U_m \approx U_{rm}\left[\underbrace{1-\frac{1}{4}(\frac{U_{sm}}{U_{rm}})^2+\frac{U_{sm}}{U_{rm}}\cos\Omega t}_{\text{AM波}}-\underbrace{\frac{1}{4}(\frac{U_{sm}}{U_{rm}})^2\cos 2\Omega t}_{\text{二次谐波}}\right]$$

$$u(t)=U_{\mathrm{m}}\cos(\omega_{\mathrm{c}}t+\varphi)$$

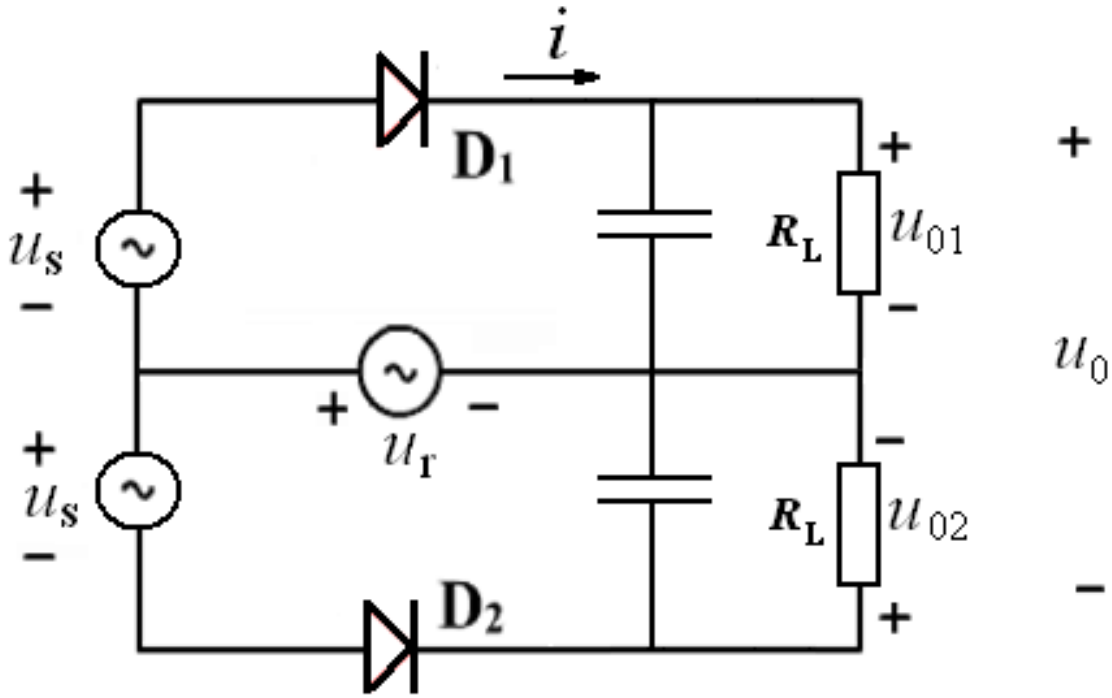
$$\approx U_{\mathrm{rm}}\left[1-\frac{1}{4}\left(\frac{U_{\mathrm{sm}}}{U_{\mathrm{rm}}}\right)^2+\frac{U_{\mathrm{sm}}}{U_{\mathrm{rm}}}\cos\Omega t-\frac{1}{4}\left(\frac{U_{\mathrm{sm}}}{U_{\mathrm{rm}}}\right)^2\cos 2\Omega t\right]\cos(\omega_{\mathrm{c}}t+\varphi)$$

$$=U_{\mathrm{rm}}\left[1-\frac{1}{4}\left(\frac{U_{\mathrm{sm}}}{U_{\mathrm{rm}}}\right)^2+\frac{U_{\mathrm{sm}}}{U_{\mathrm{rm}}}\cos\Omega t\right]\cos(\omega_{\mathrm{c}}t+\varphi)$$

$$-\frac{1}{4}\left(\frac{U_{\mathrm{sm}}}{U_{\mathrm{rm}}}\right)^2\cos 2\Omega t\cos(\omega_{\mathrm{c}}t+\varphi)$$

二次谐波产生的频率： $\omega_{\mathrm{c}}\pm 2\Omega$

平衡式的同步检波：



抵消 2Ω 以及高次偏移谐波分量

$$u_{D1} = u_s + u_r \qquad u_{D2} = u_r - u_s$$

$$u_{01}(t) = K_{\mathrm{d}} U_{1\mathrm{m}}(t)$$

$$U_{1\mathrm{m}}(t) = U_{\mathrm{rm}} \sqrt{1 + 2\frac{U_{\mathrm{sm}}}{U_{\mathrm{rm}}} \cos \Omega t}$$

$$u_{02}(t) = K_{d} U_{2\mathrm{m}}(t)$$

$$U_{2\mathrm{m}}(t) = U_{\mathrm{rm}} \sqrt{1 - 2\frac{U_{\mathrm{sm}}}{U_{\mathrm{rm}}} \cos \Omega t}$$

根据 $\sqrt{1+x}=1+\frac{x}{2}-\frac{x^2}{8}$ $\sqrt{1-x}=1-\frac{x}{2}-\frac{x^2}{8}$

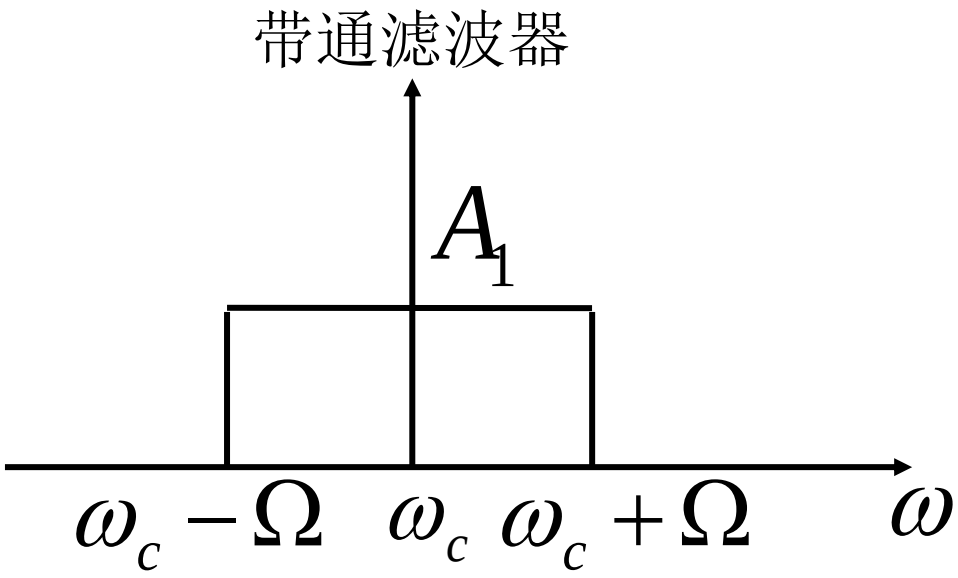
得 $\sqrt{1+x}-\sqrt{1-x}=x$

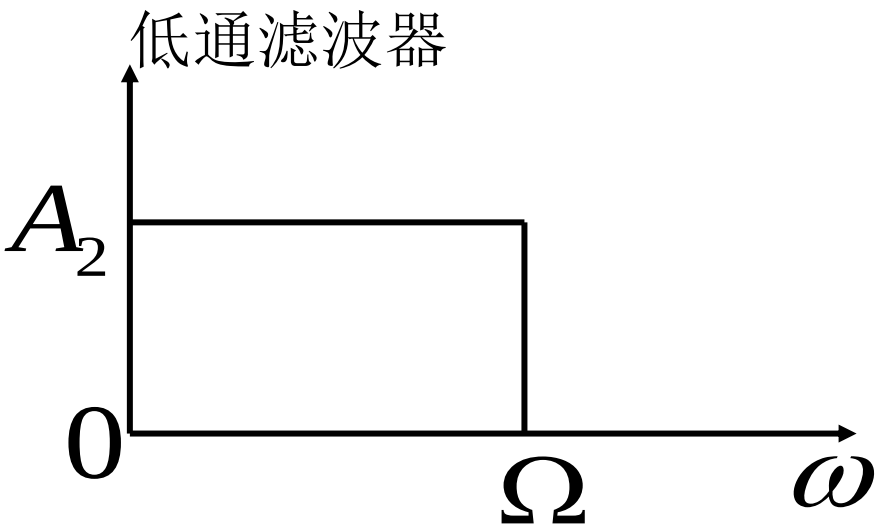
$$\therefore\quad u_0(t)=u_{01}(t)-u_{02}(t)$$

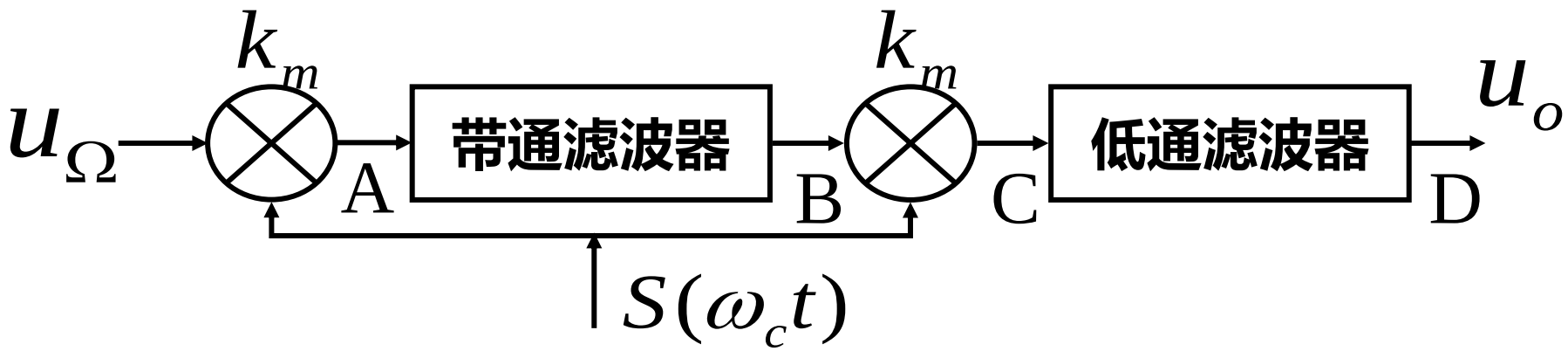
$$=K_d\cdot 2\frac{U_{\mathrm{sm}}}{U_{\mathrm{rm}}}U_{\mathrm{rm}}\cos\Omega t$$

$$=2K_{\mathrm{d}}U_{\mathrm{sm}}\cos\Omega t$$

例:


带通滤波器
A_1
$\omega_c-\Omega$
ω_c
$\omega_c+\Omega$
ω


低通滤波器
A_2
0
Ω
ω



其中 $u_\Omega = U_{\Omega m}\cos\Omega t$

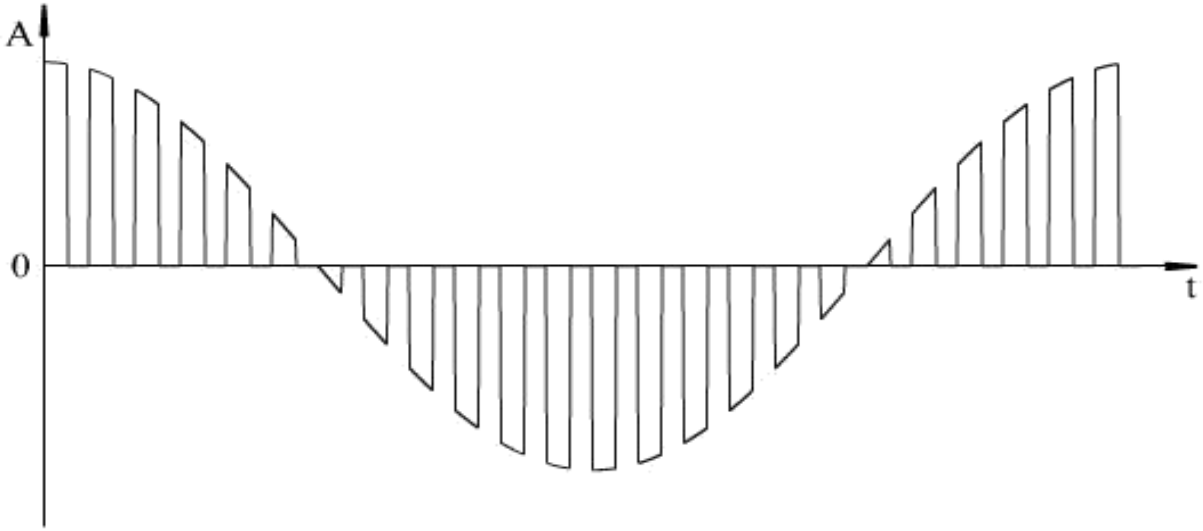
试画出A，B，C，D，各点波形

解

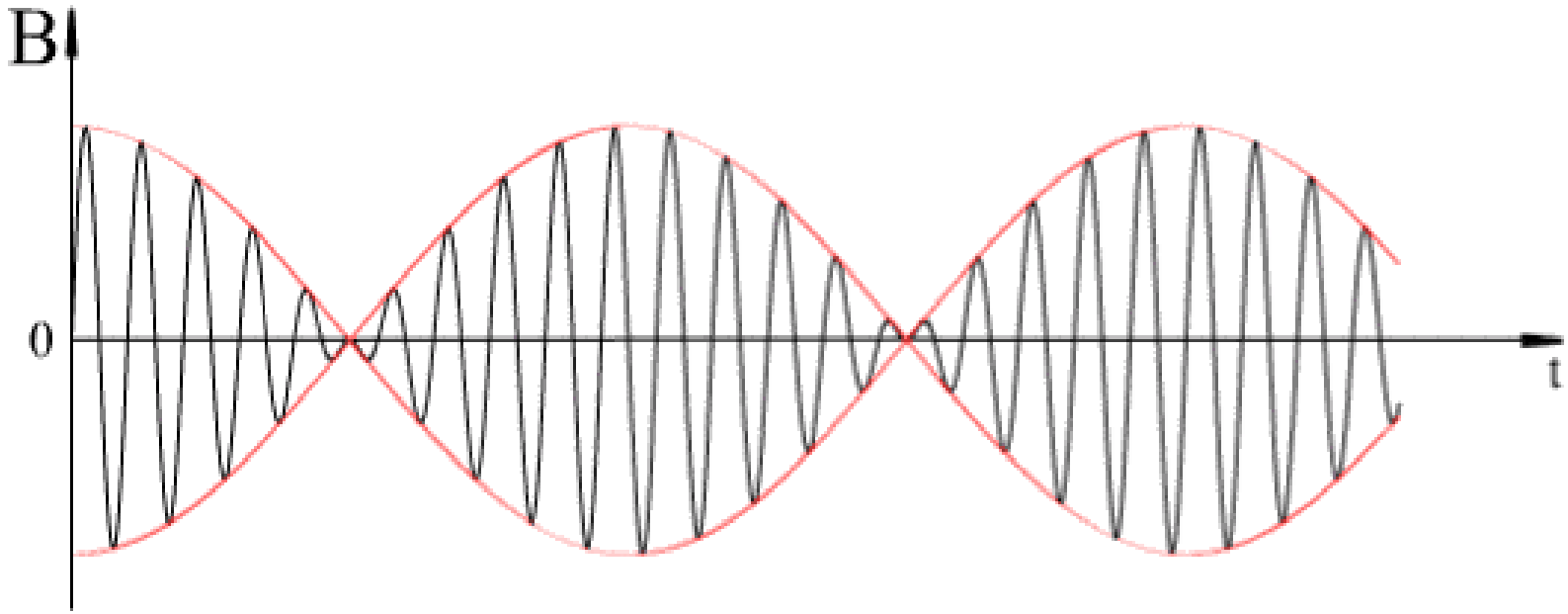
$$u_A = k_m u_\Omega S(\omega_c t) = k_m U_{\Omega m} \cos \Omega t \cdot S(\omega_c t)$$

$$= k_m U_{\Omega m} \cos \Omega t \left(\frac{1}{2} + \frac{2}{\pi} \cos \omega_c t - \frac{2}{3\pi} \cos 3\omega_c t + \cdots\right)$$

其中频率成分是 Ω，$\omega_c \pm \Omega$，$3\omega_c \pm \Omega$



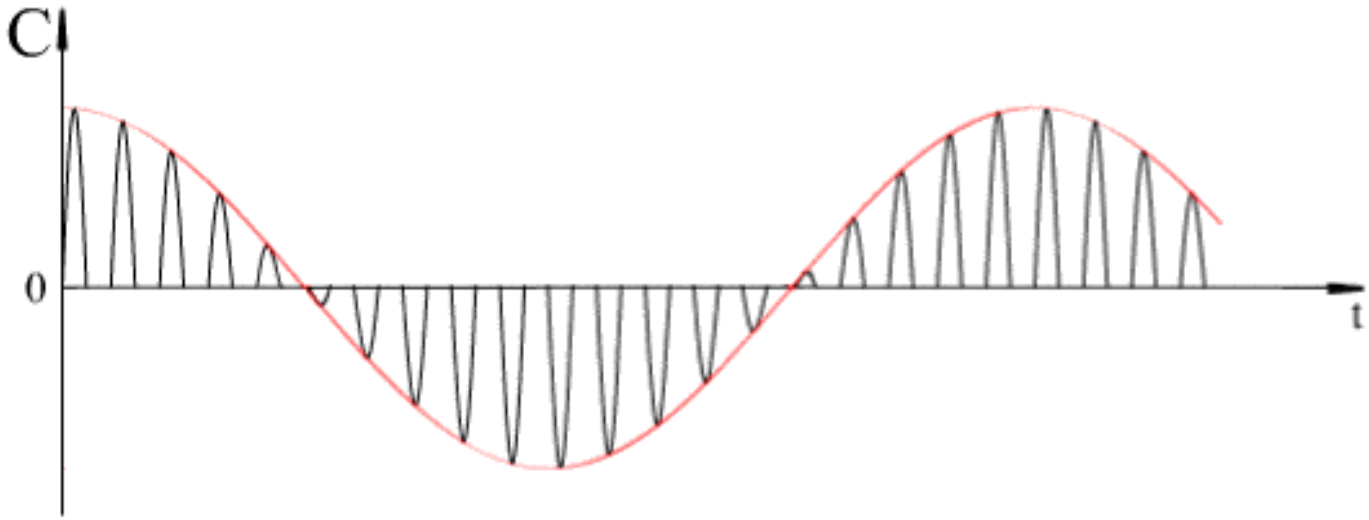
$$u_B = A_1 k_m U_{\Omega m} \frac{2}{\pi} \cos \Omega t \cos \omega_c t$$



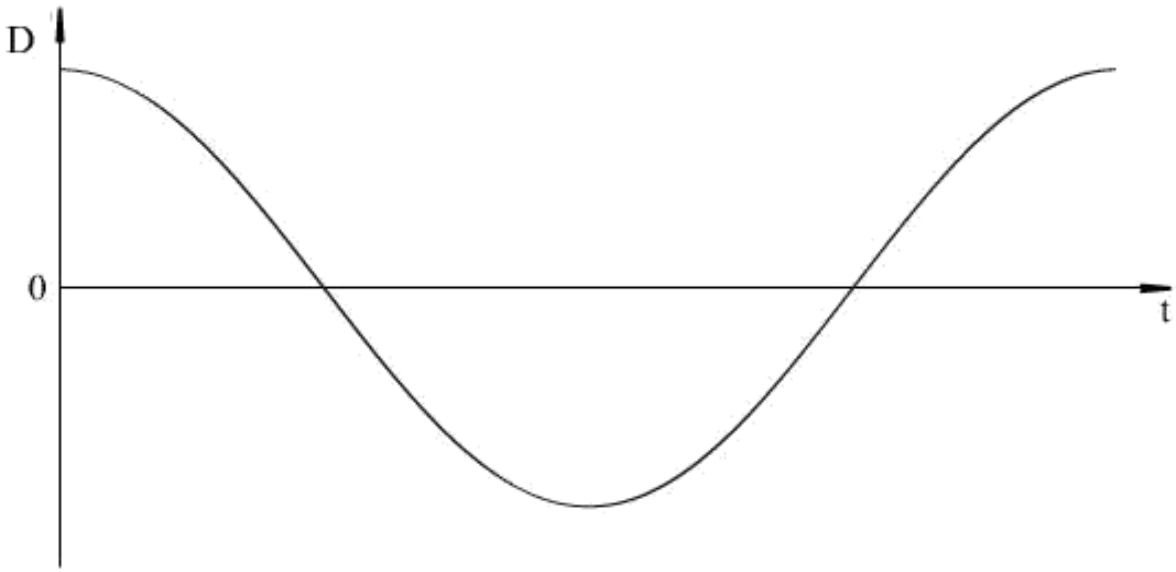
$$u_C = k_m u_B S(\omega_c t)$$

$$= \frac{2}{\pi} k_m^2 A_1 \cdot U_{\Omega m} \cos \Omega t \cos \omega_c t \left(\frac{1}{2} + \frac{2}{\pi} \cos \omega_c t - \cdots\right)$$

其中频率成分是 $\Omega, \omega_c \pm \Omega, 2\omega_c \pm \Omega, \cdots\cdots, 2n\omega_c \pm \Omega$



$$u_D = \frac{2}{\pi^2} A_1 A_2 k_m^2 \cos \Omega t = U_{Dm} \cos \Omega t$$



第五章　　总结

5.1　调幅信号的基本特性

5.2　乘法器

5.3　平衡调制器

5.4　高电平调幅

5.5　包络检波

5.6　同步检波